

# *SQANTI3 report*

*Unique Genes: 667*  
*Unique Isoforms: 3925*

### *Transcript Classification*

| Category         | Isoforms, count |
|------------------|-----------------|
| FSM              | 539             |
| ISM              | 1138            |
| NIC              | 744             |
| NNC              | 1139            |
| Genic<br>Genomic | 109             |
| Antisense        | 31              |
| Fusion           | 38              |
| Intergenic       | 40              |
| Genic<br>Intron  | 147             |

### *Gene Classification*

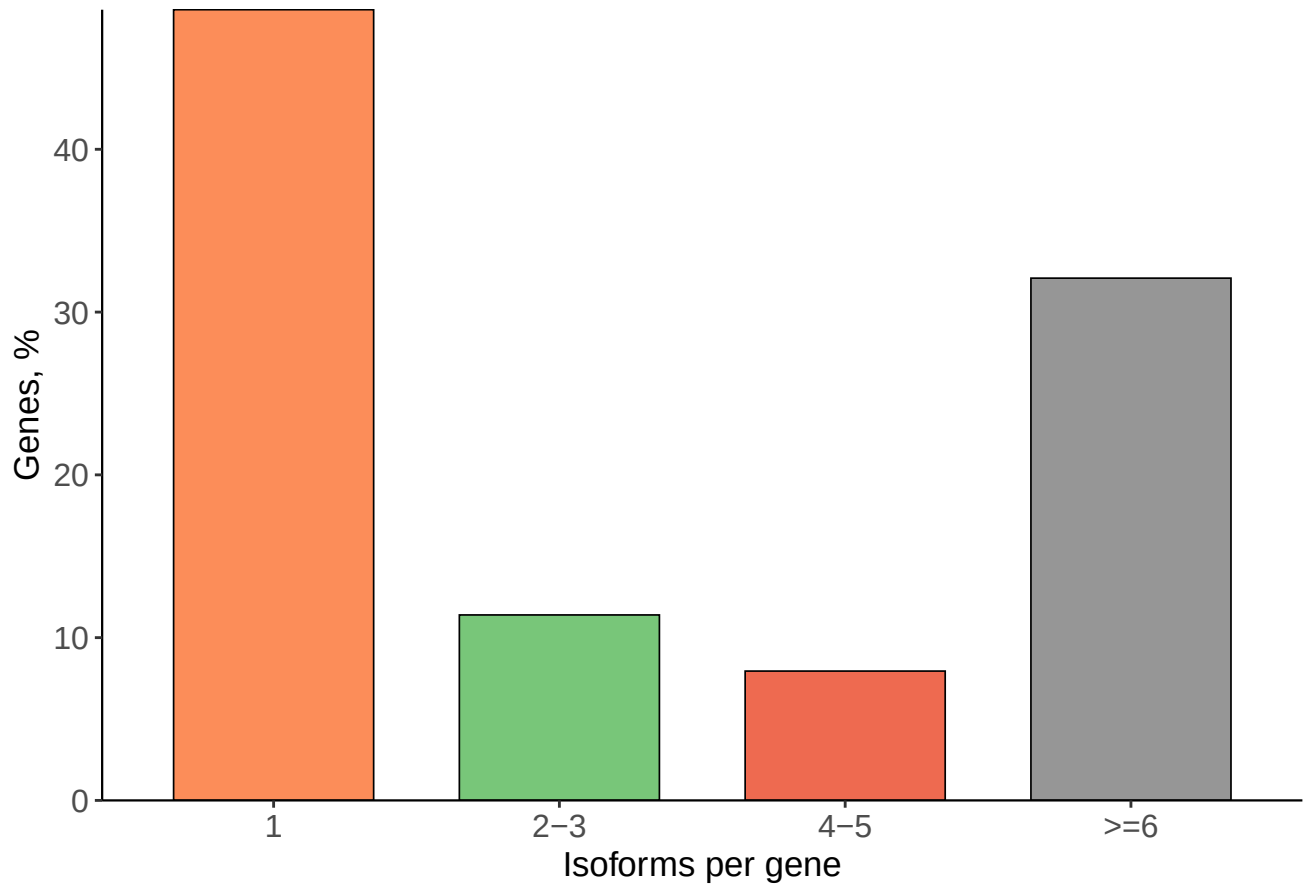
| Category        | Genes, count |
|-----------------|--------------|
| Annotated Genes | 451          |
| Novel Genes     | 216          |

### *Splice Junction Classification*

| Category            | SJs, count | Percent |
|---------------------|------------|---------|
| Known canonical     | 3374       | 69.74   |
| Known Non-canonical | 4          | 0.08    |
| Novel canonical     | 1195       | 24.70   |
| Novel Non-canonical | 265        | 5.48    |

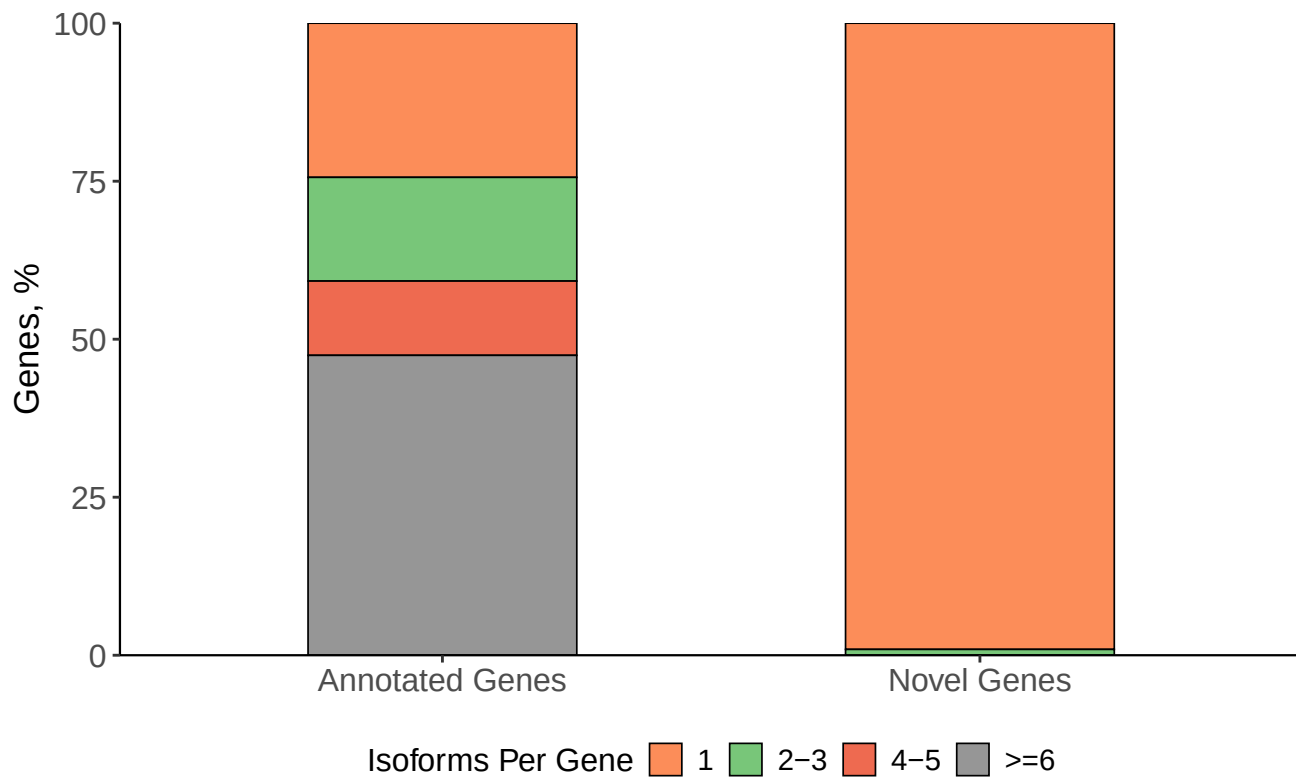
## *Gene Characterization*

## Number of Isoforms per Gene

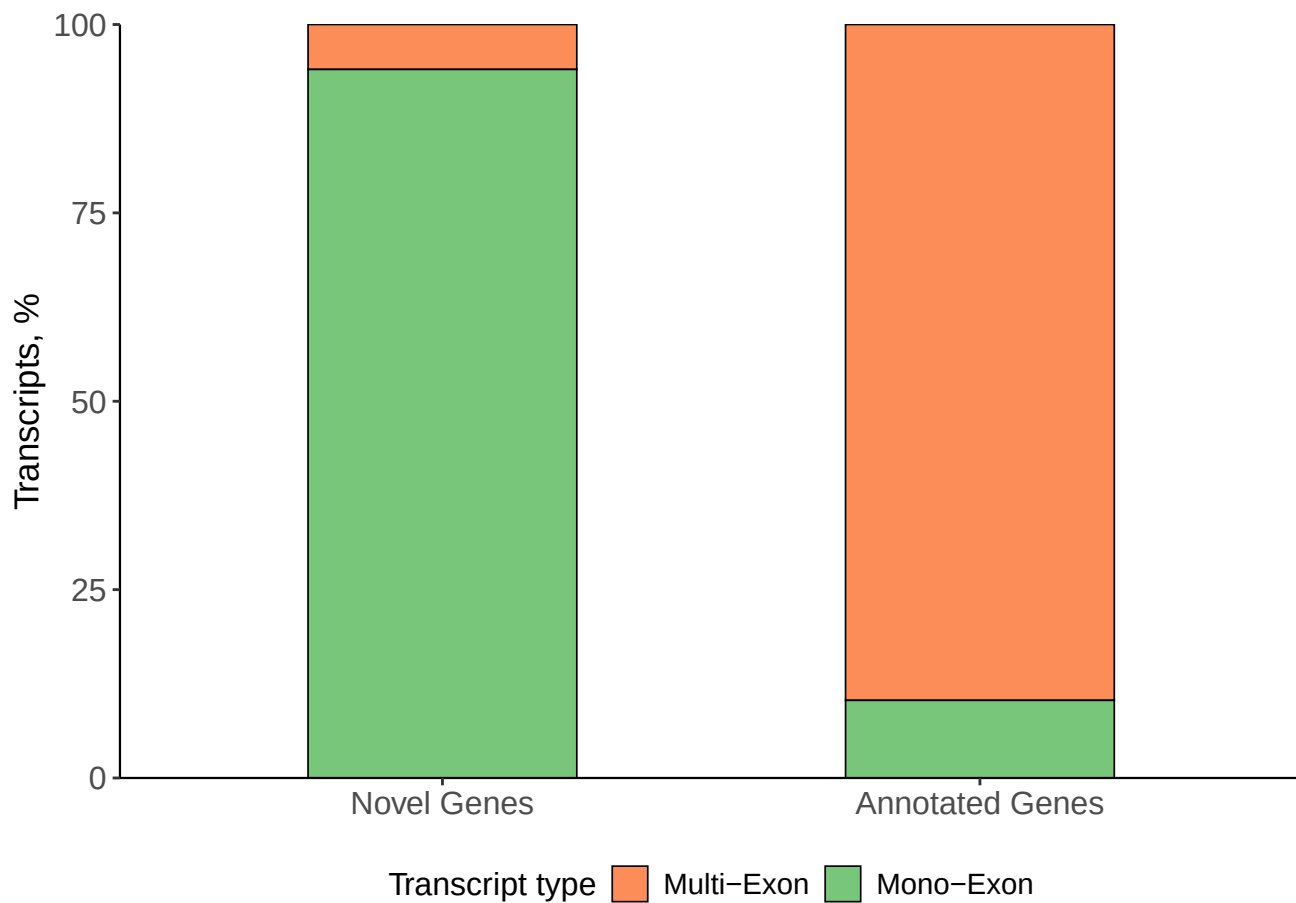


# Number of Isoforms per Gene

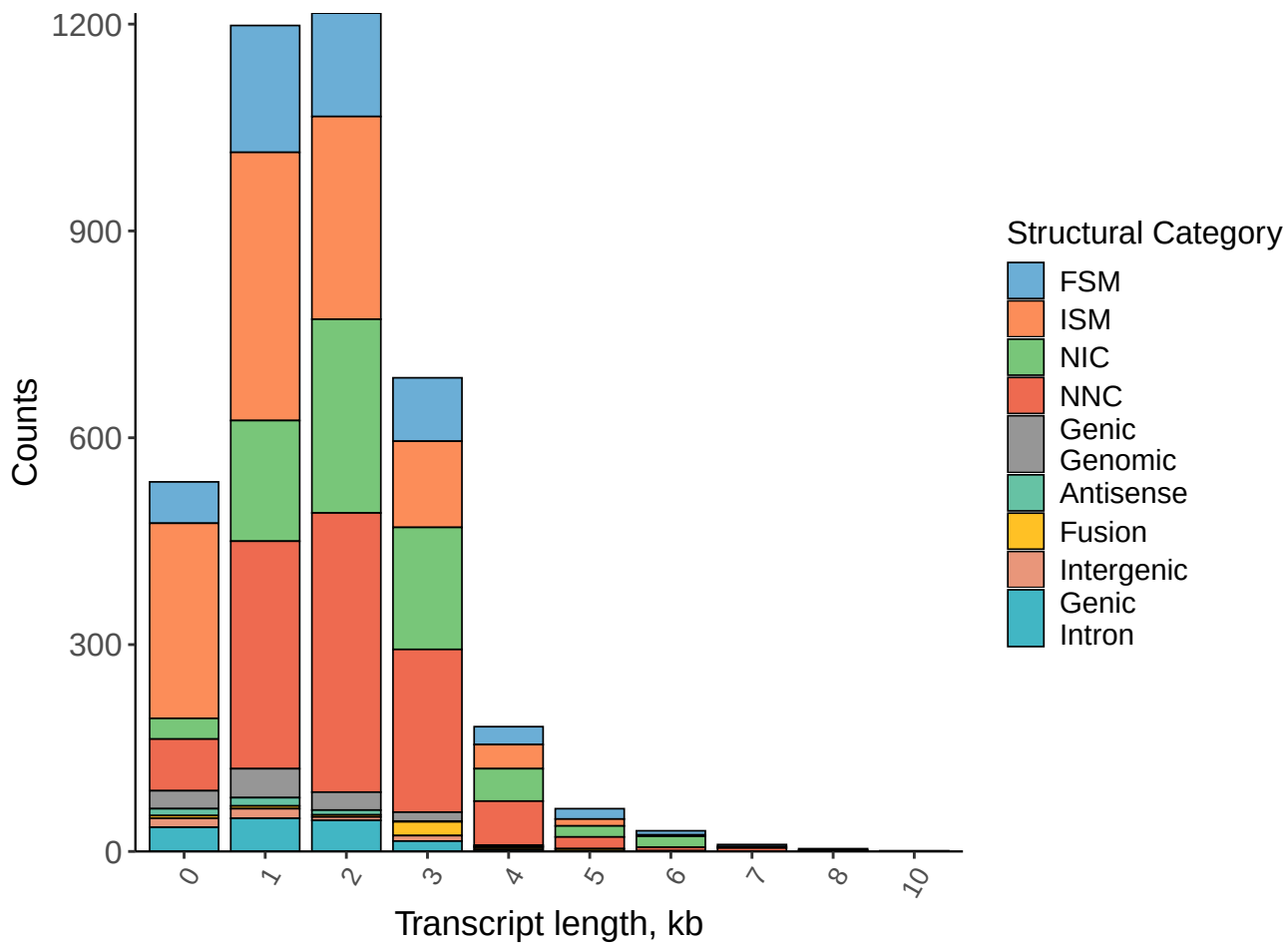
Known vs Novel Genes



Distribution of Mono- vs Multi-Exon Transcripts

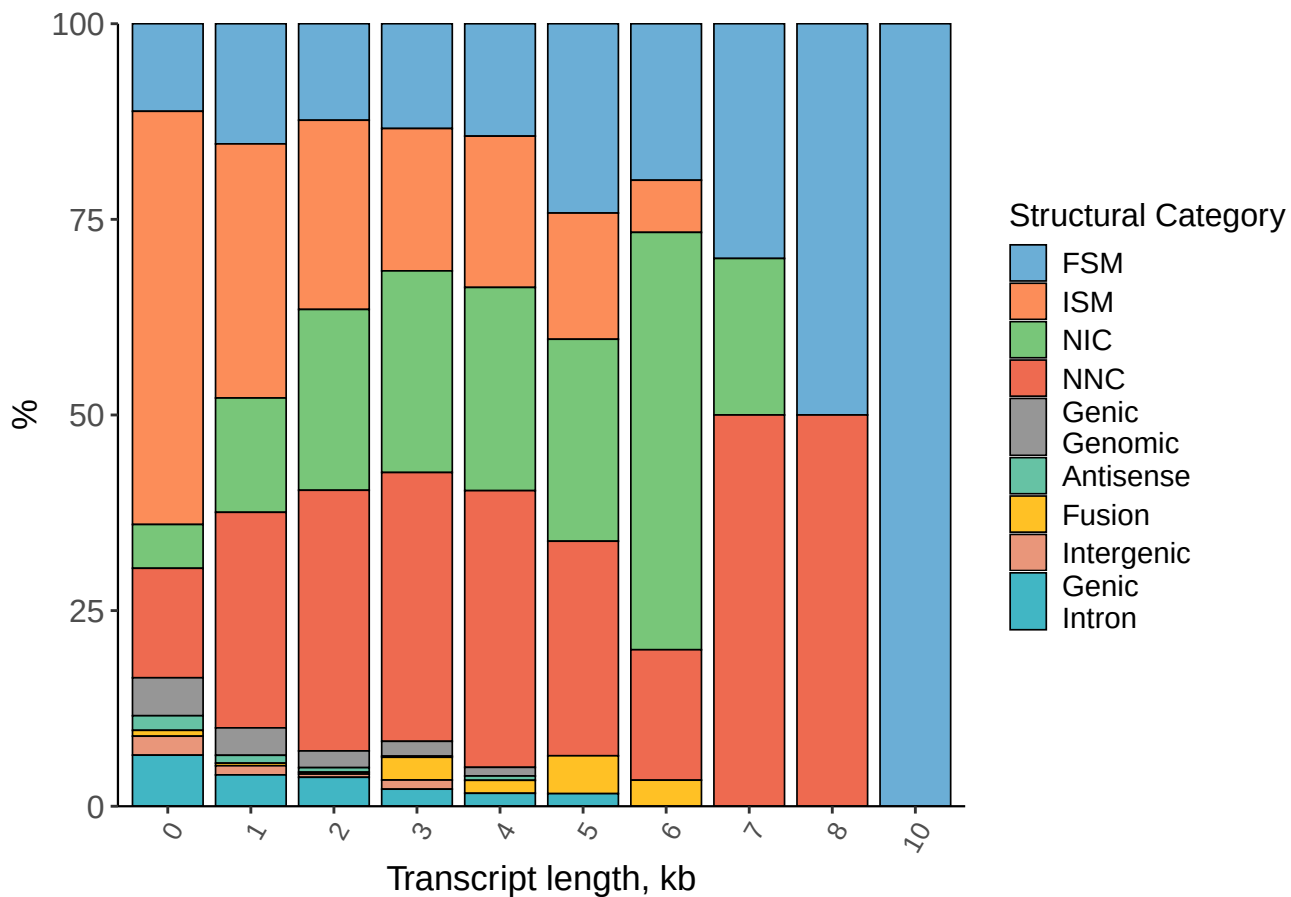


## Structural Categories by Transcript Length

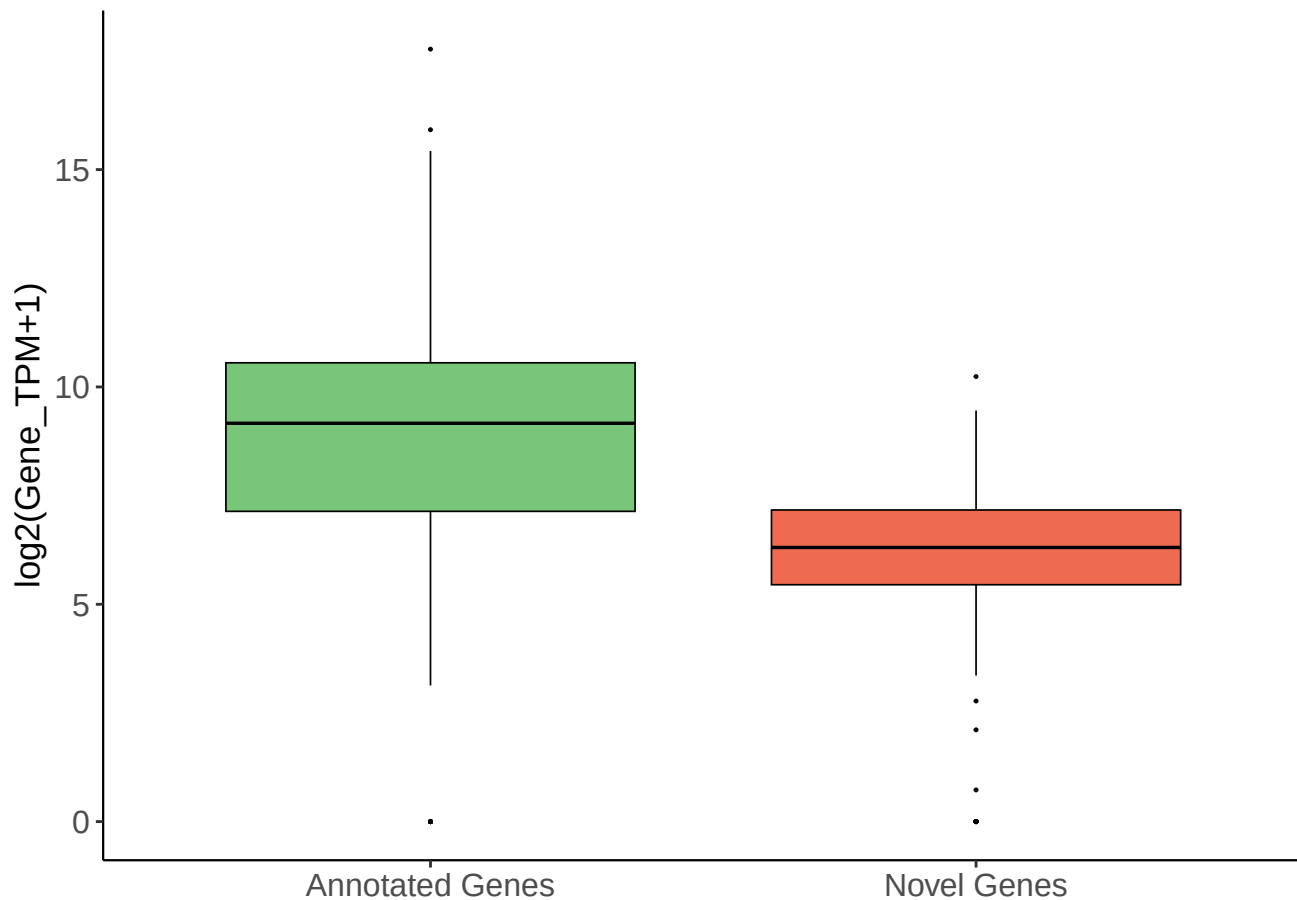




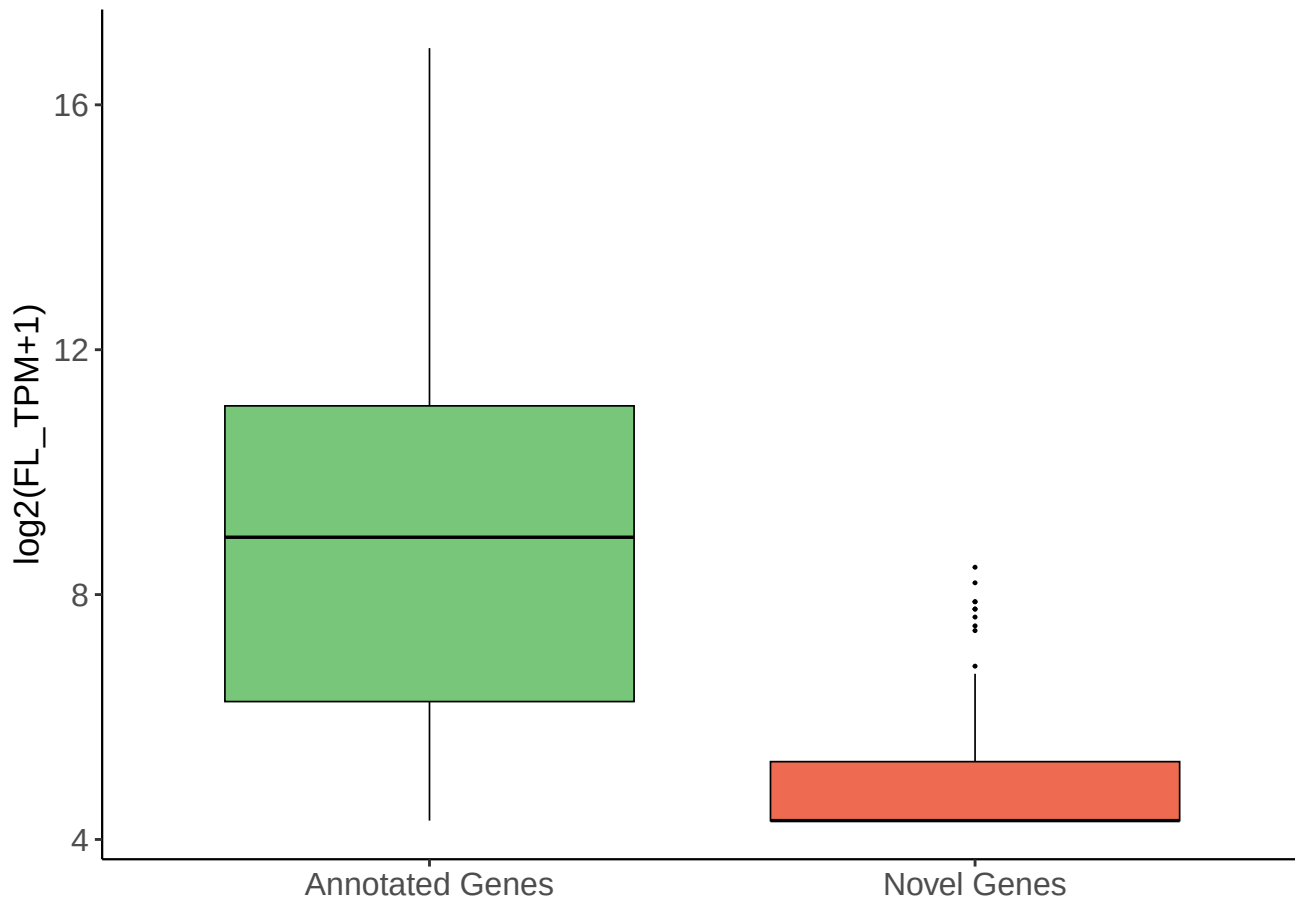
## Structural Categories by Transcript Length



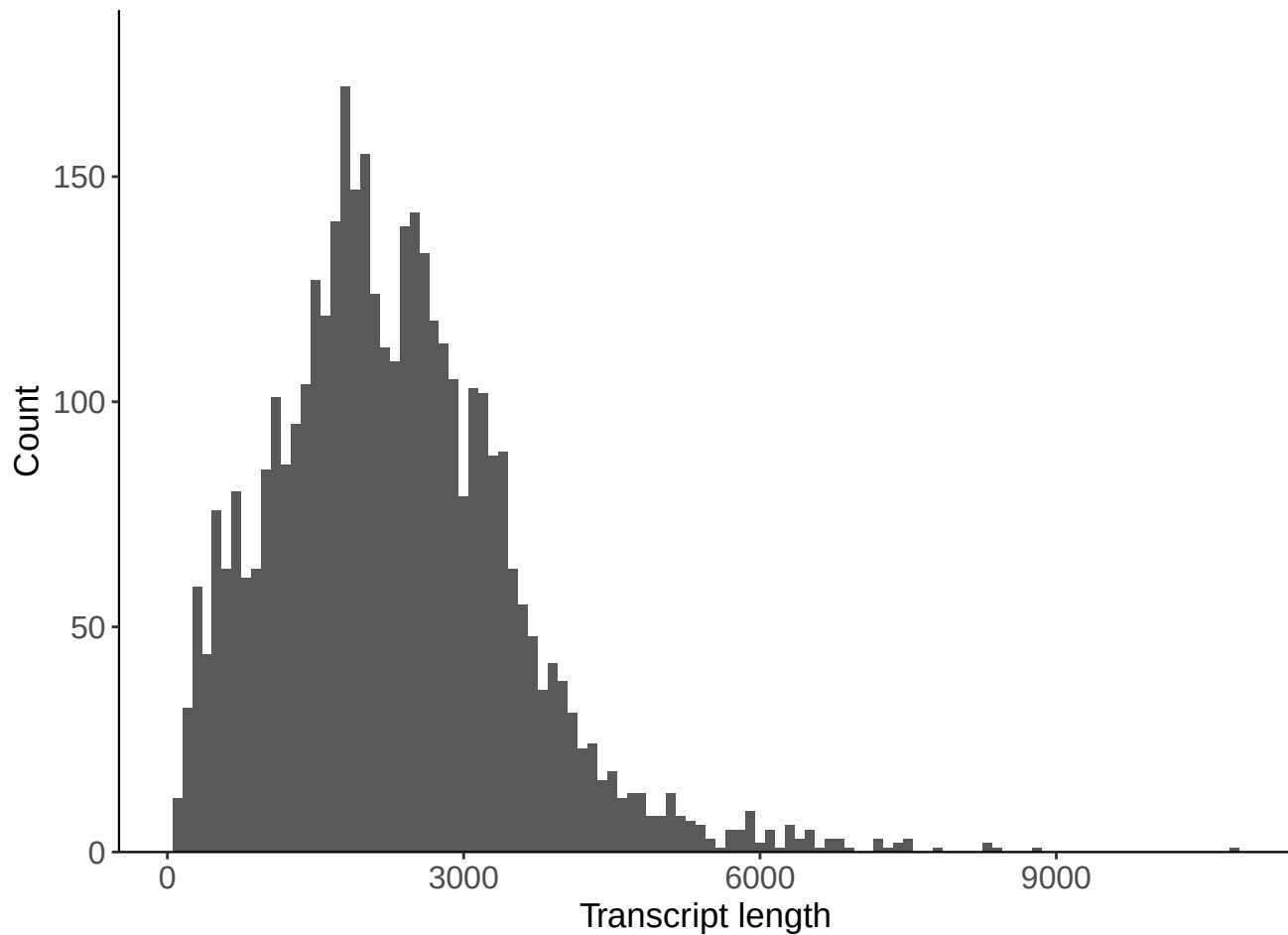
## Annotated vs Novel Gene Expression



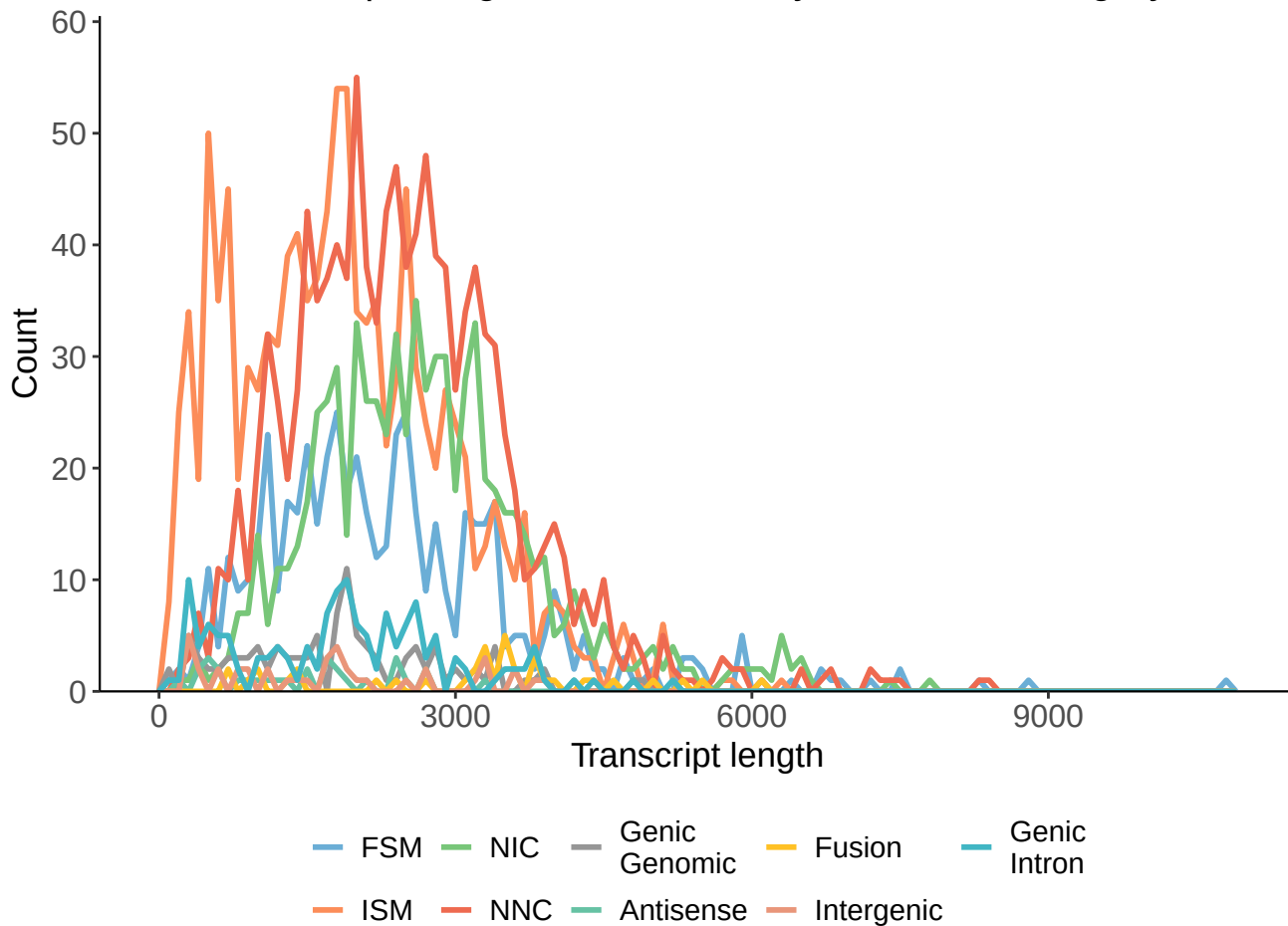
Number of FL reads per Gene by Type of Gene Annotation



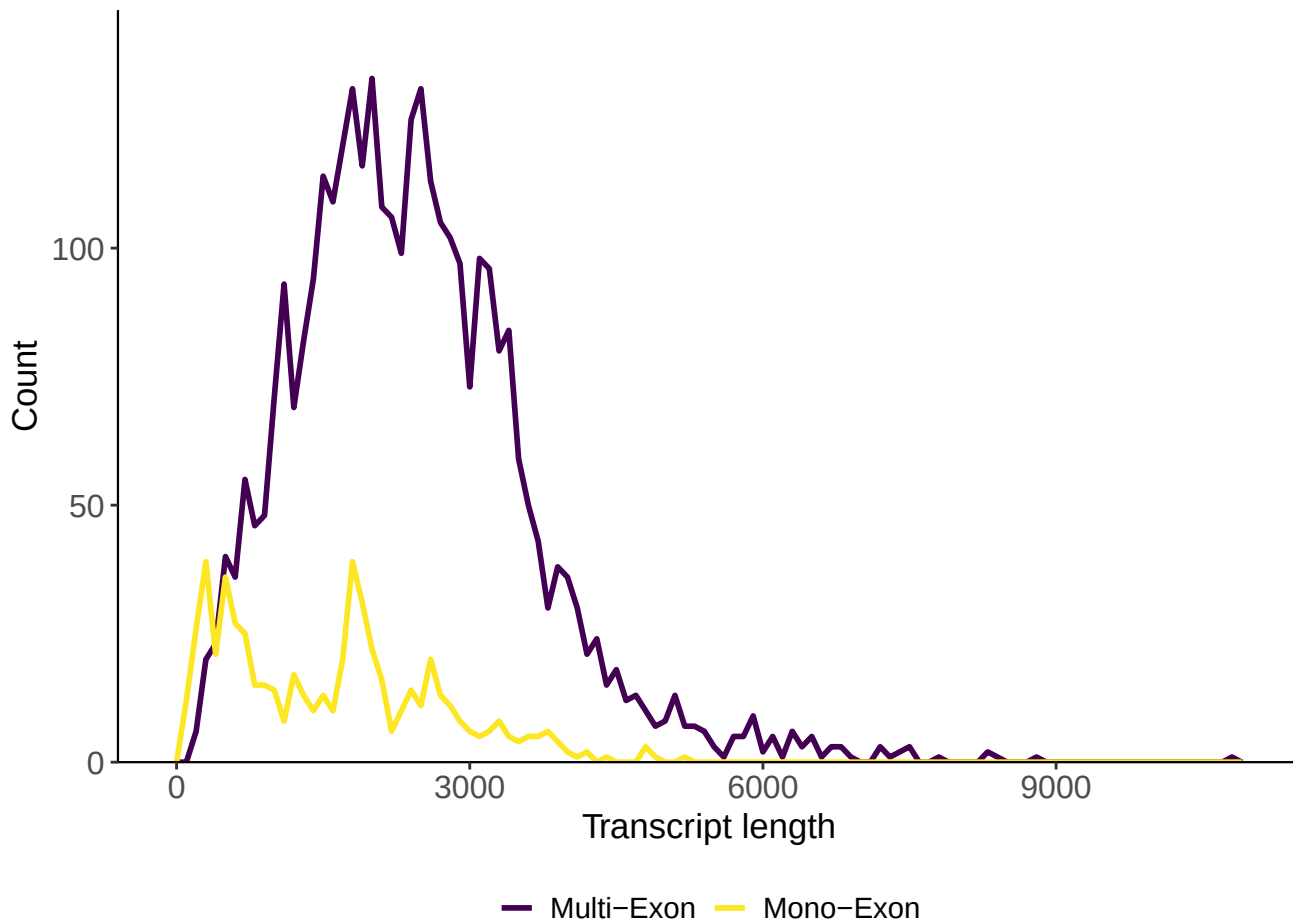
All Transcript Lengths Distribution



## Transcript Lengths Distribution by Structural Category

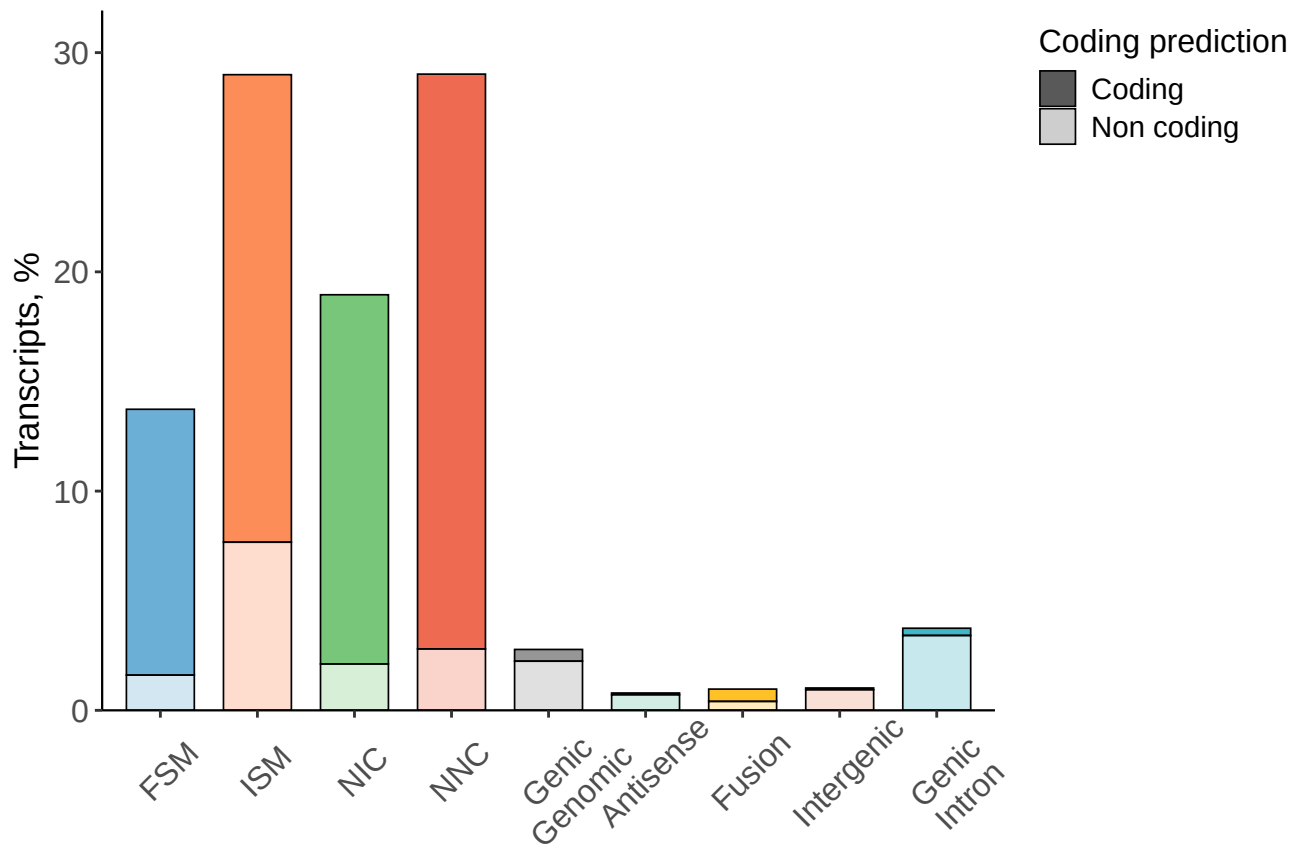


Mono- vs Multi- Exon Transcript Lengths Distribution



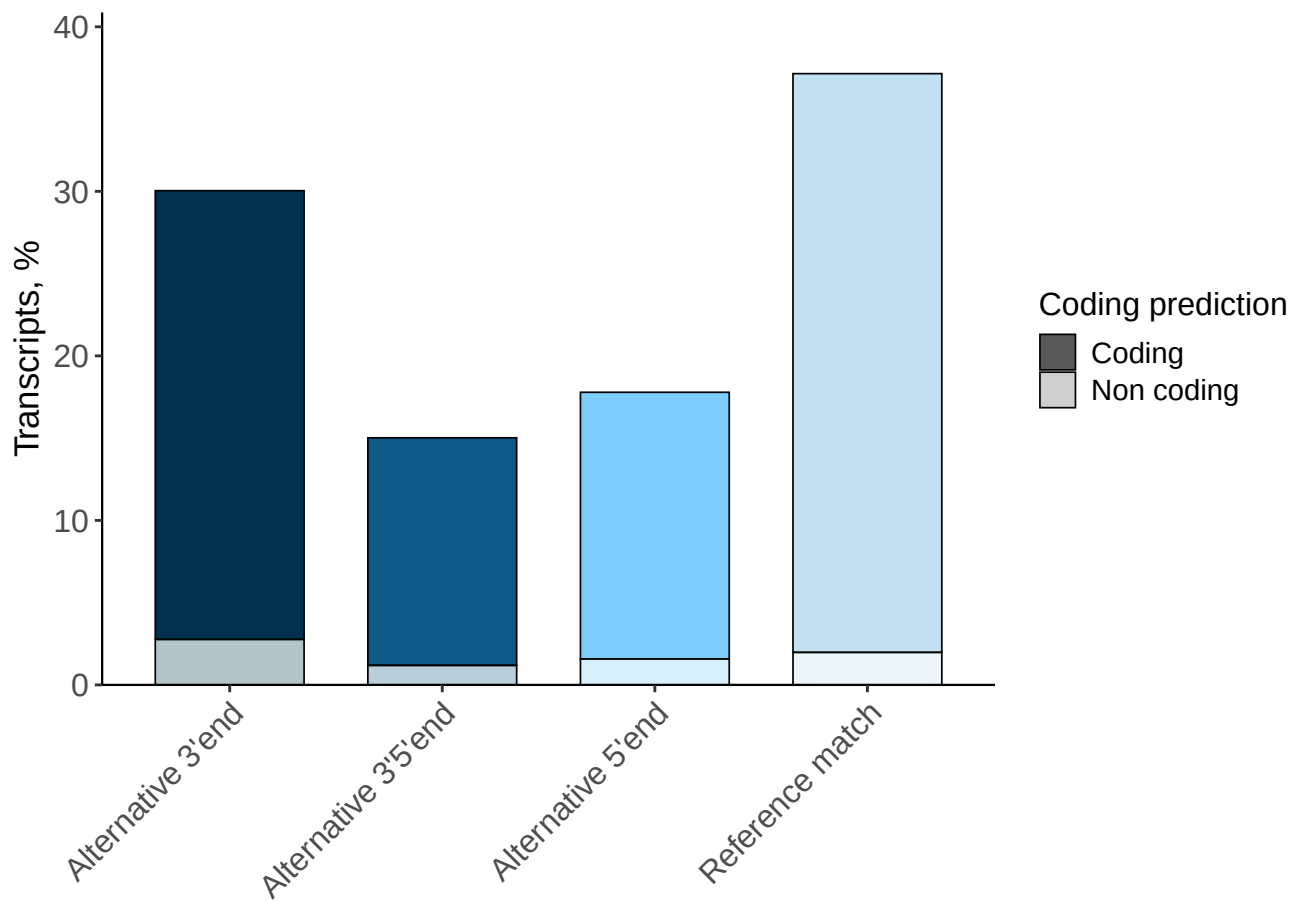
## *Structural Isoform Characterization*

## Isoform Distribution Across Structural Categories

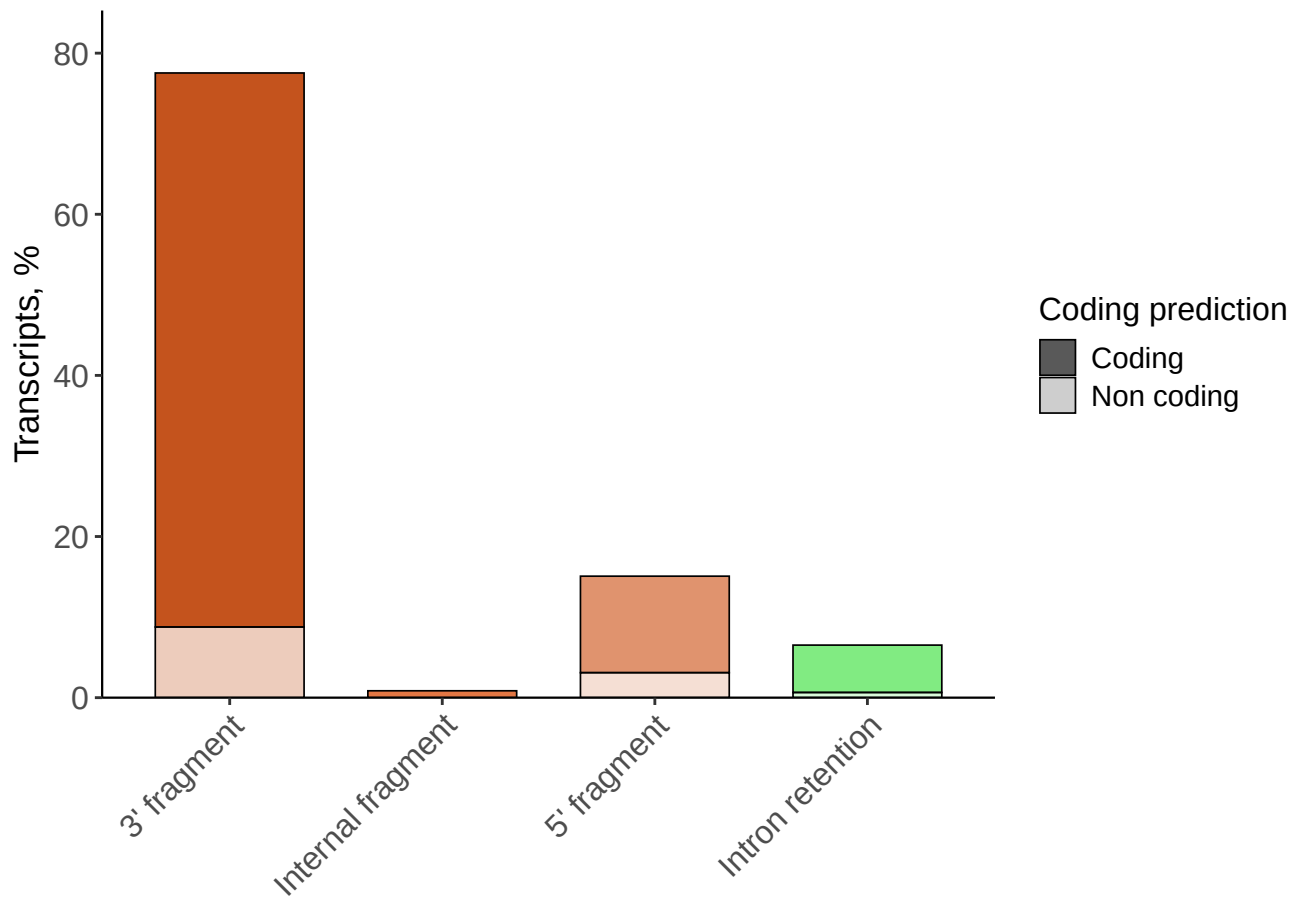




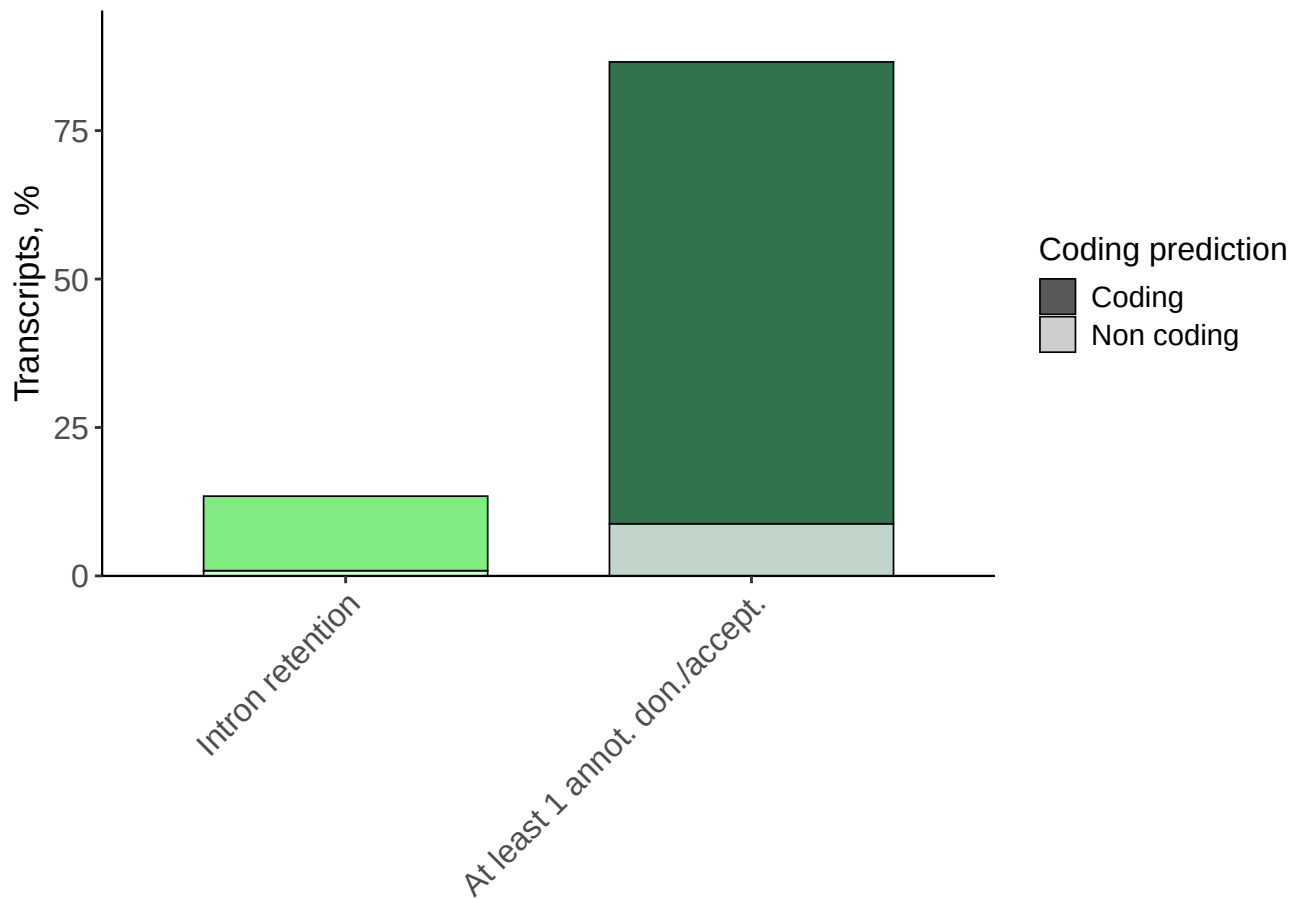
## Isoform Distribution Across FSM



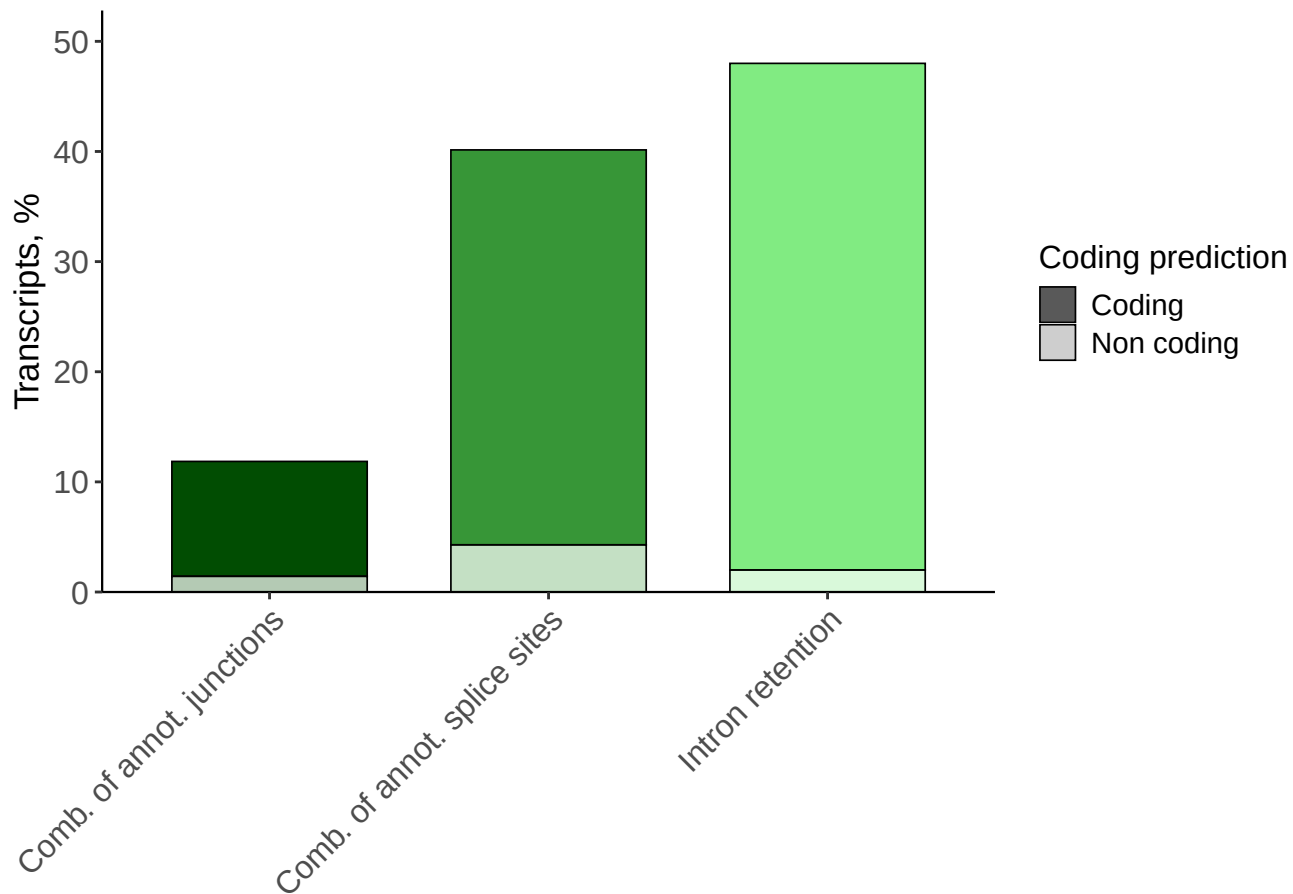
## Isoform Distribution Across ISM



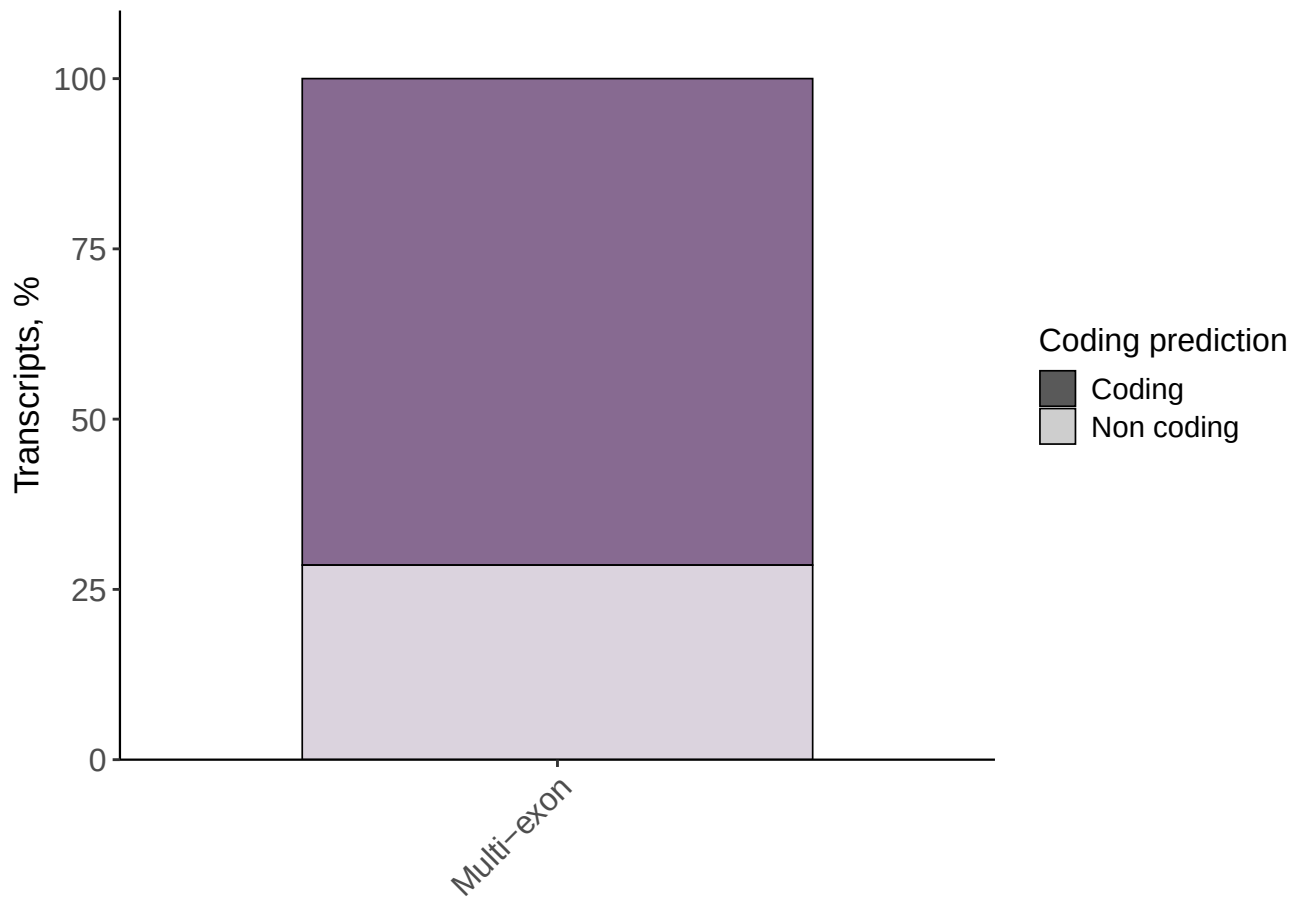
## Isoform Distribution Across NNC



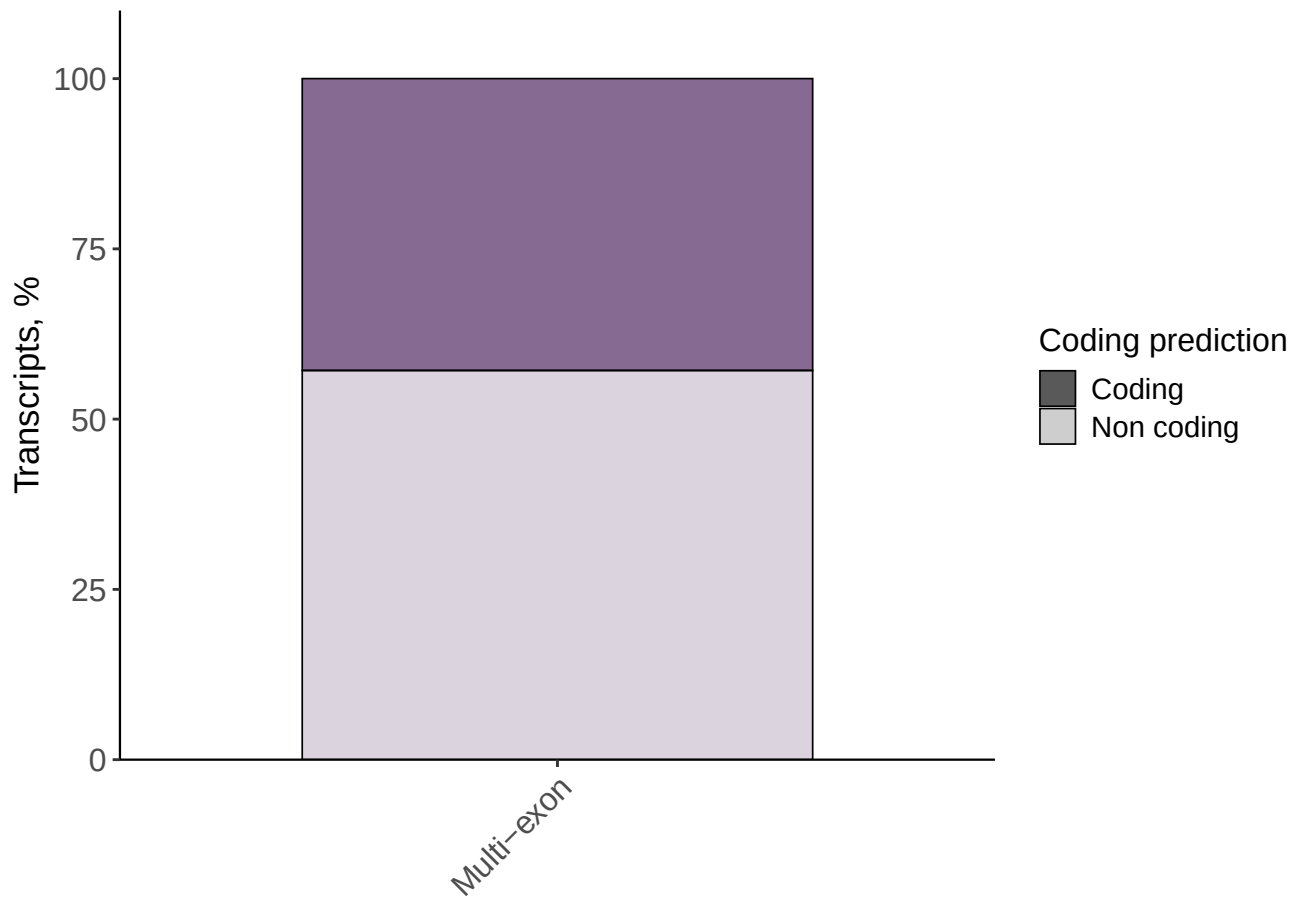
## Isoform Distribution Across NIC



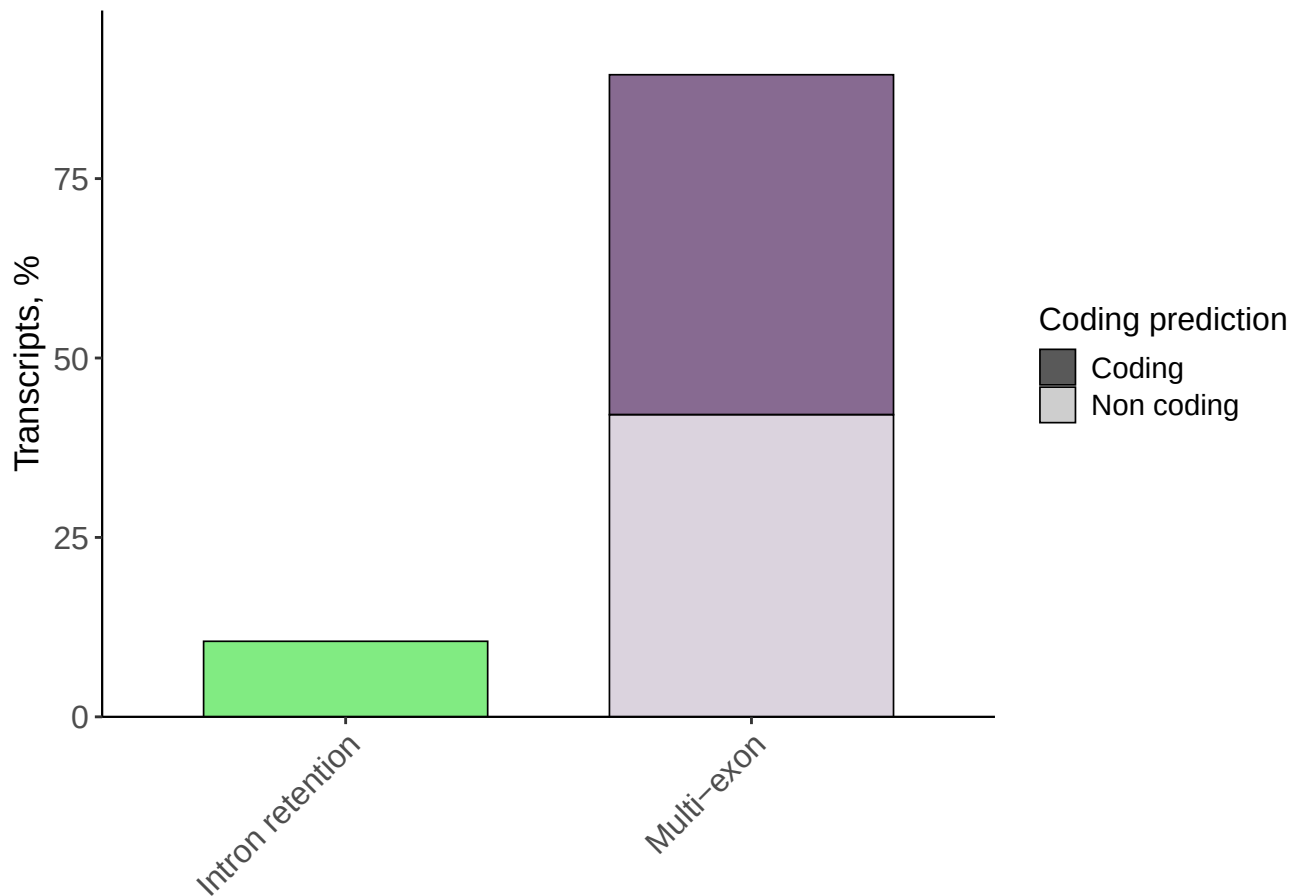
## Isoform Distribution Across Genic Genomic



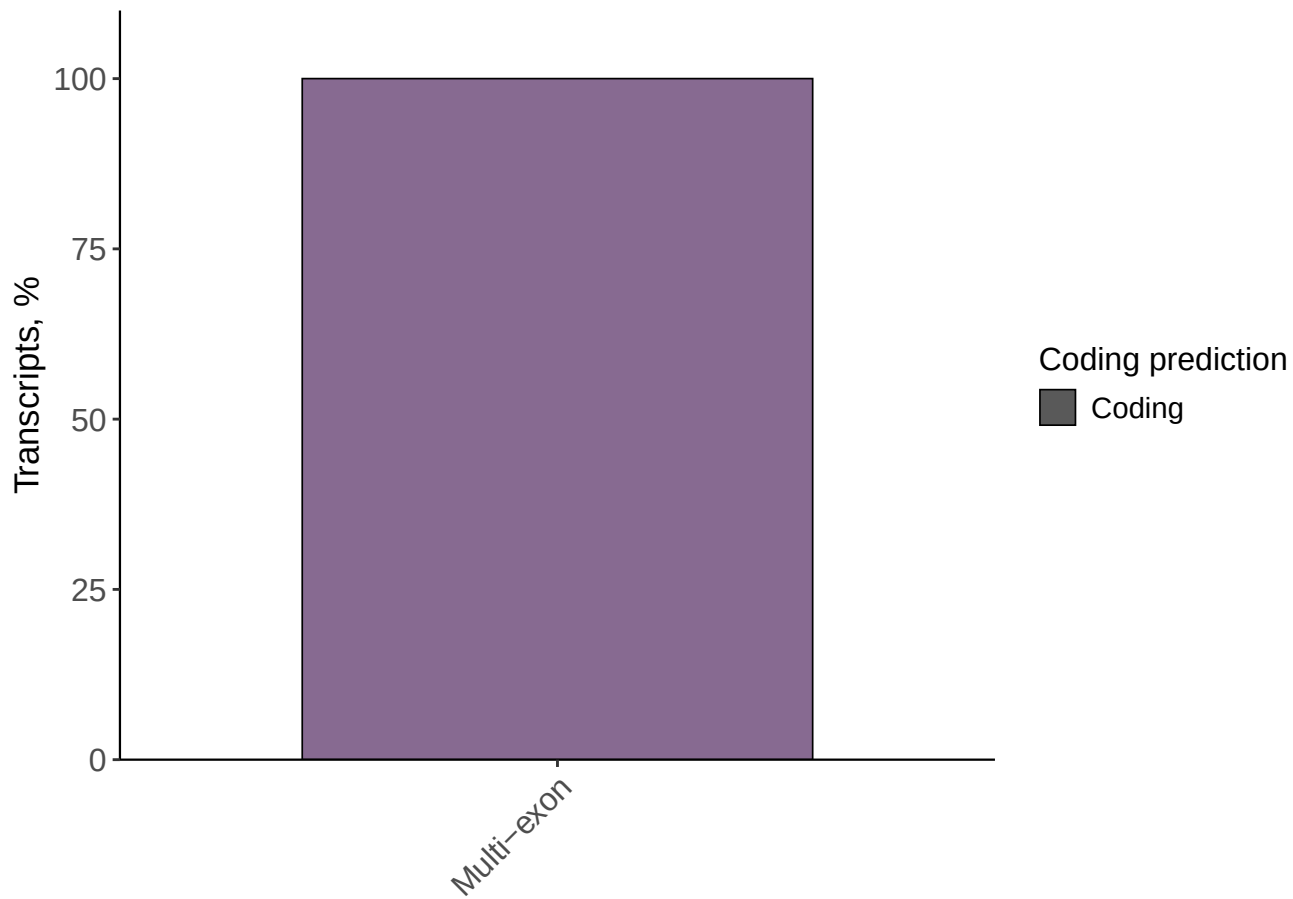
## Isoform Distribution Across Antisense



## Isoform Distribution Across Fusion

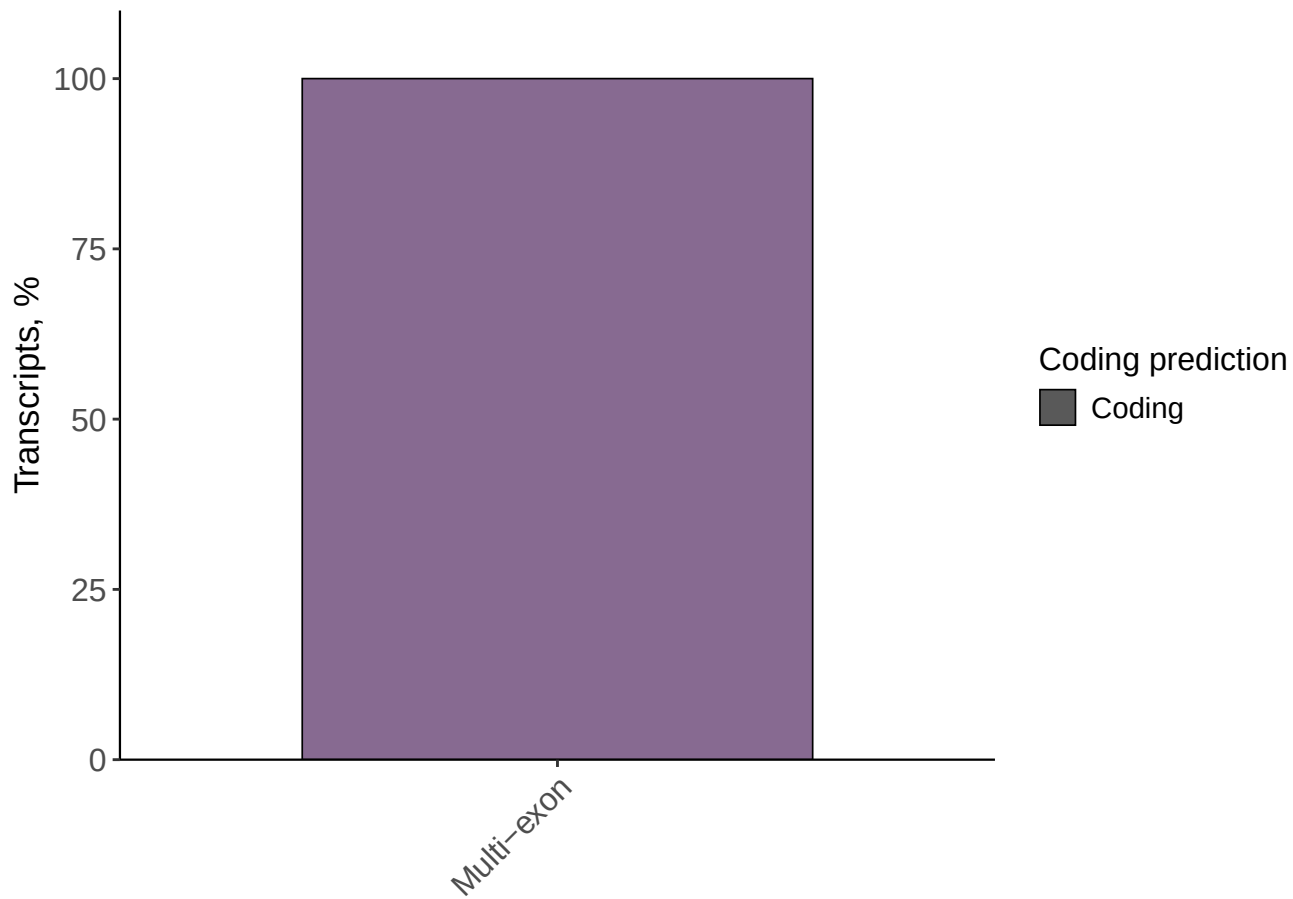


## Isoform Distribution Across Intergenic

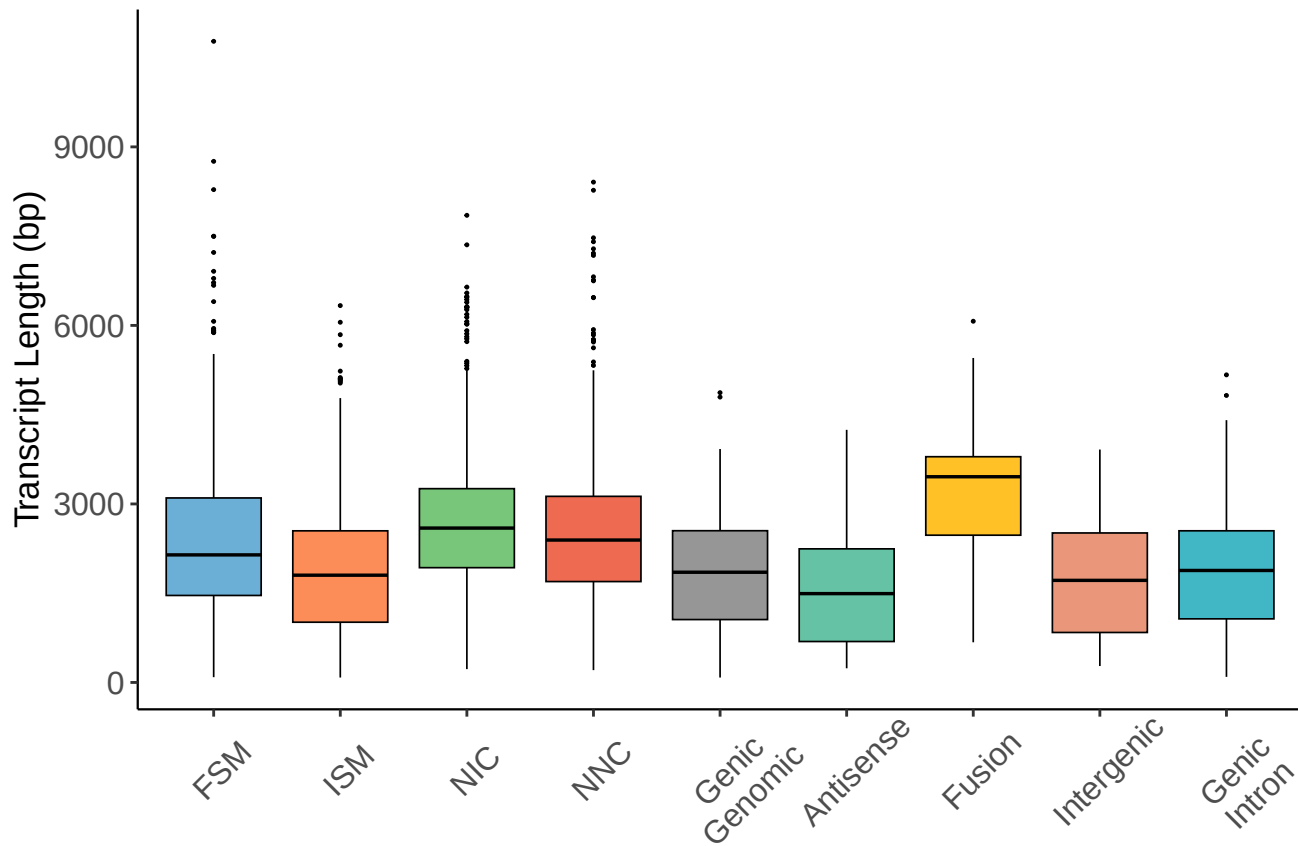




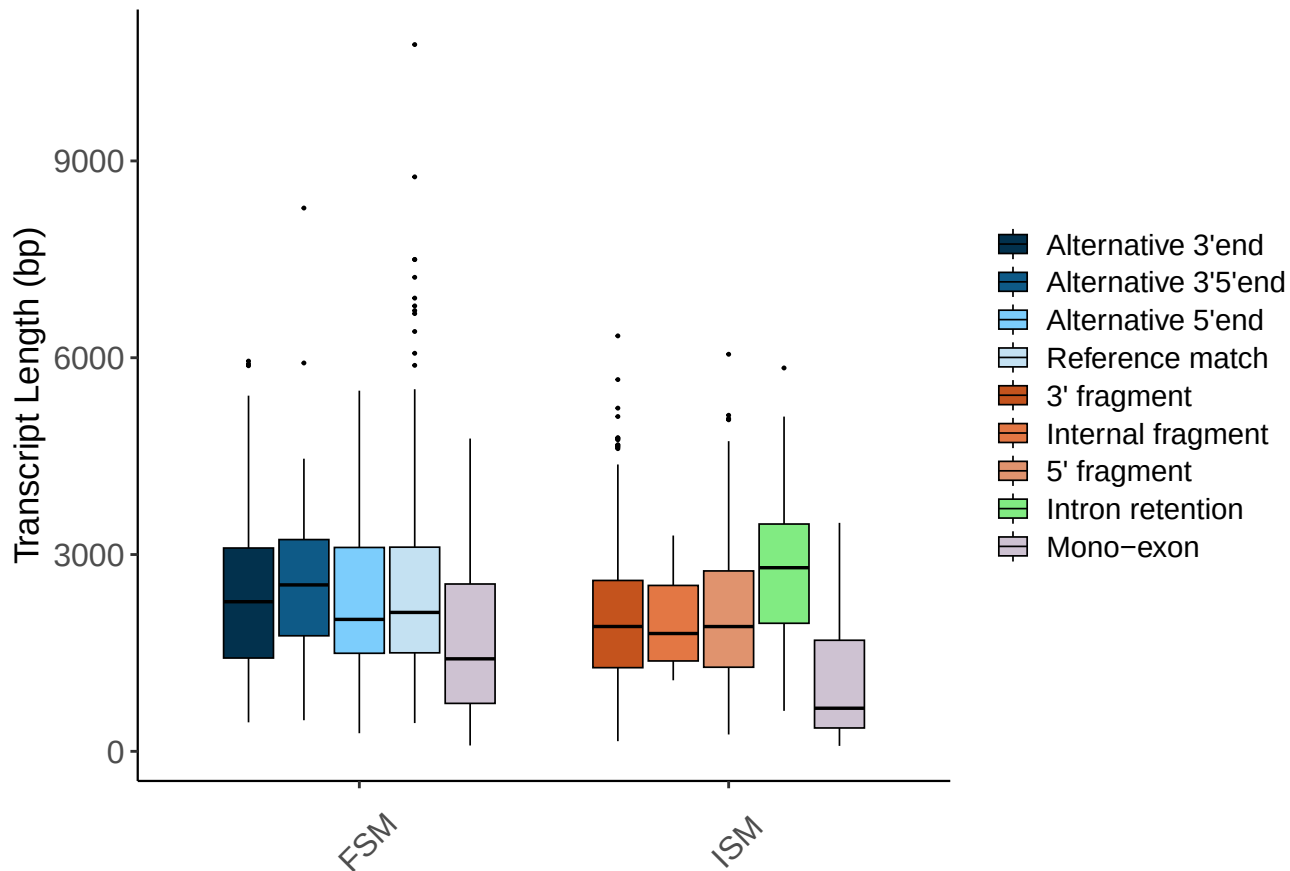
## Isoform Distribution Across Genic Intron



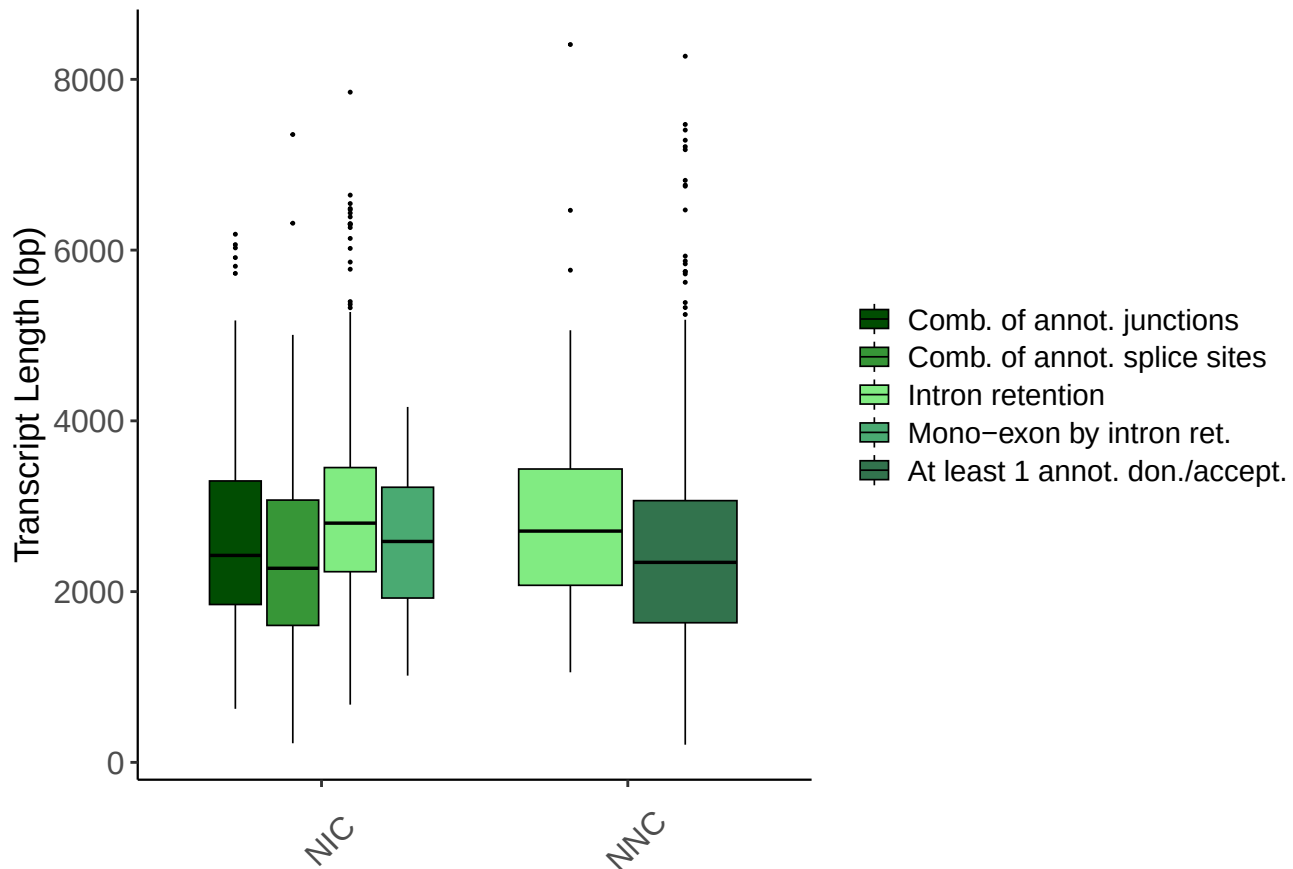
Transcript Lengths by Structural Classification



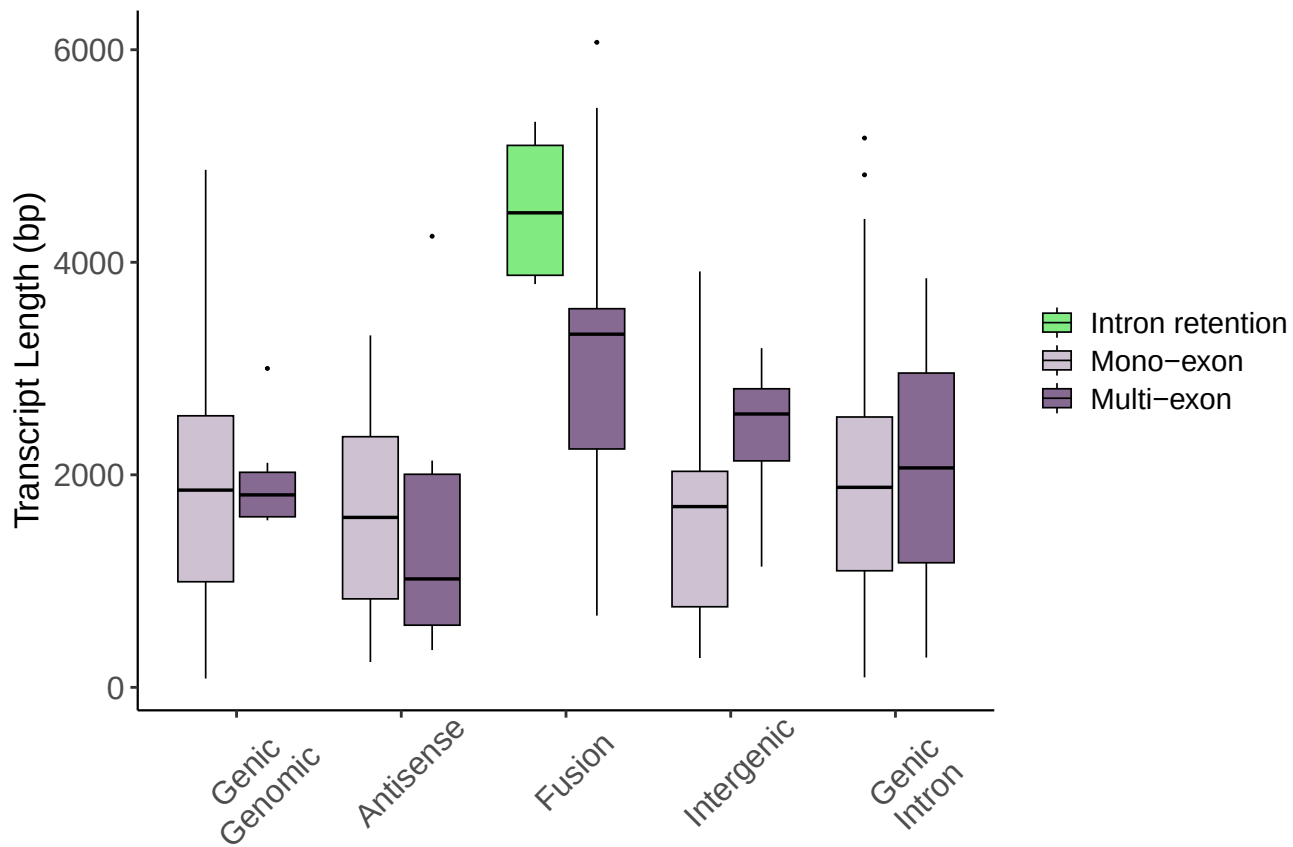
Transcript Lengths by Subcategory



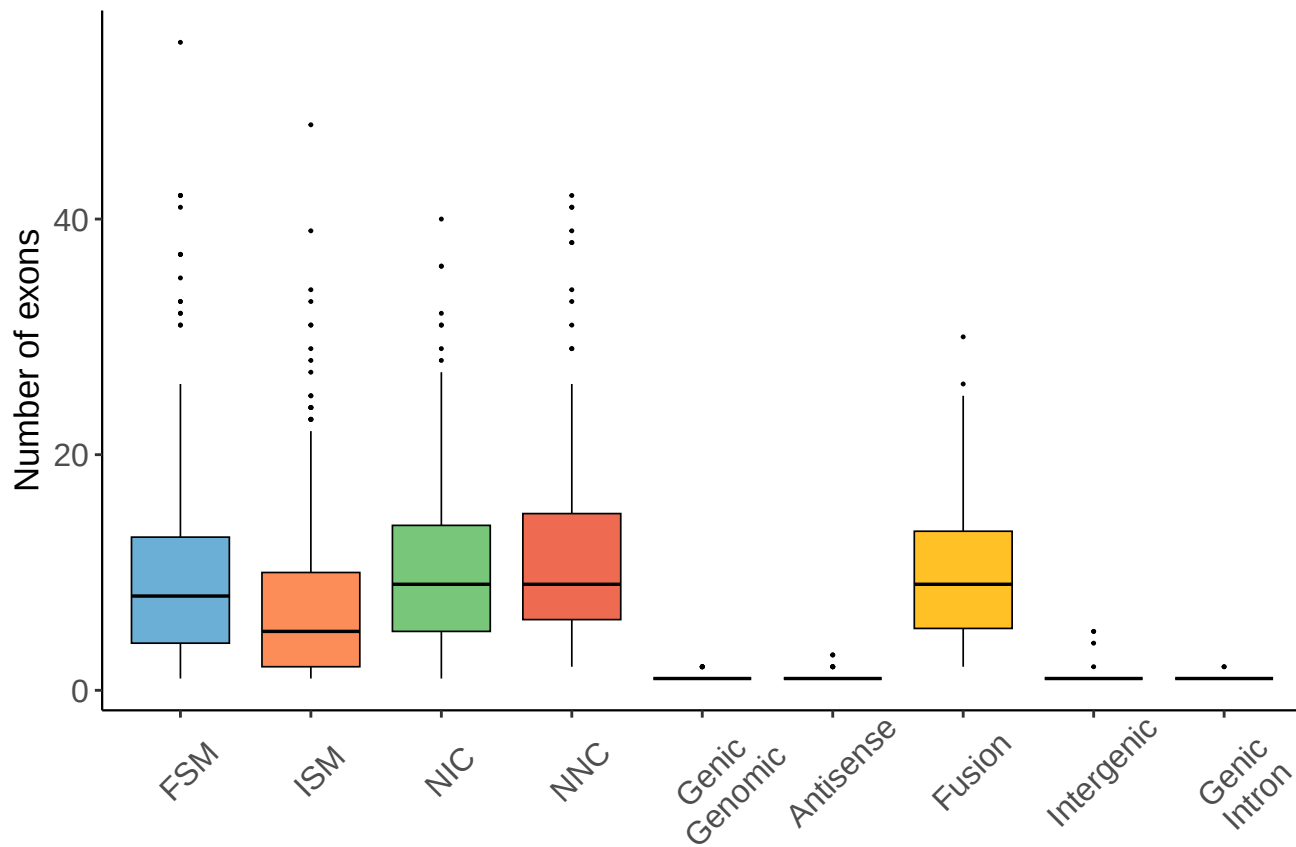
Transcript Lengths by Subcategory



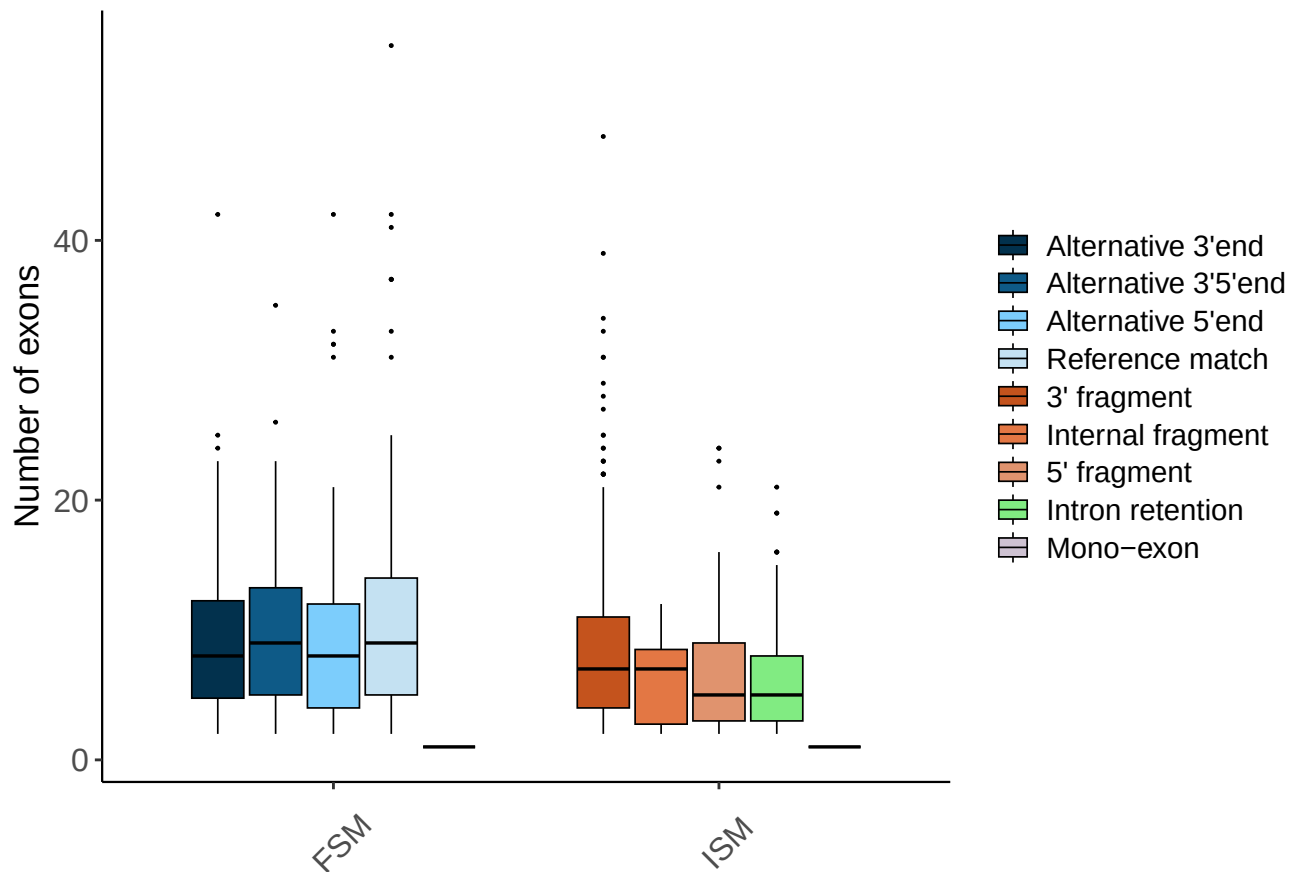
Transcript Lengths by Subcategory



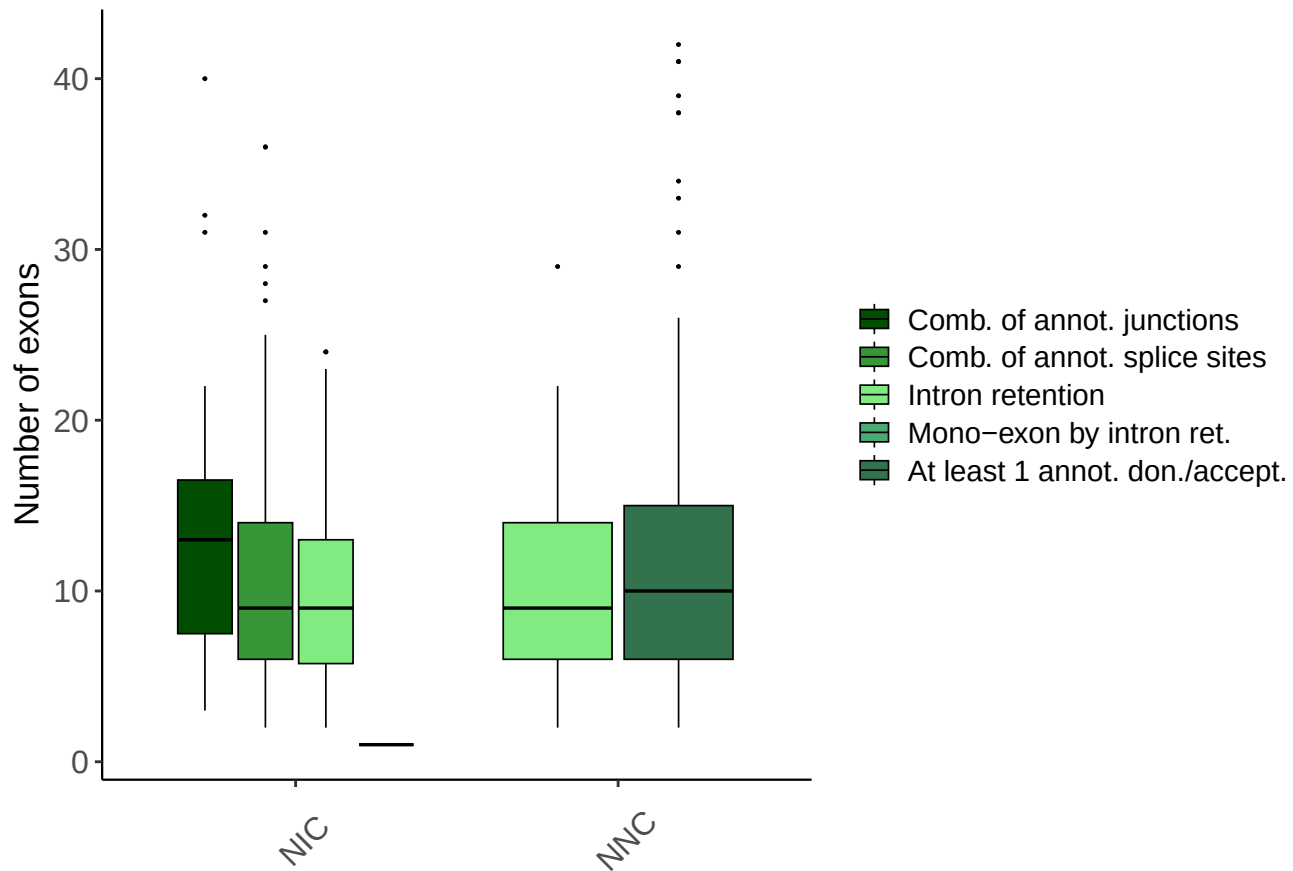
Exon Counts by Structural Classification



Exon Counts by Subcategory

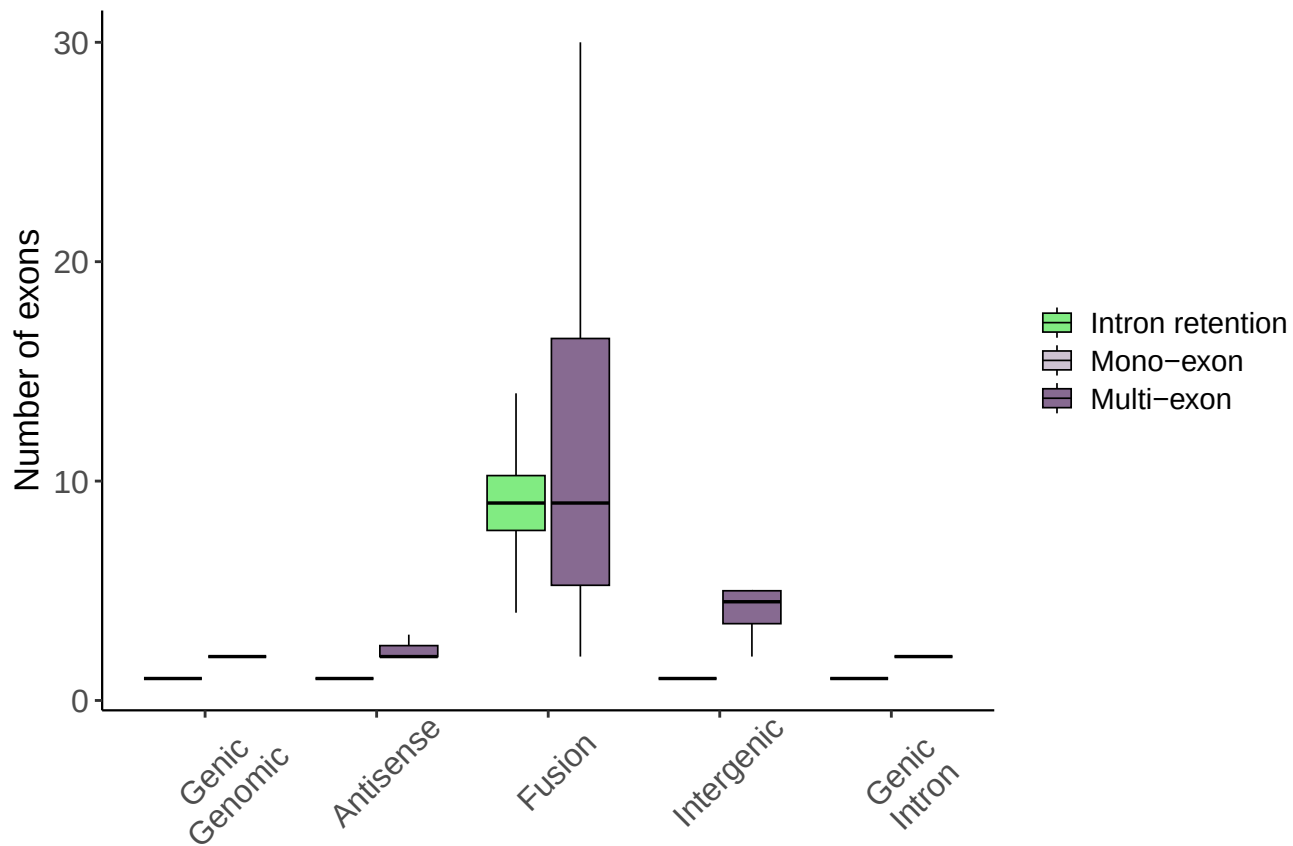


Exon Counts by Subcategory

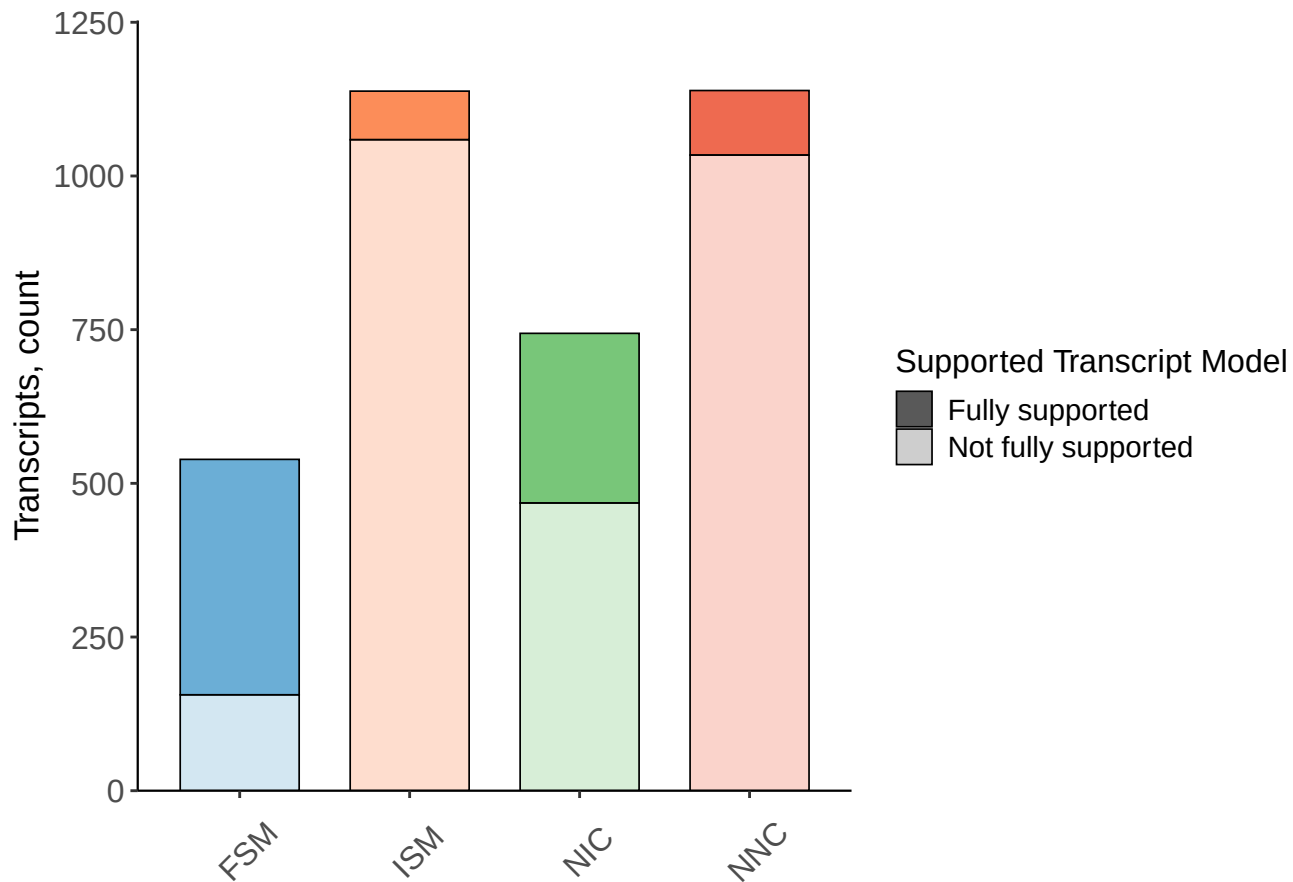




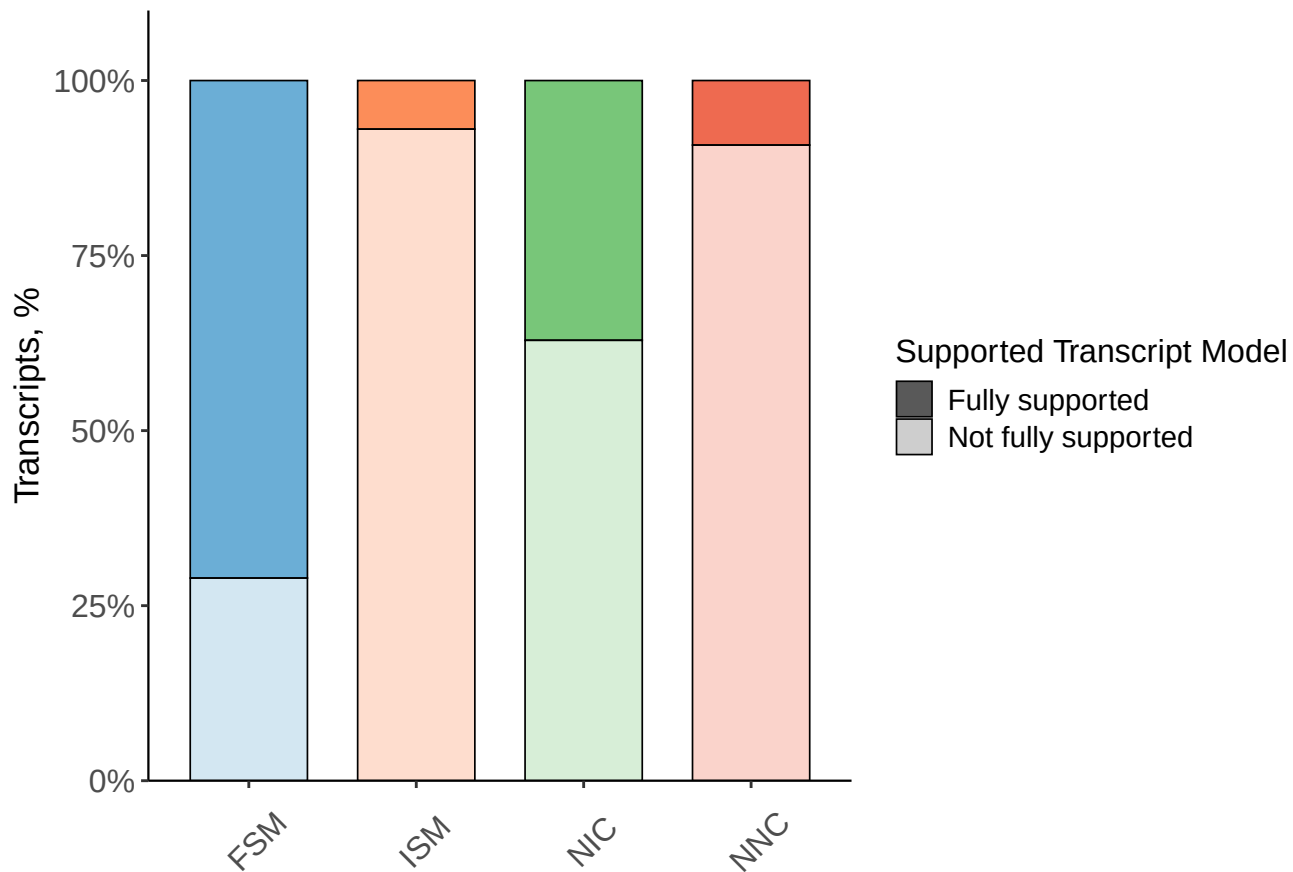
Exon Counts by Subcategory



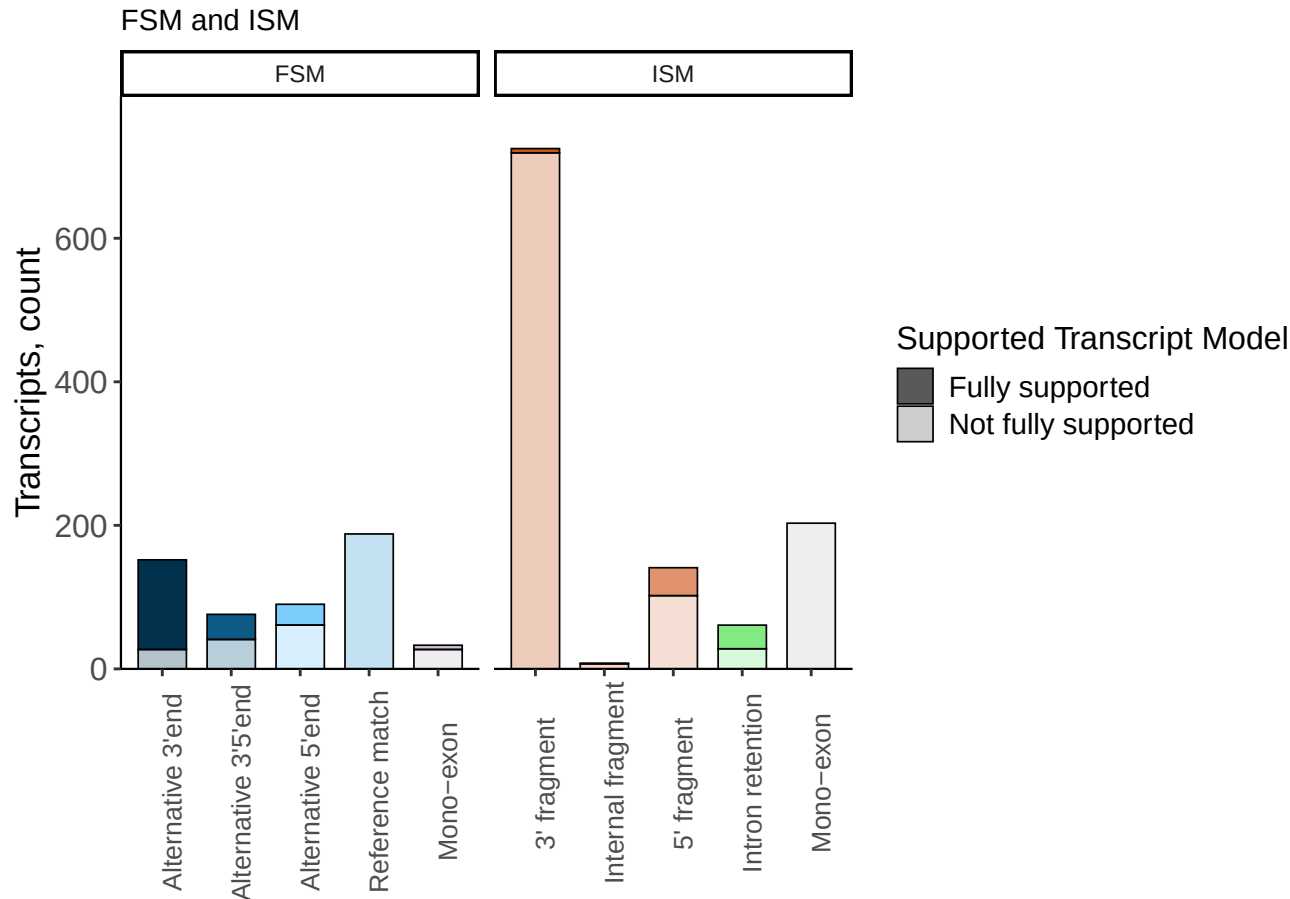
## Isoform Distribution Across Structural Categories



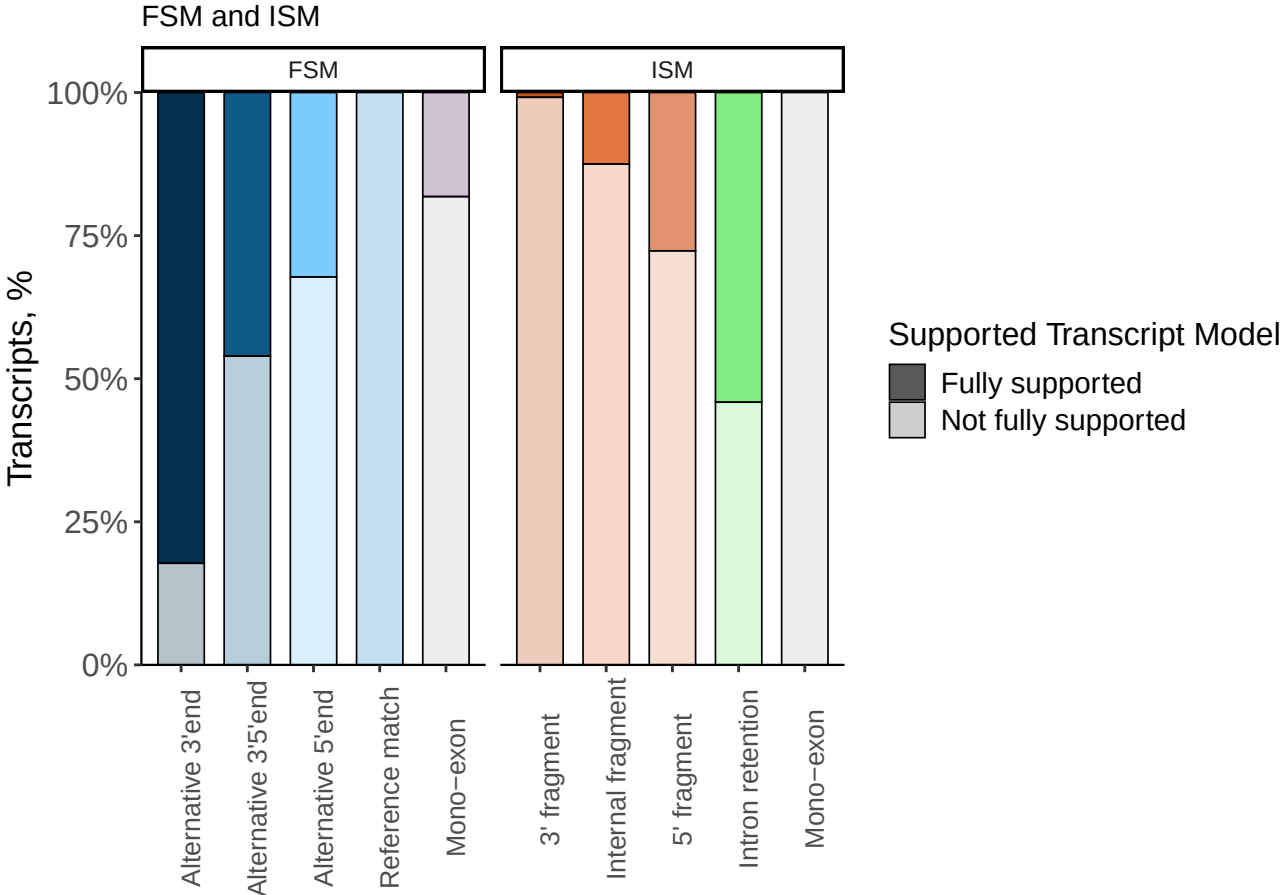
## Isoform Distribution Across Structural Categories



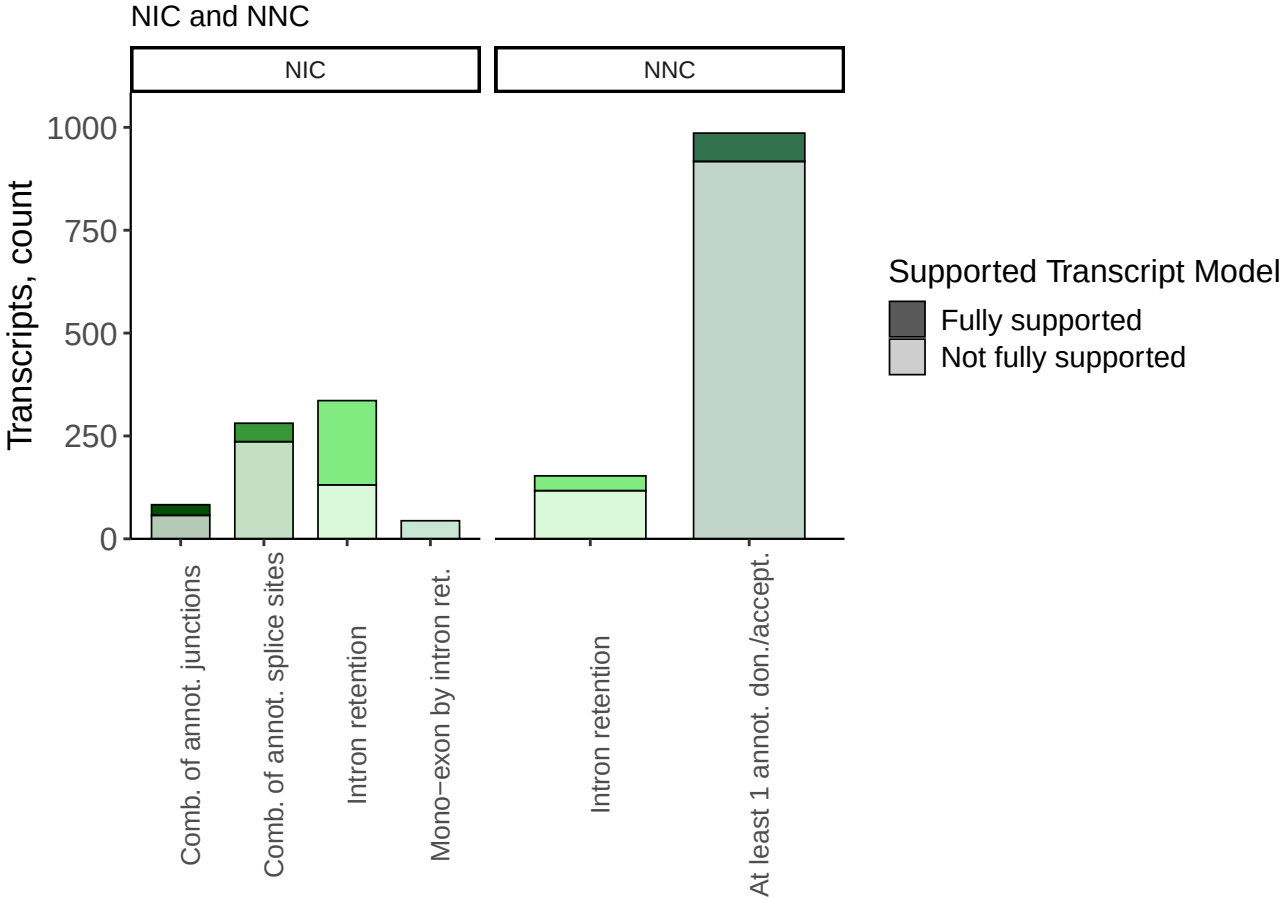
# Isoform Distribution Across Structural Subcategories



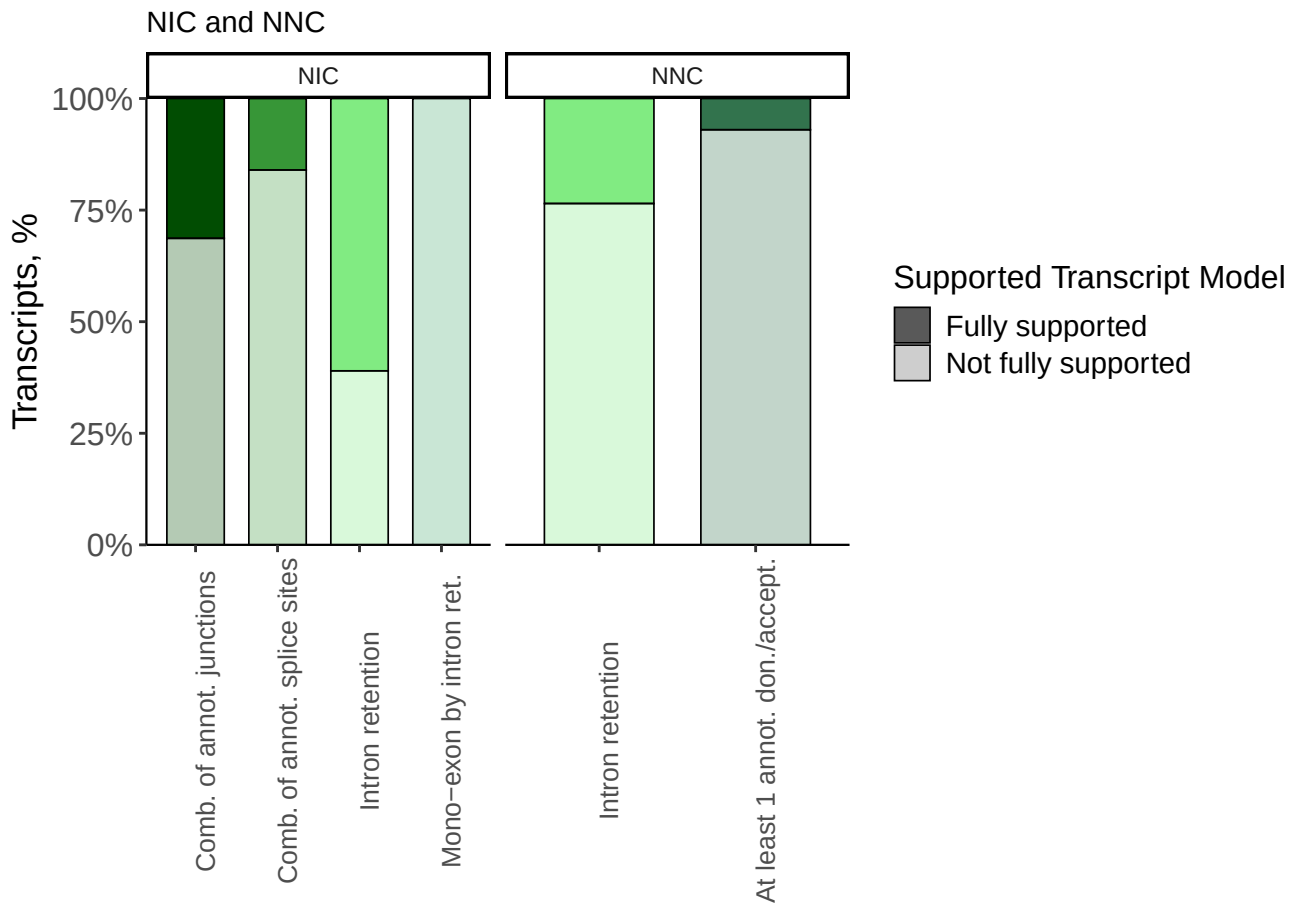
# Isoform Distribution Across Structural Subcategories



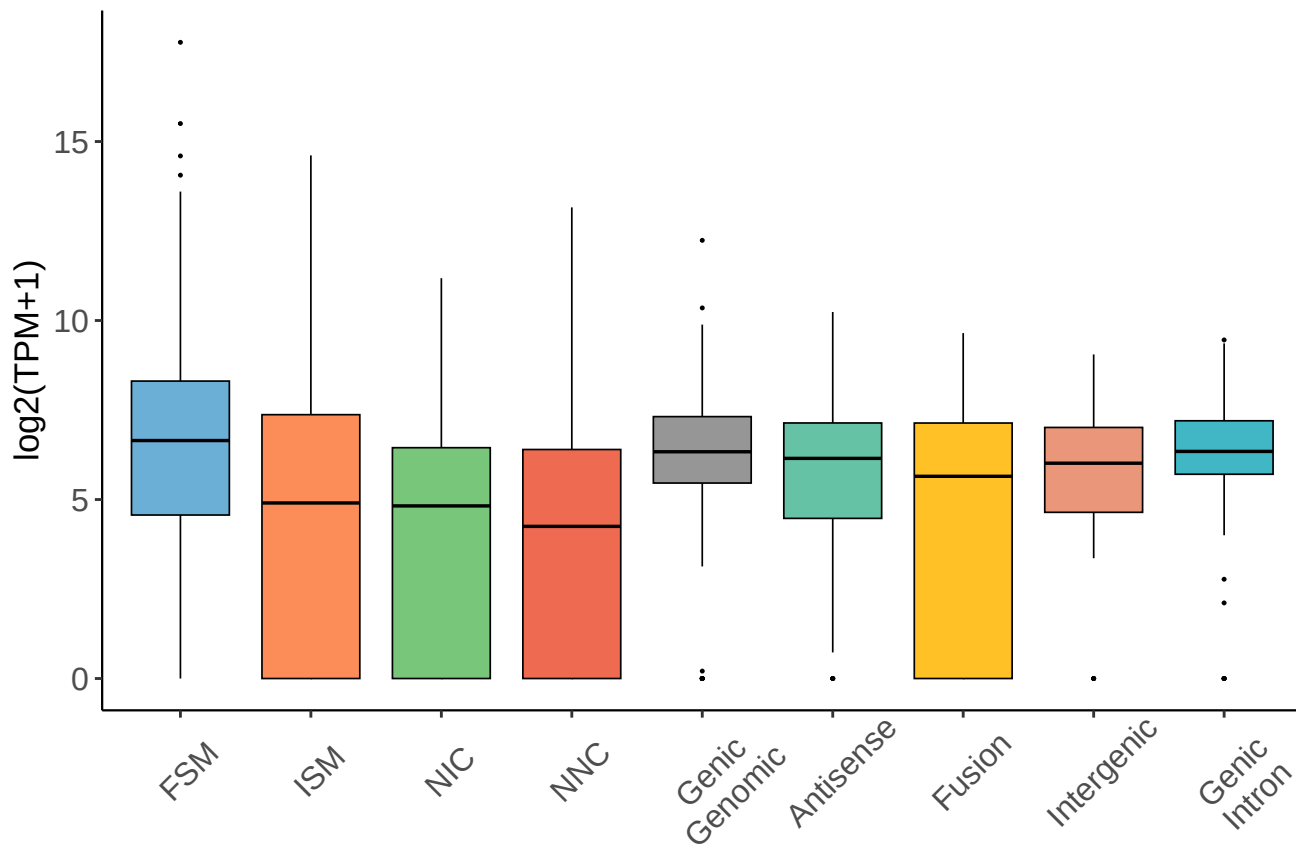
# Isoform Distribution Across Structural Subcategories



## Isoform Distribution Across Structural Subcategories

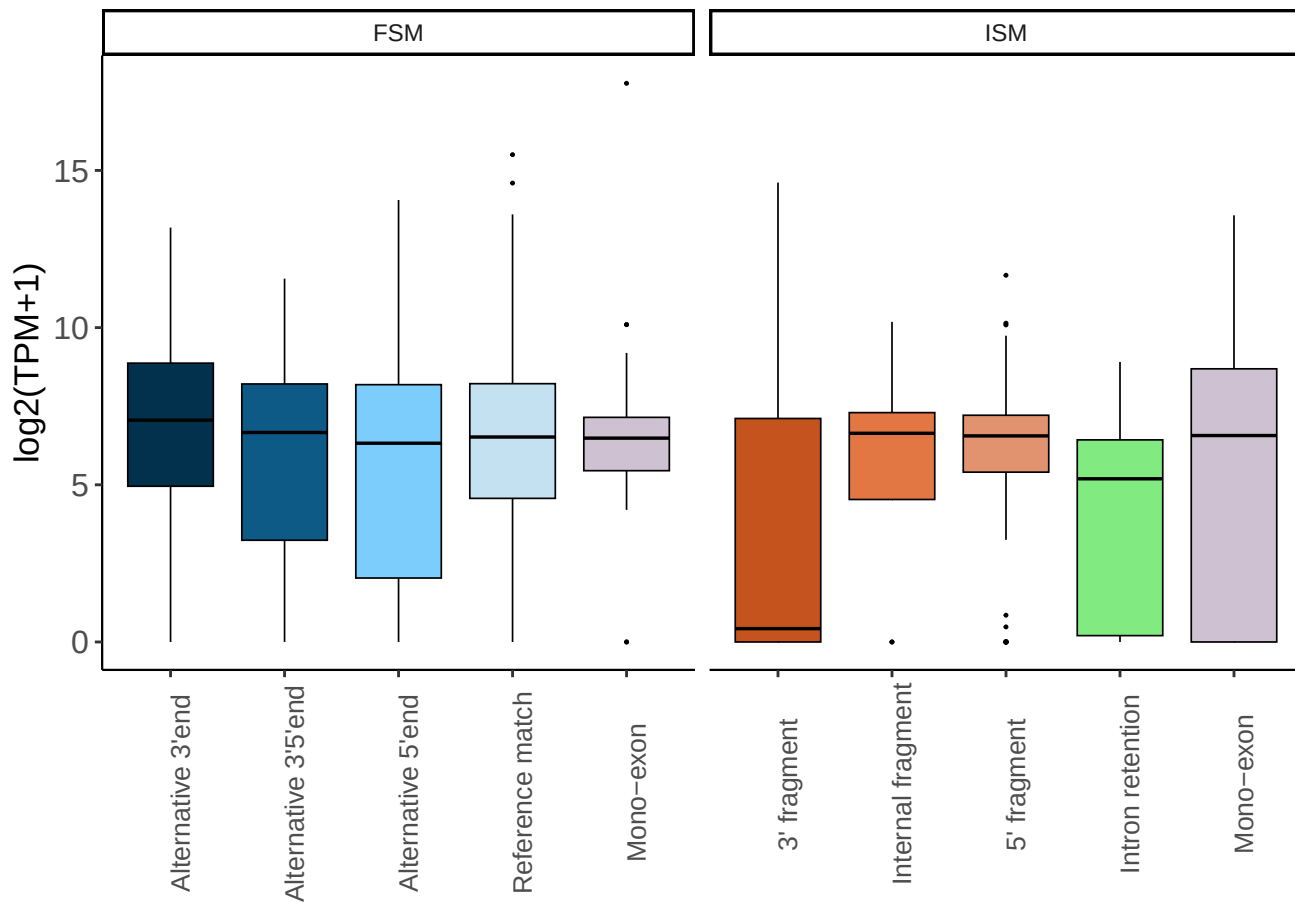


Transcript Expression by Structural Category

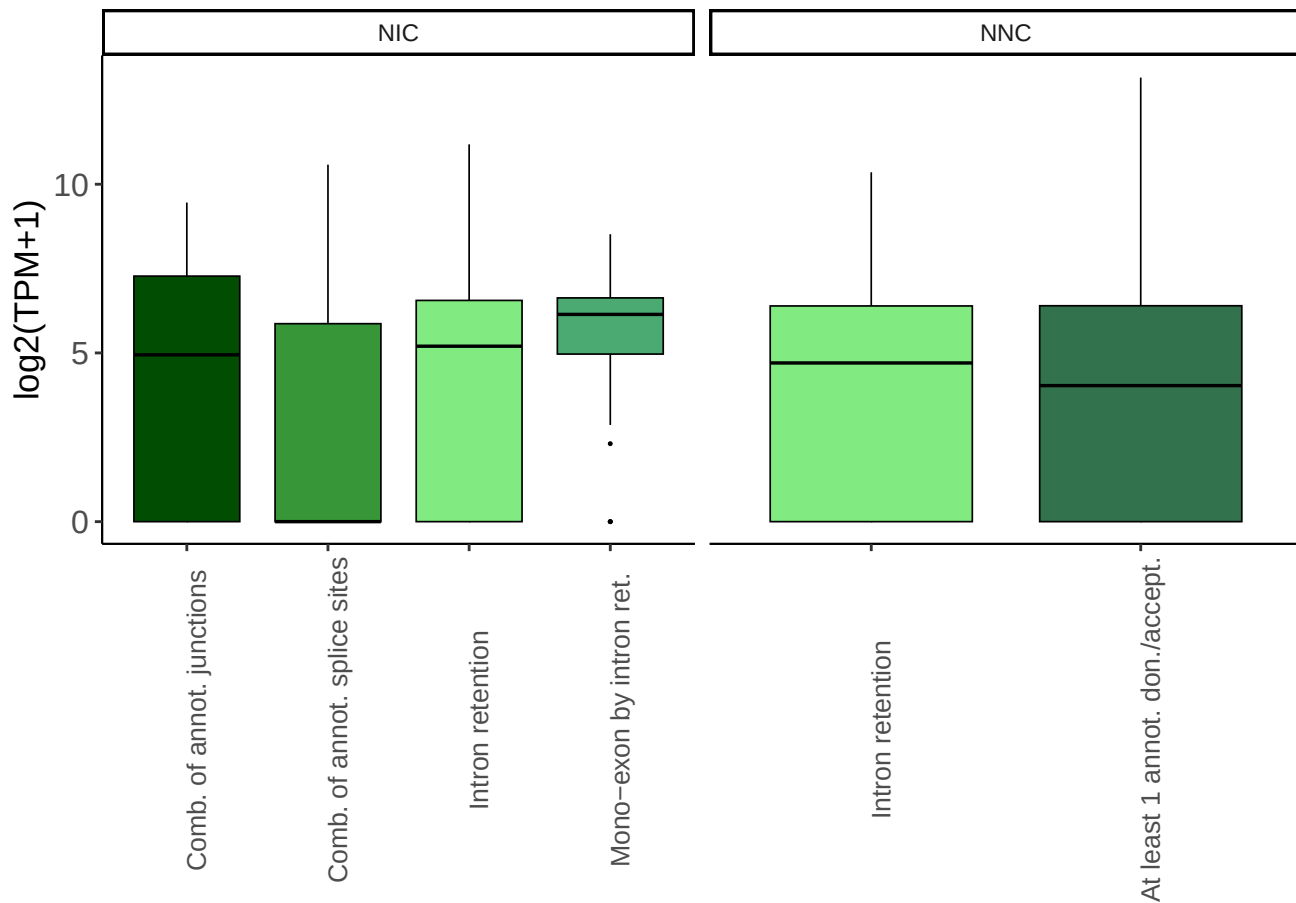




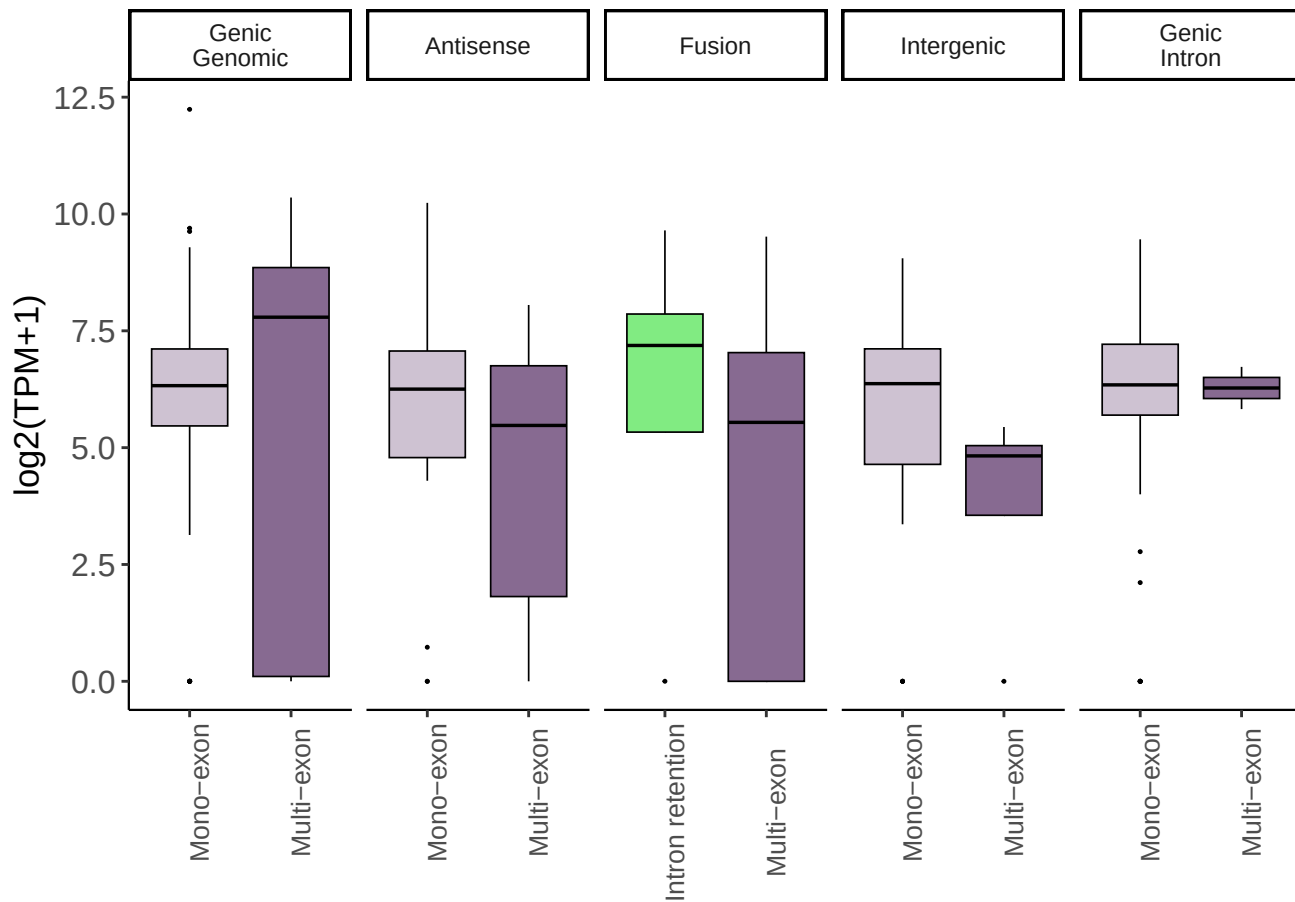
## Transcript Expression by Subcategory



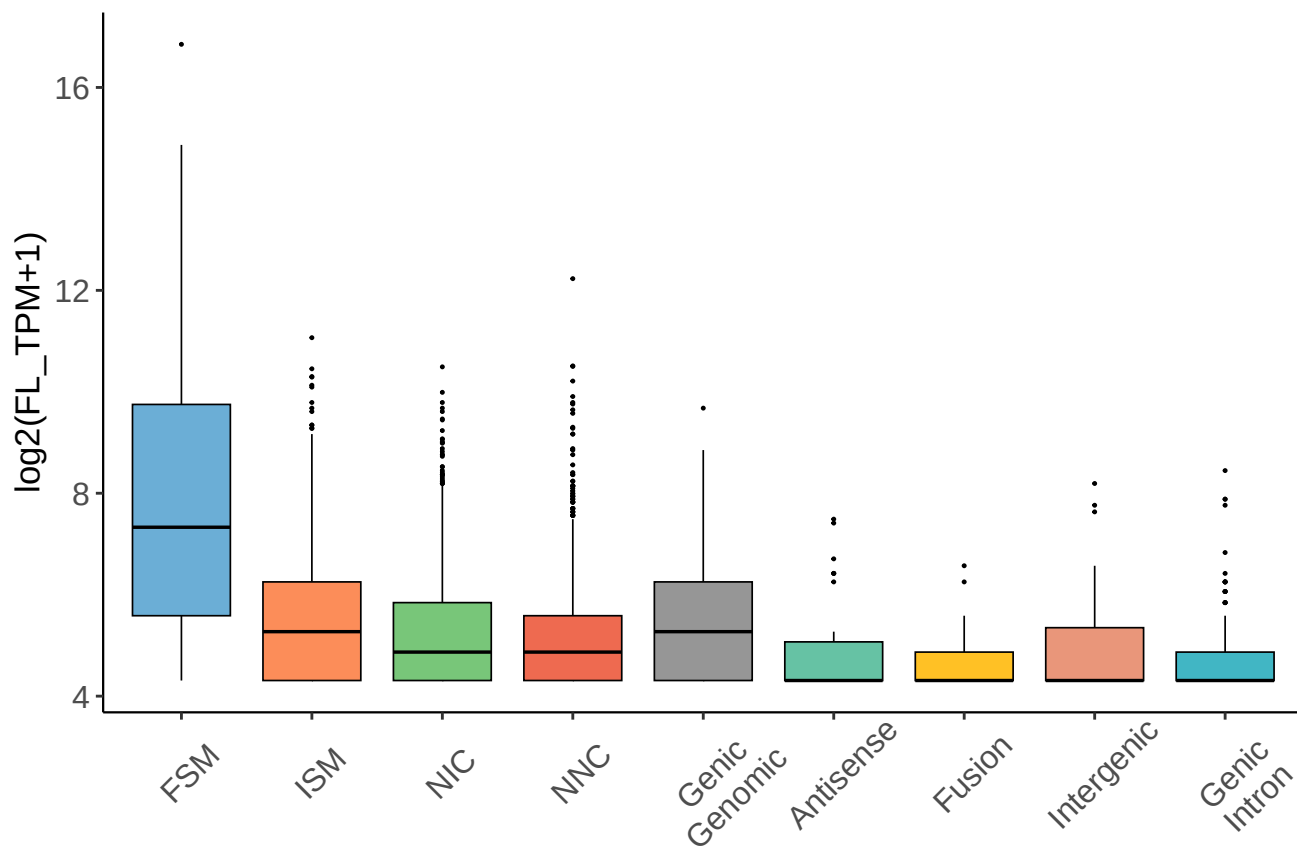
## Transcript Expression by Subcategory



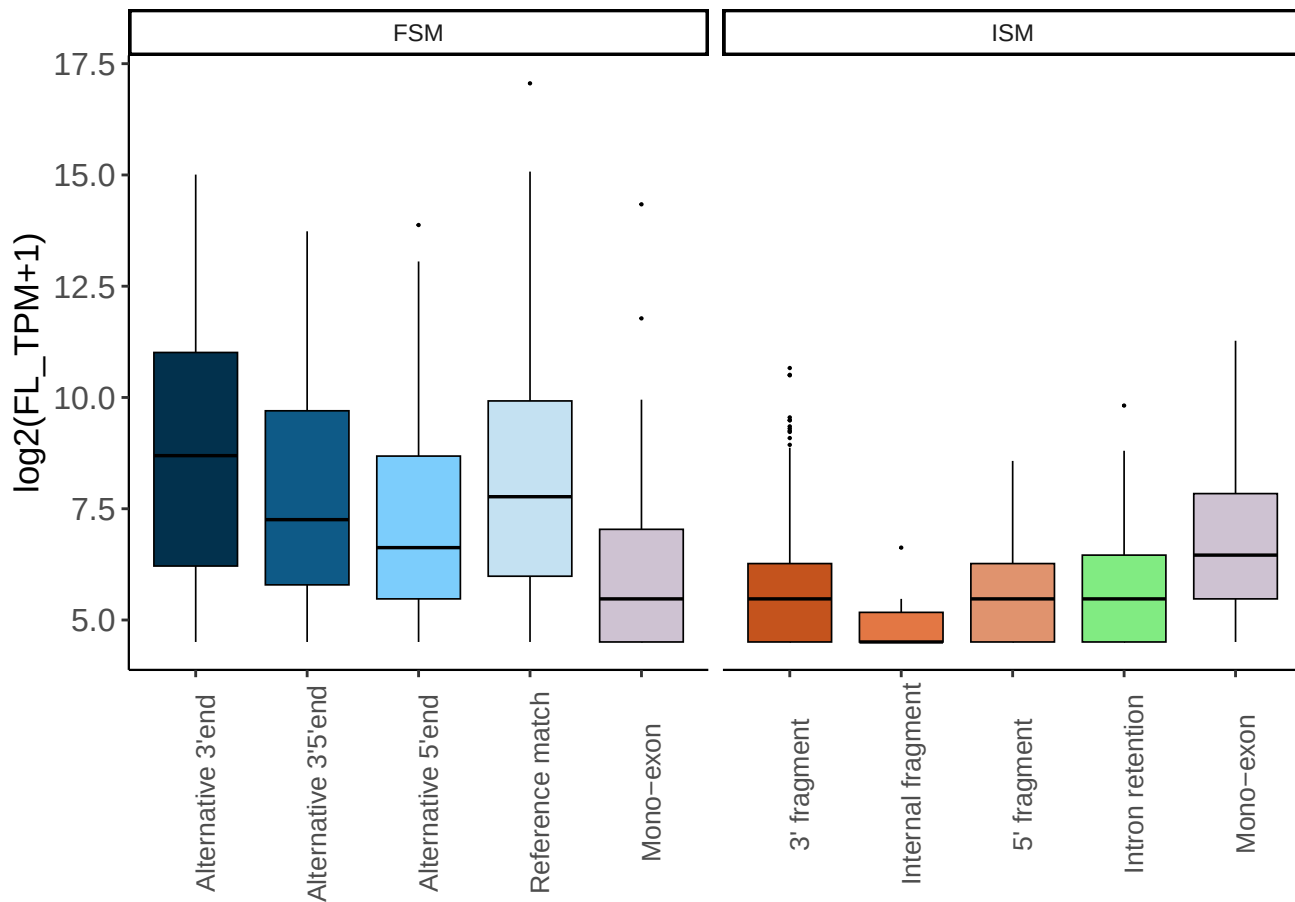
## Transcript Expression by Subcategory



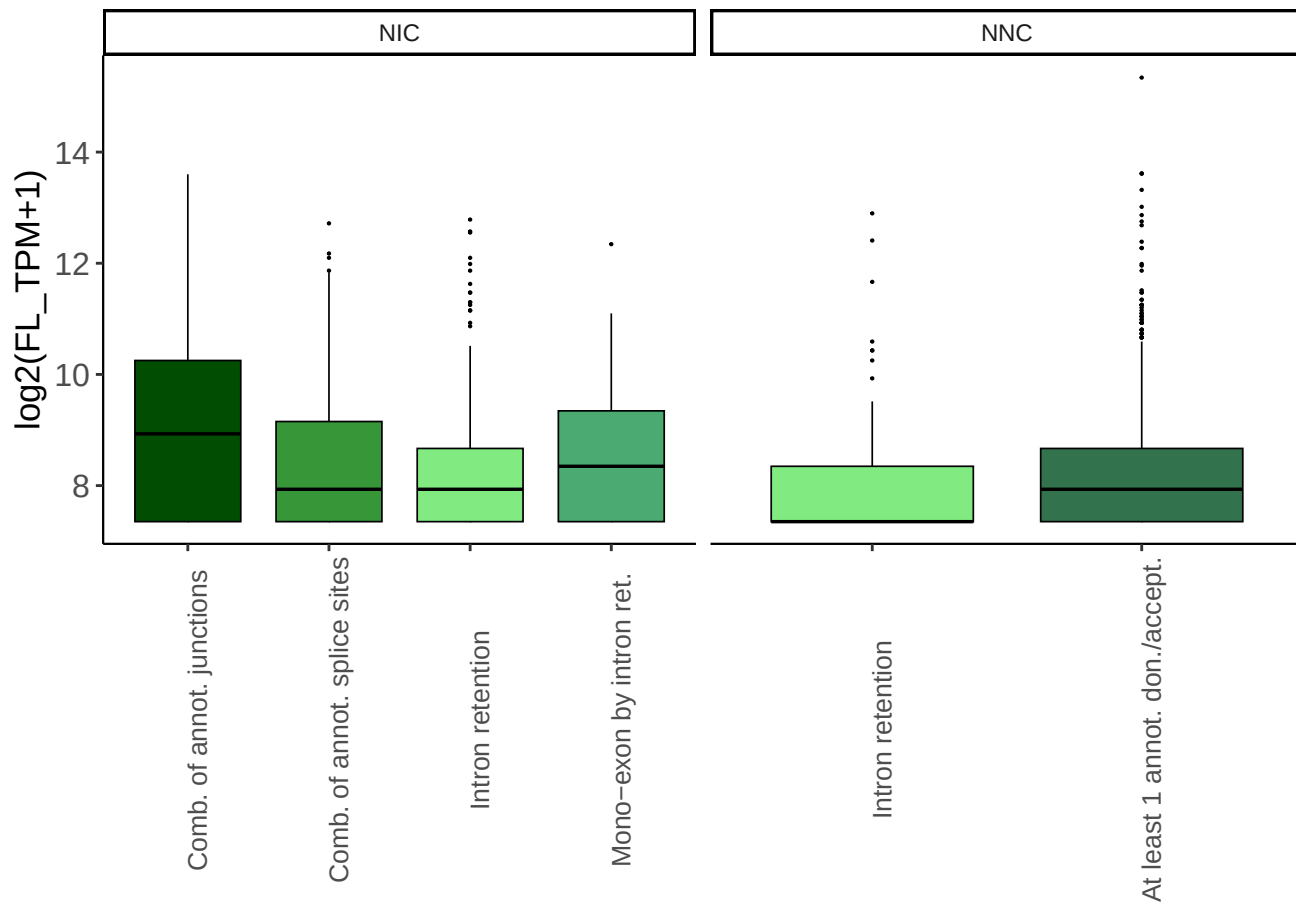
Long Reads Count by Structural Category



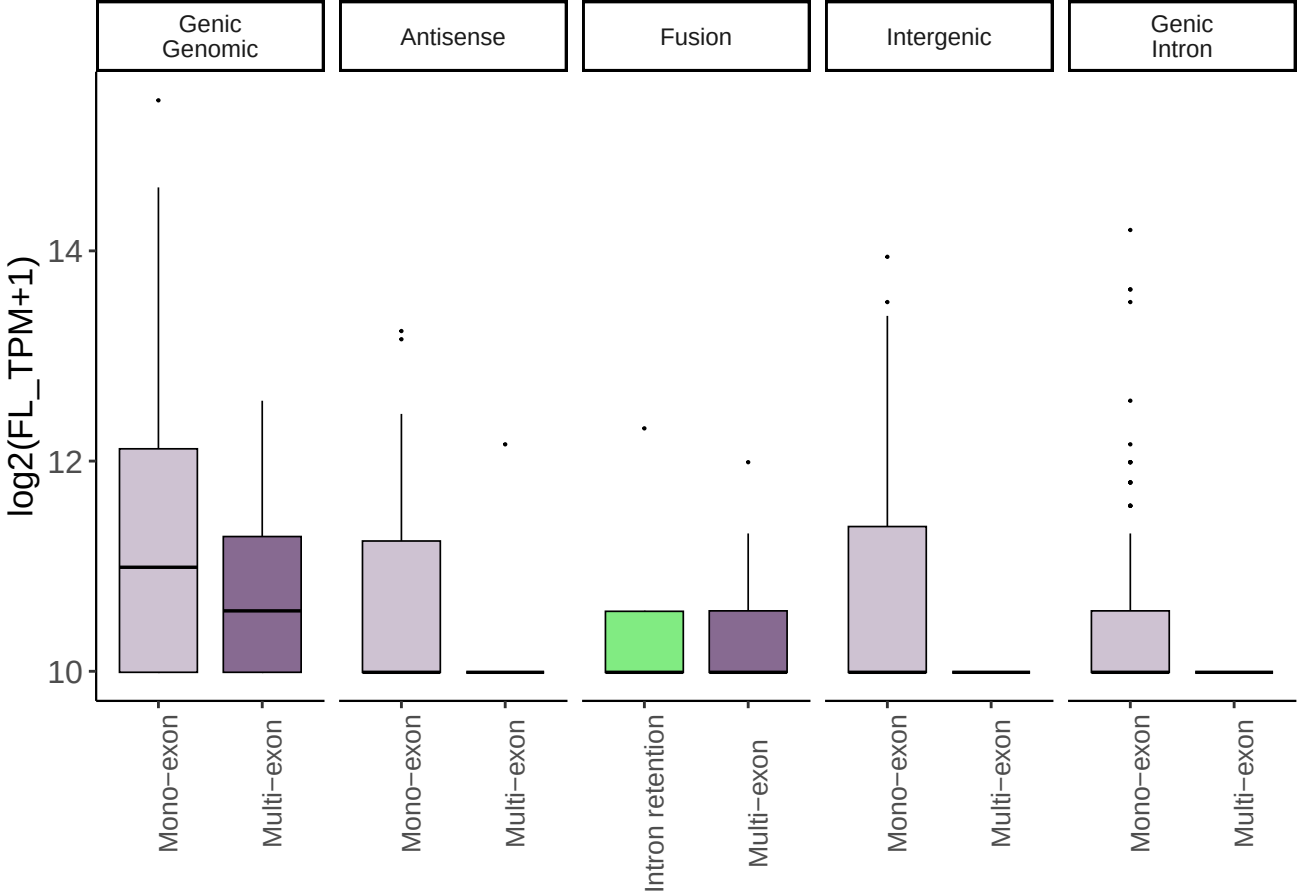
Long Reads Count by Subcategory



# Long Reads Count by Subcategory

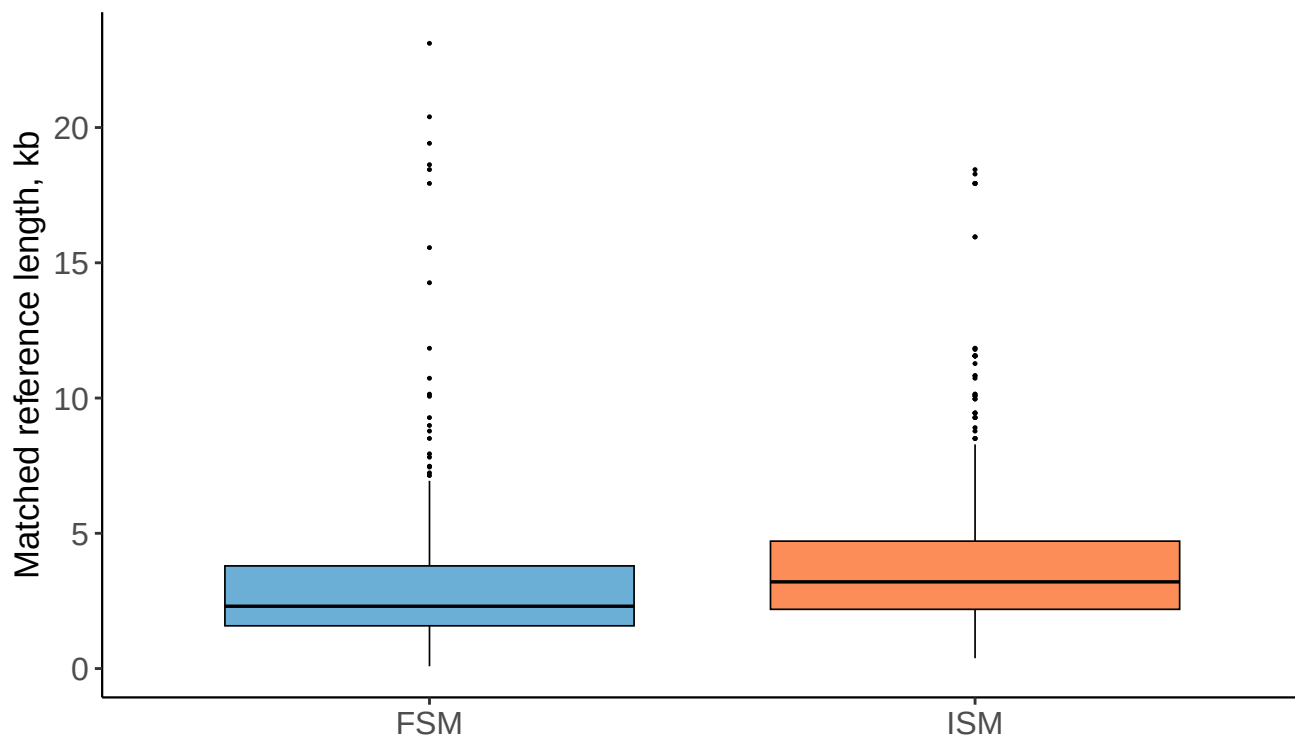


Long Reads Count by Subcategory



# Length Distribution of Matched Reference Transcripts

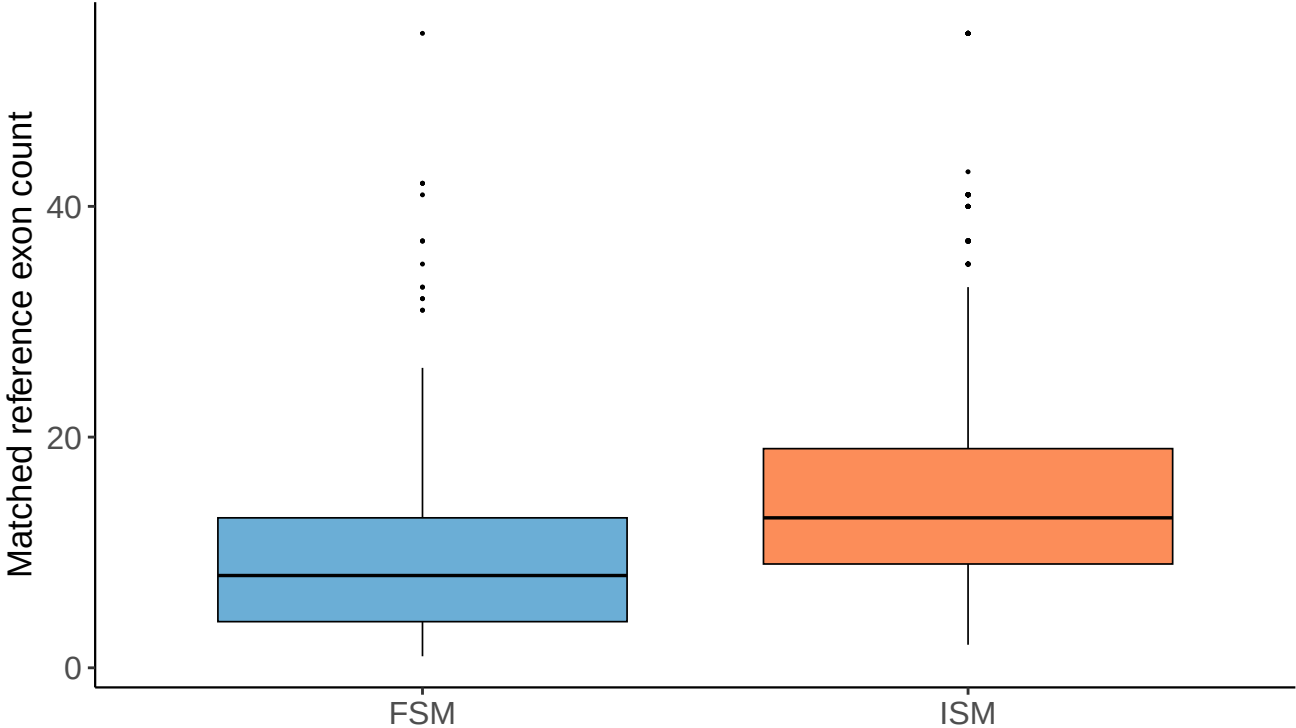
Applicable Only to FSM and ISM Categories



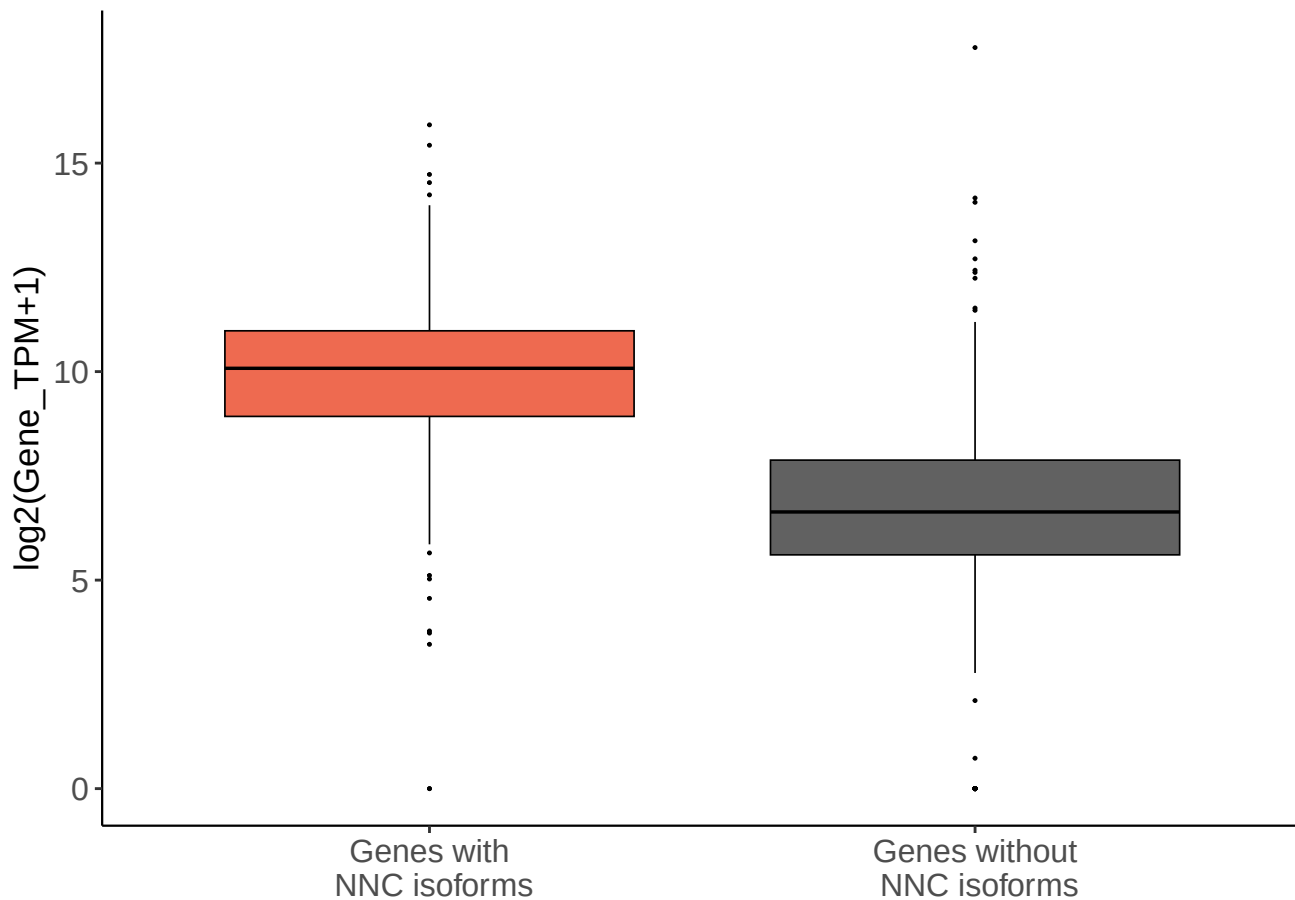


# Exon Count Distribution of Matched Reference Transcripts

Applicable Only to FSM and ISM Categories

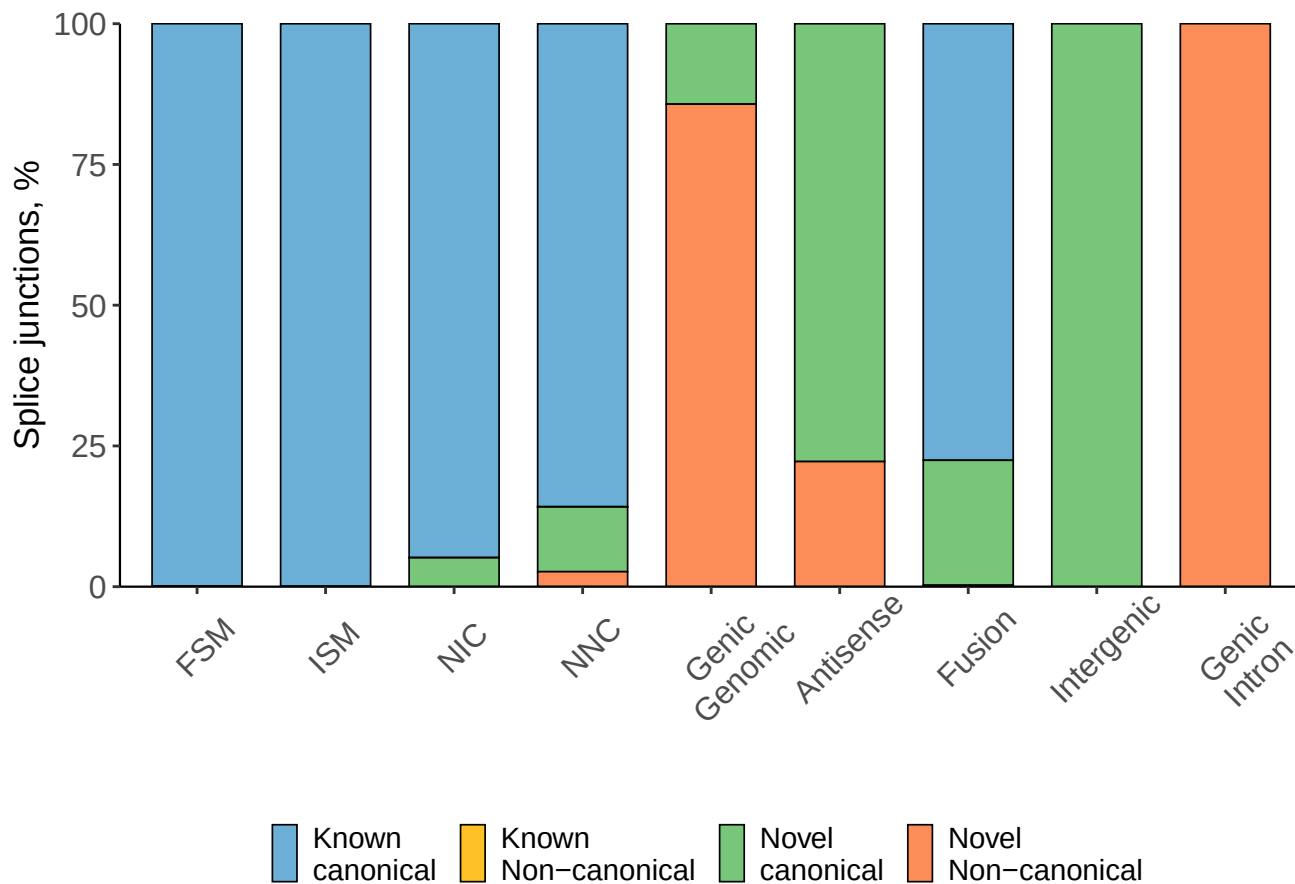


## Gene Expression of NNC And Not NNC Containing Genes

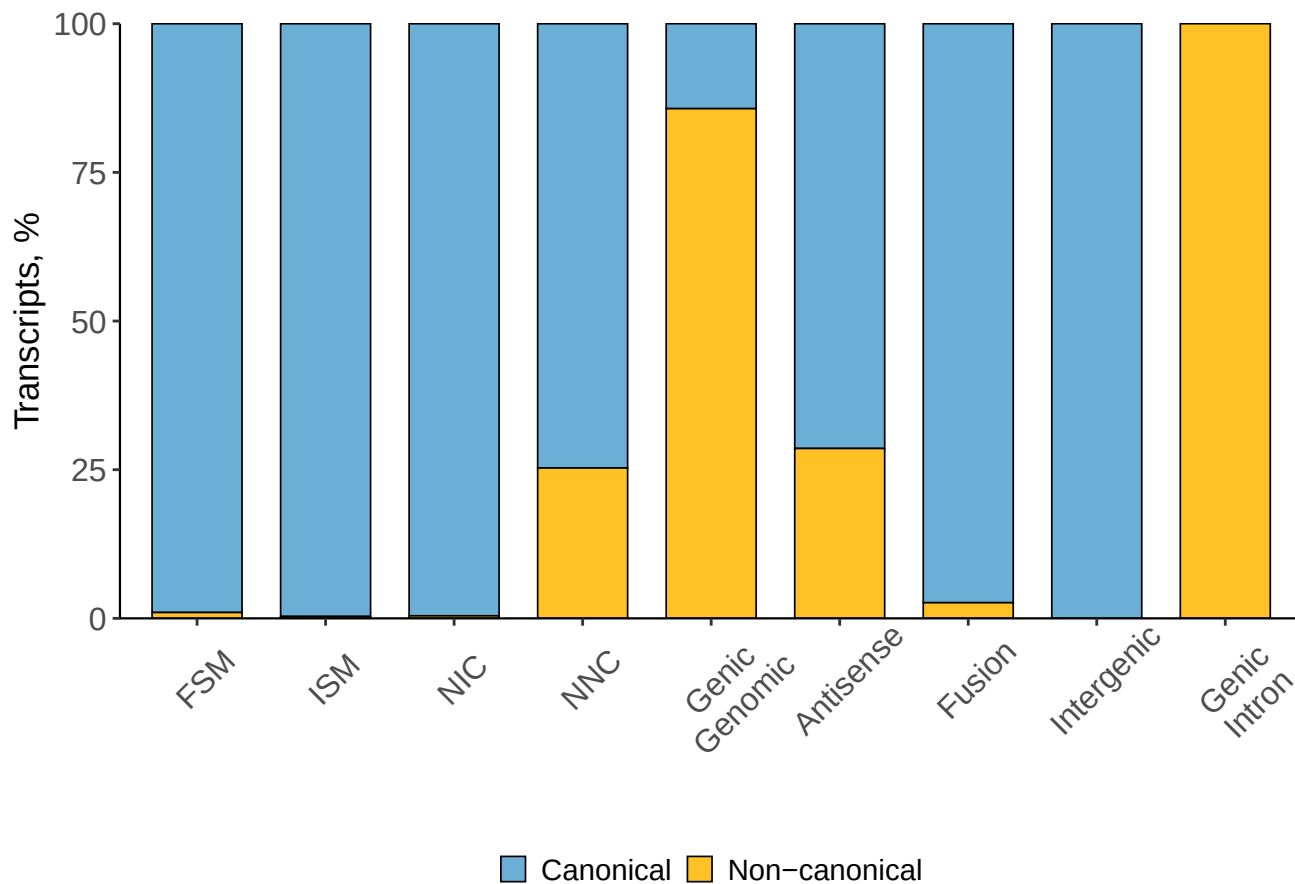


## *Splice Junction Characterization*

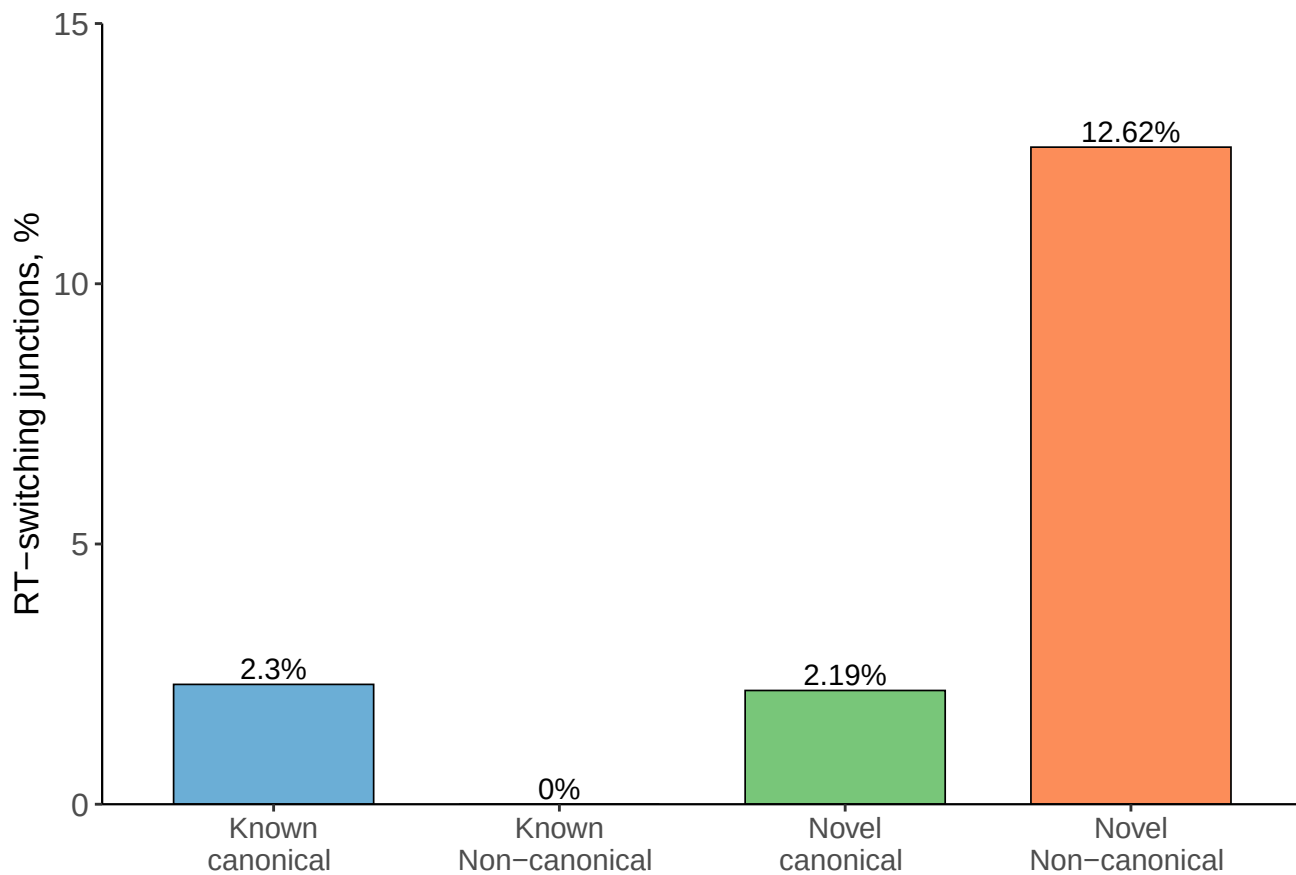
# Distribution of Splice Junctions by Structural Classification



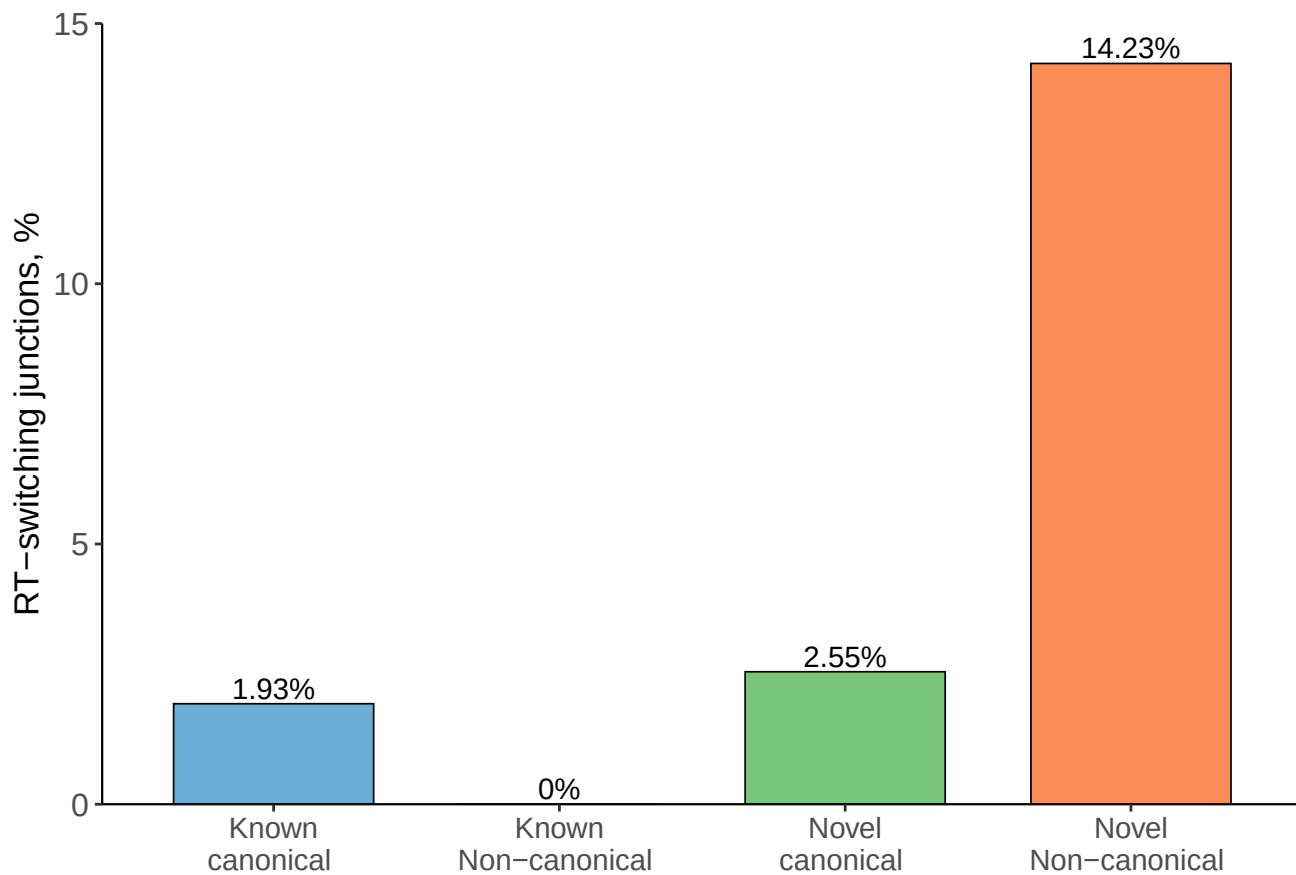
Distribution of Transcripts by Splice Junctions



## RT-Switching All Junctions



## Unique Junctions RT-switching

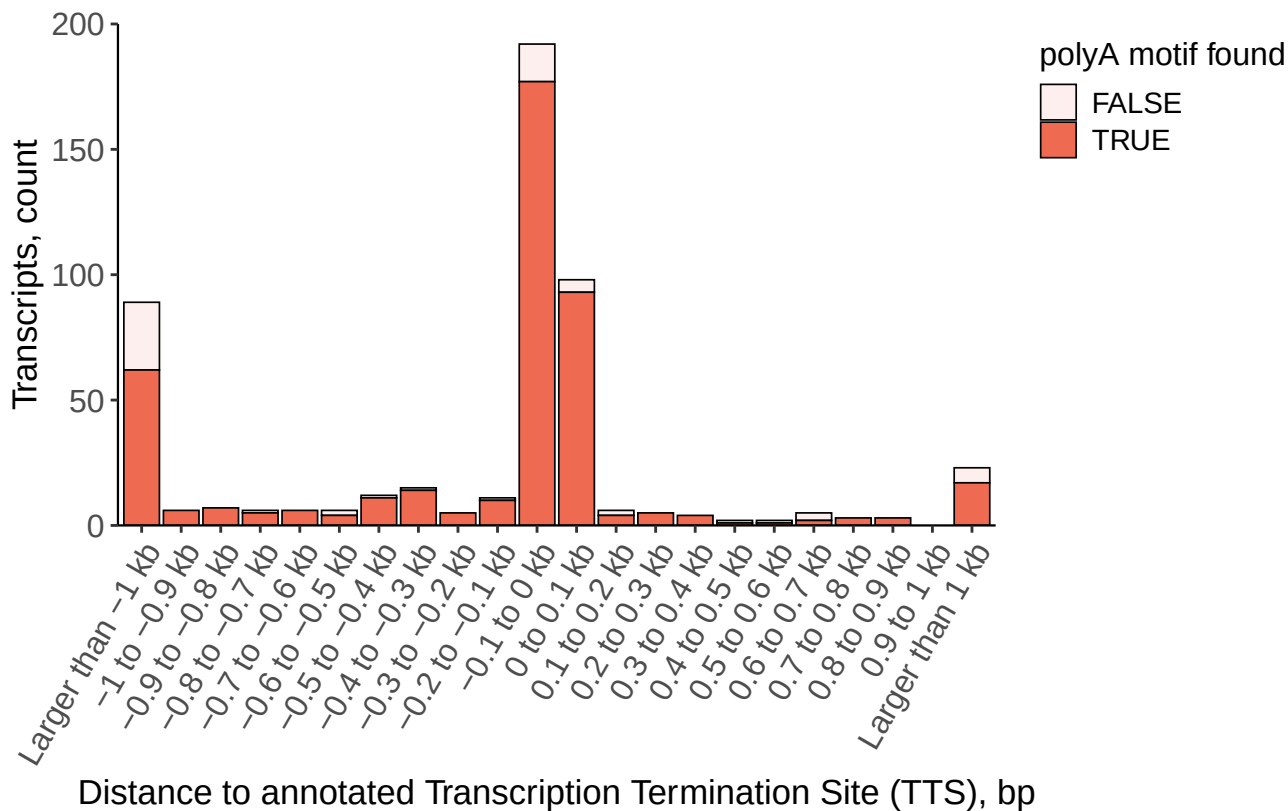


*Comparison With Annotated TSS and TTS*



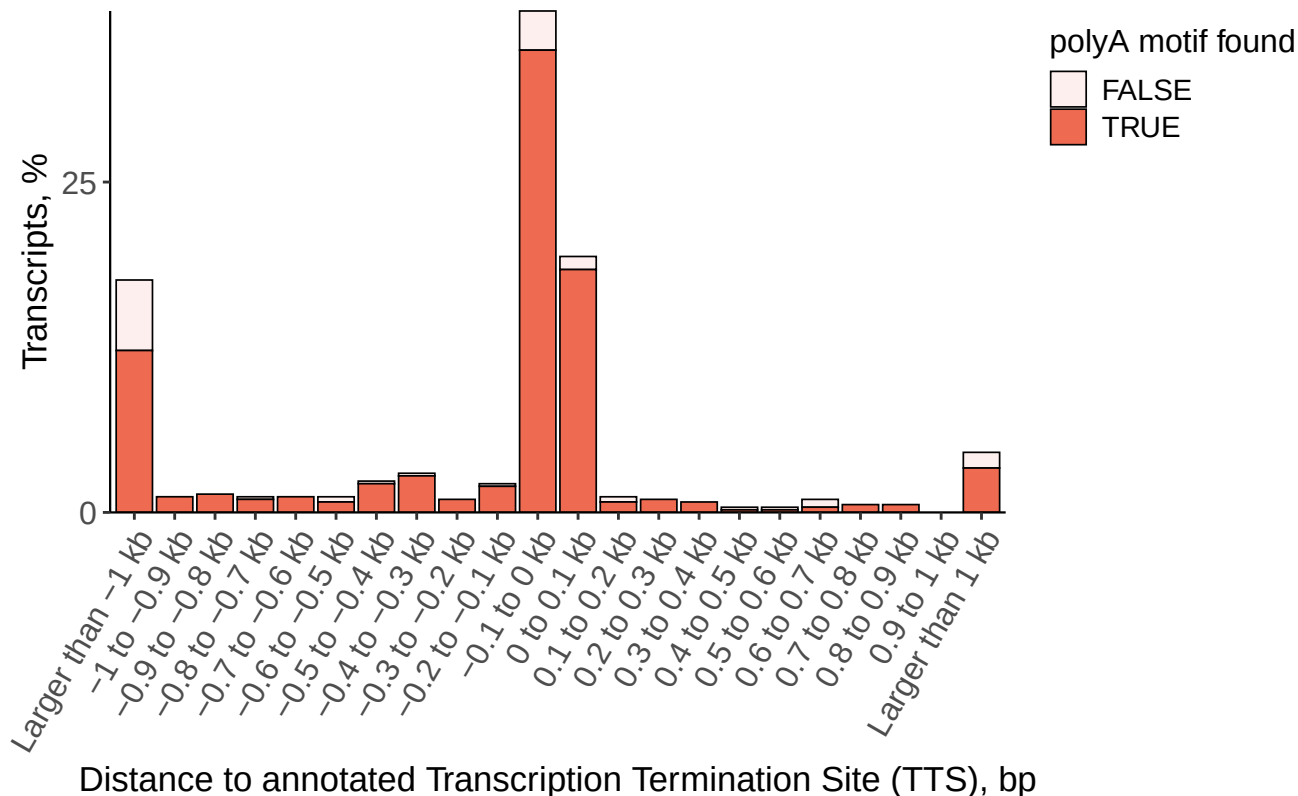
# Distance to annotated Transcription Termination Site (TTS) FSM

Negative values indicate upstream of annotated termination site



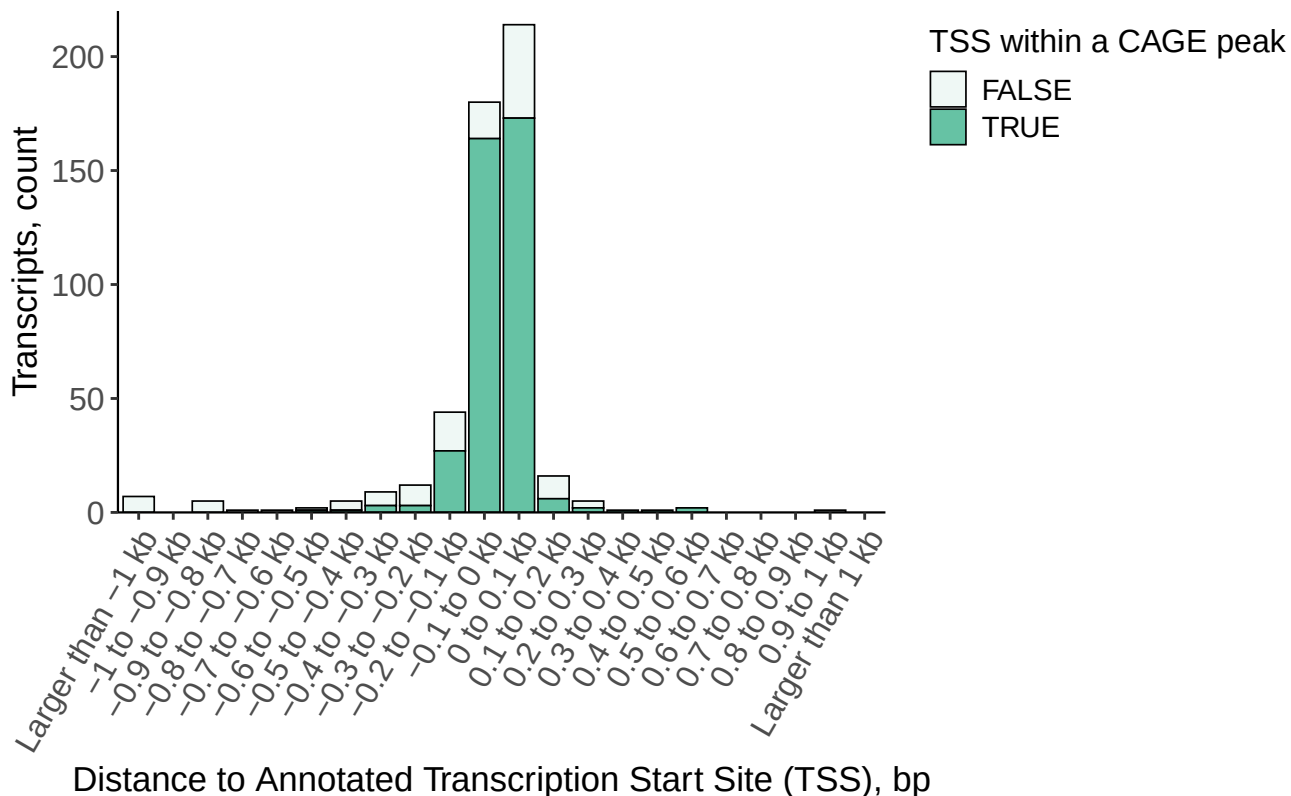
# Distance to annotated Transcription Termination Site (TTS) FSM

Negative values indicate upstream of annotated termination site



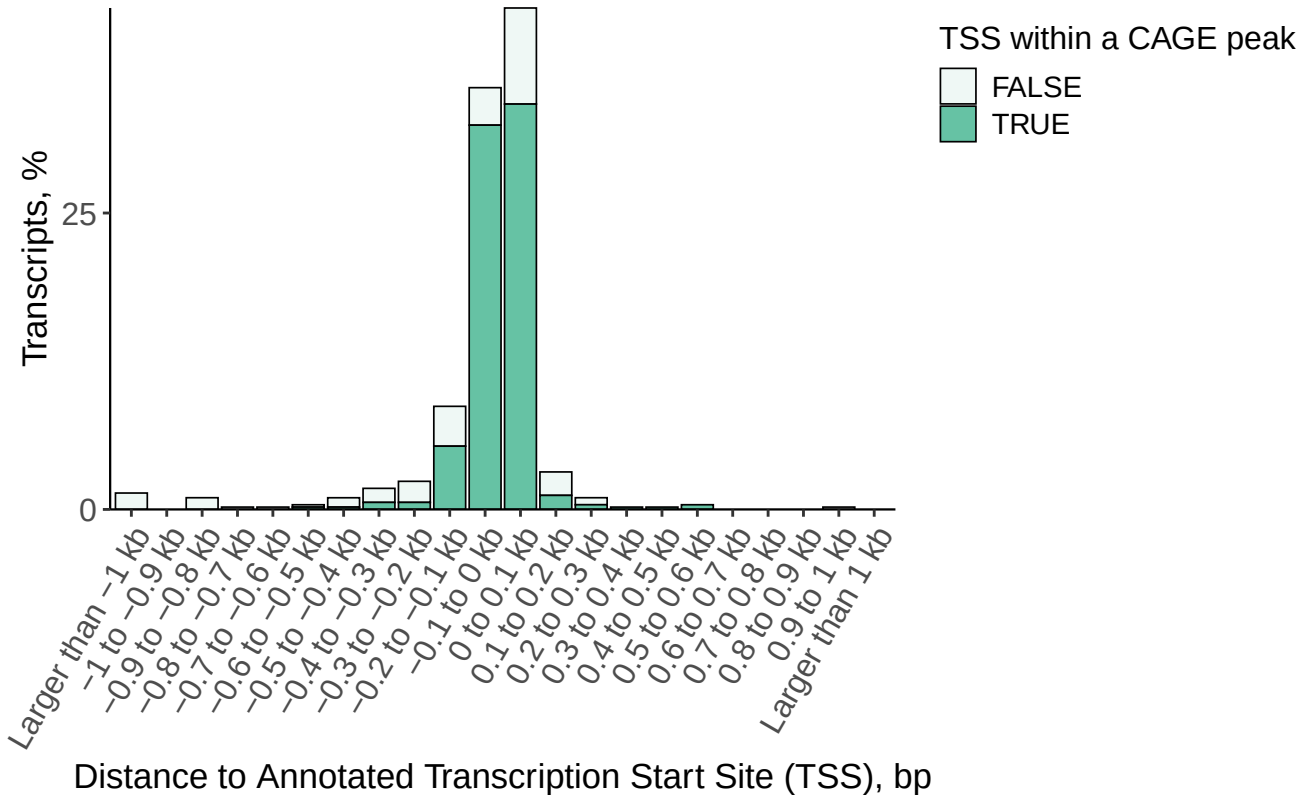
# Distance to Annotated Transcription Start Site for FSM

Negative values indicate downstream of annotated TSS



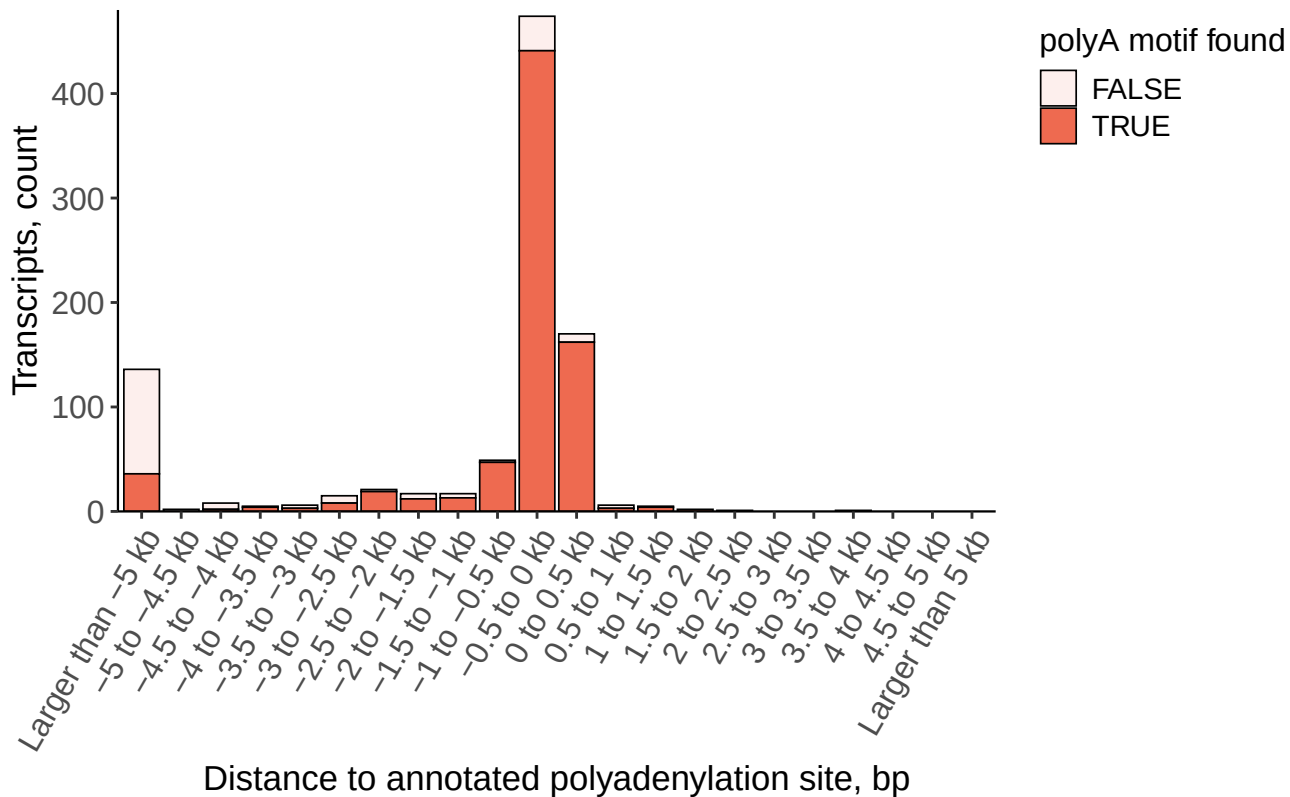
# Distance to Annotated Transcription Start Site for FSM

Negative values indicate downstream of annotated TSS



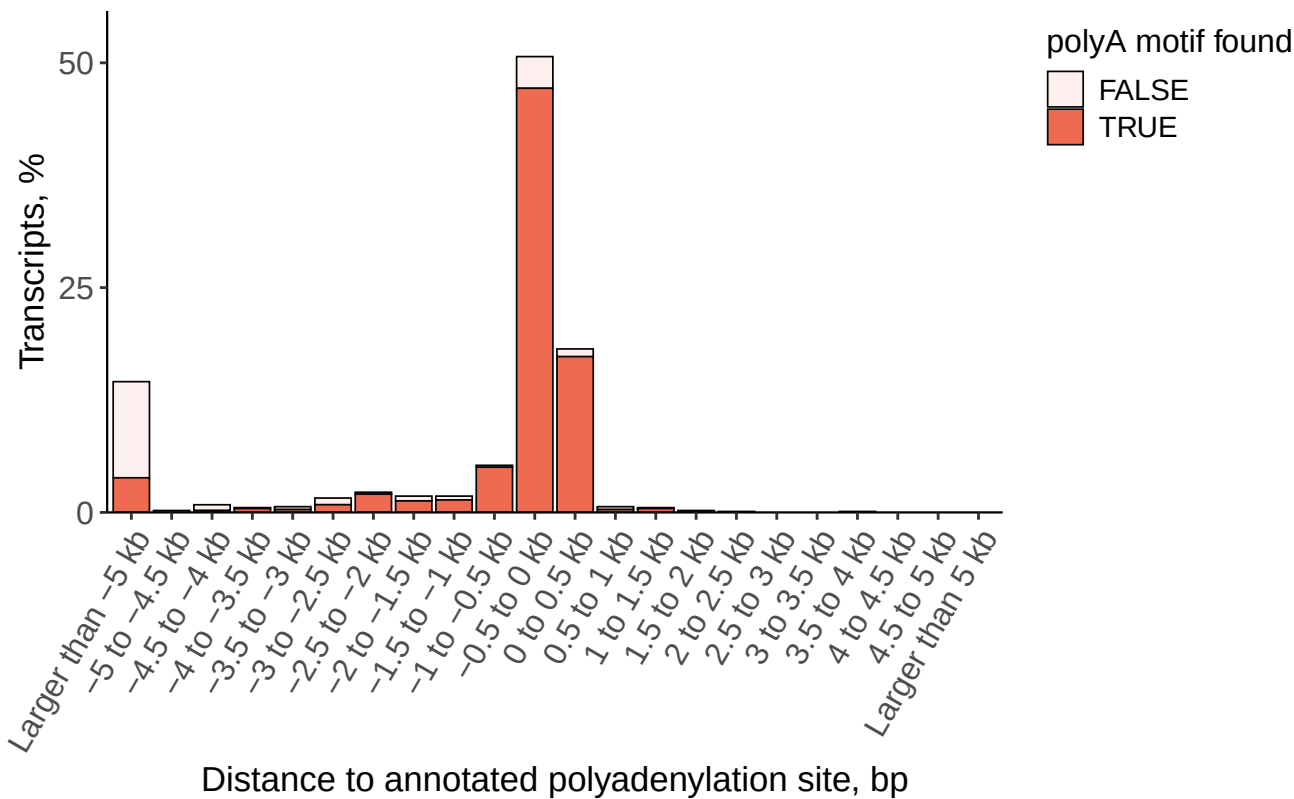
# Distance to Annotated Transcription Termination Site (TTS)

Negative values indicate upstream of annotated TTS



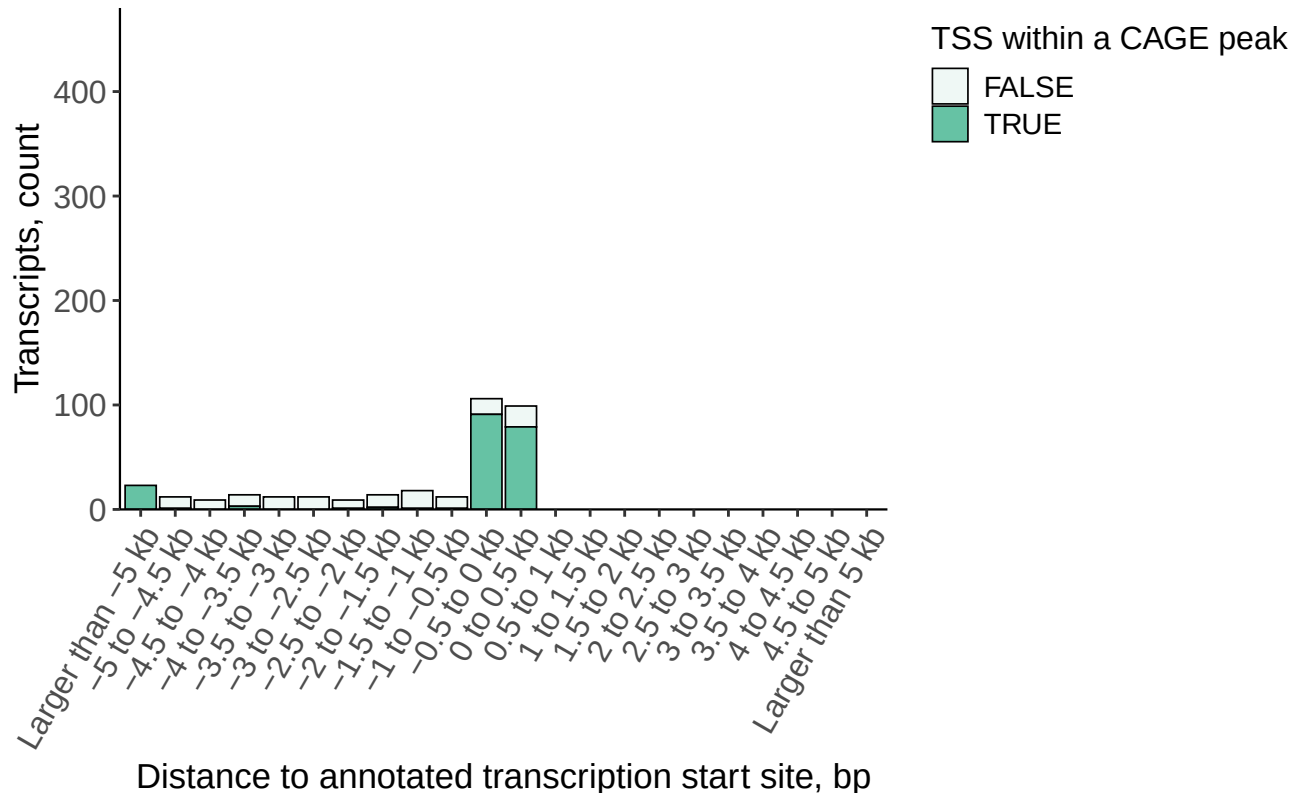
# Distance to Annotated Transcription Termination Site (TTS)

Negative values indicate upstream of annotated TTS



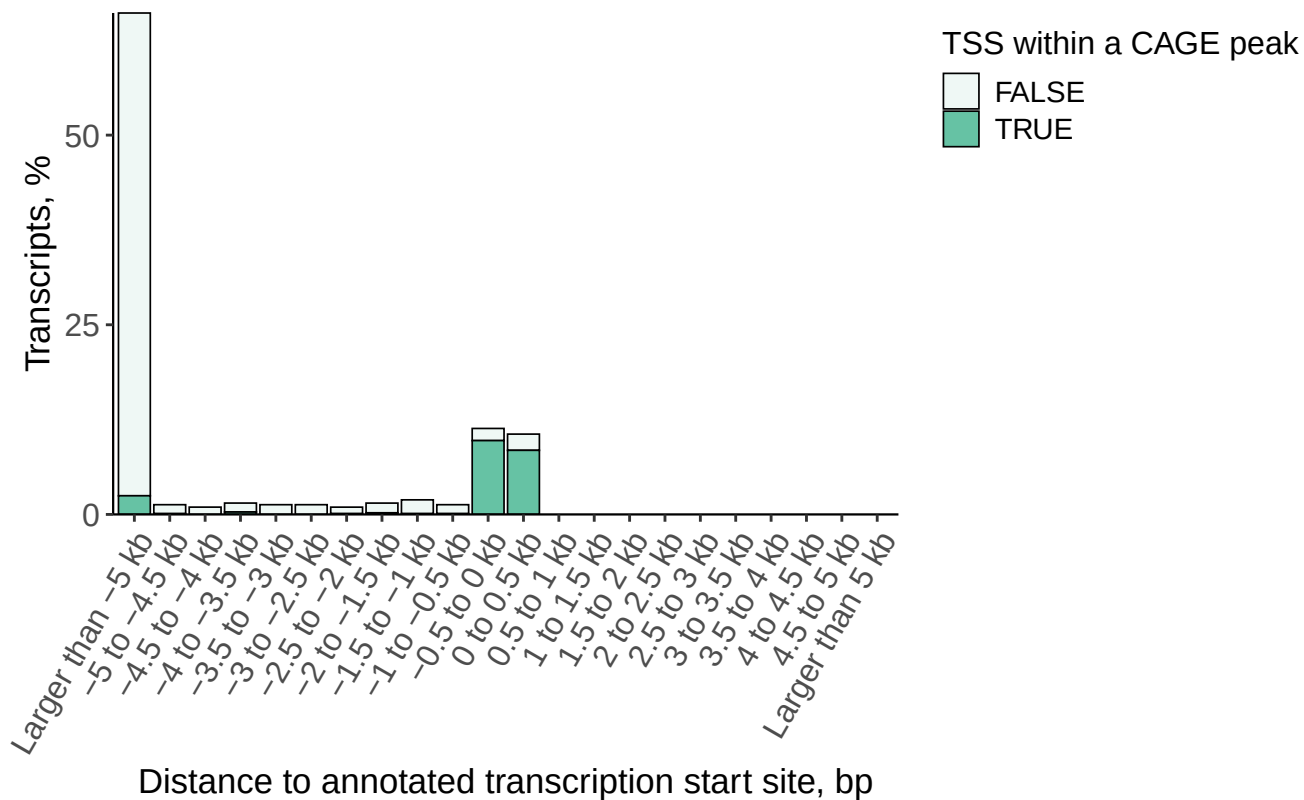
# Distance to Annotated Transcription Start Site for ISM

Negative values indicate downstream of annotated TSS



# Distance to Annotated Transcription Start Site for ISM

Negative values indicate downstream of annotated TSS

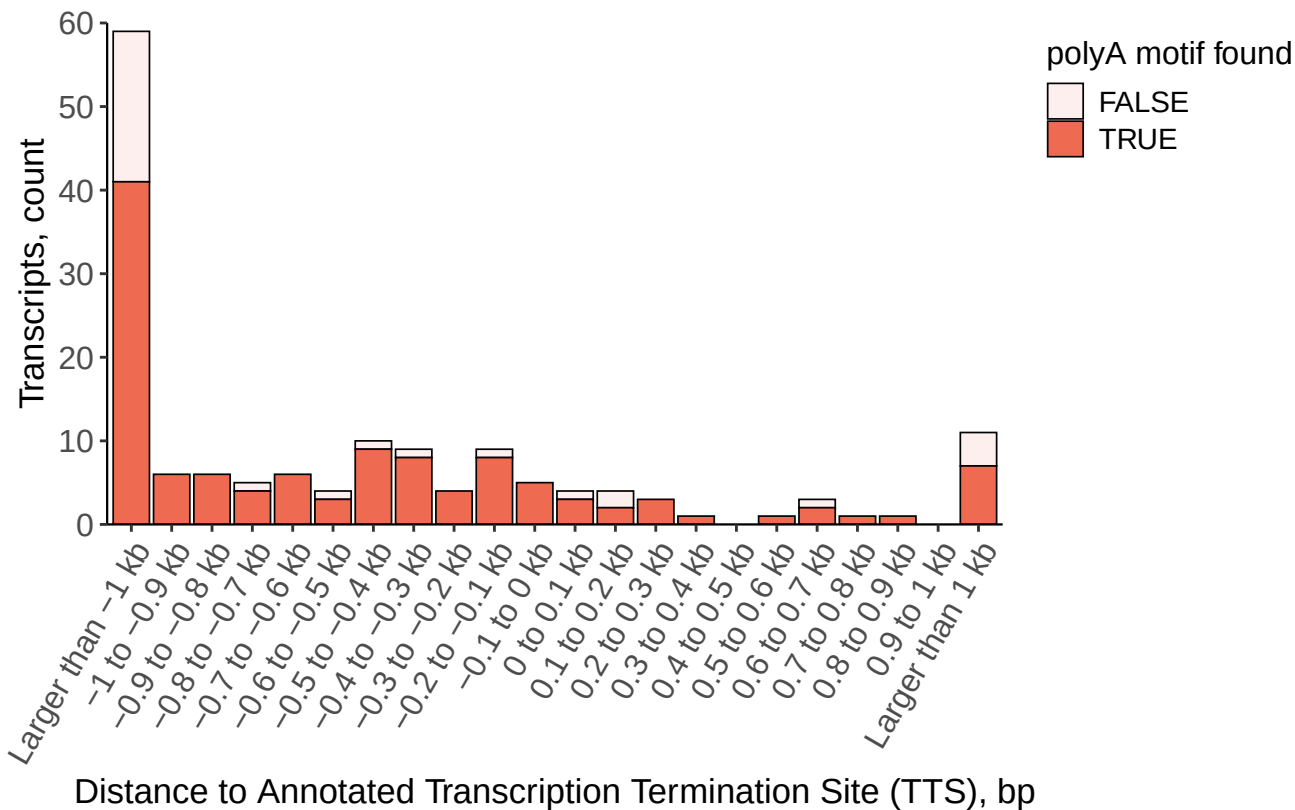




*Comparison With Annotated TSS and TTS  
by Subcategories*

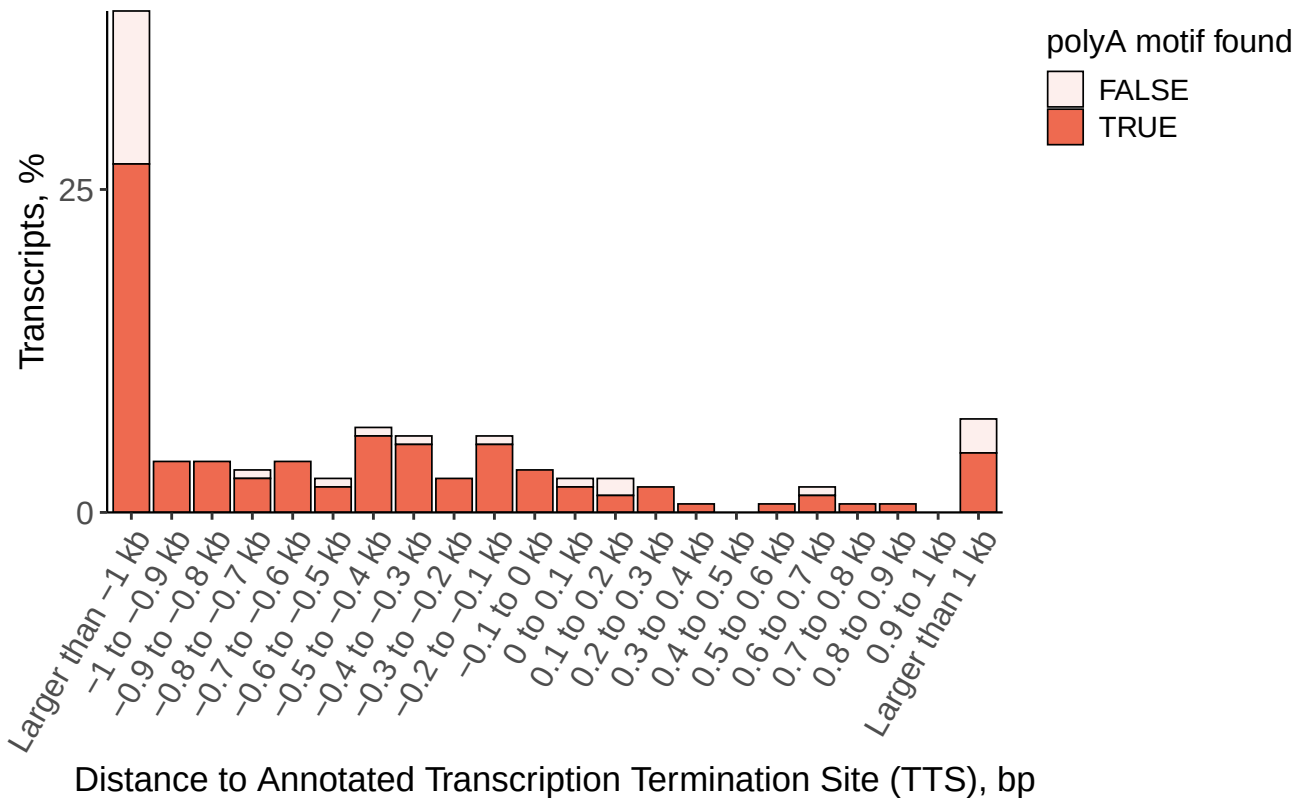
# Distance to annotated Transcription Termination Site (TTS) FSM Alternative 3'End

Negative values indicate upstream of annotated termination site



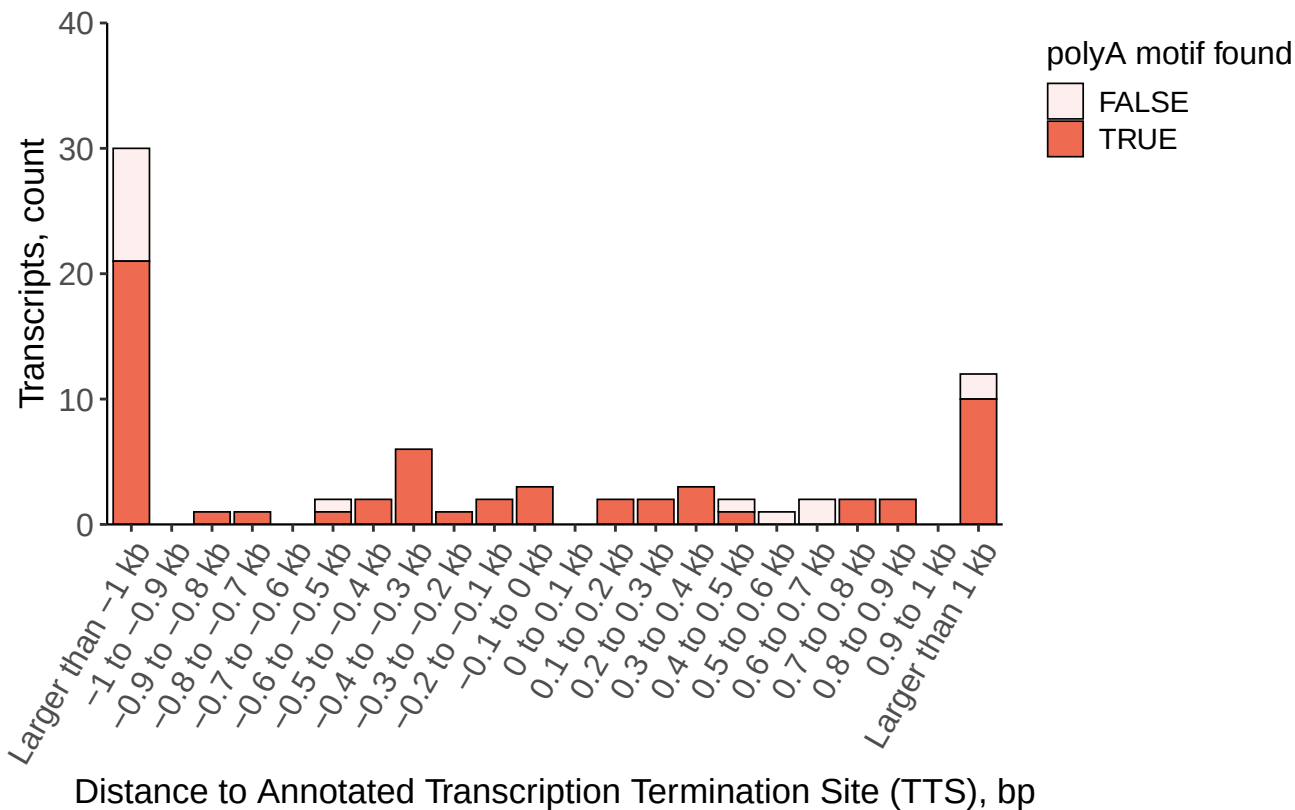
# Distance to annotated Transcription Termination Site (TTS) FSM Alternative 3'End

Negative values indicate upstream of annotated termination site



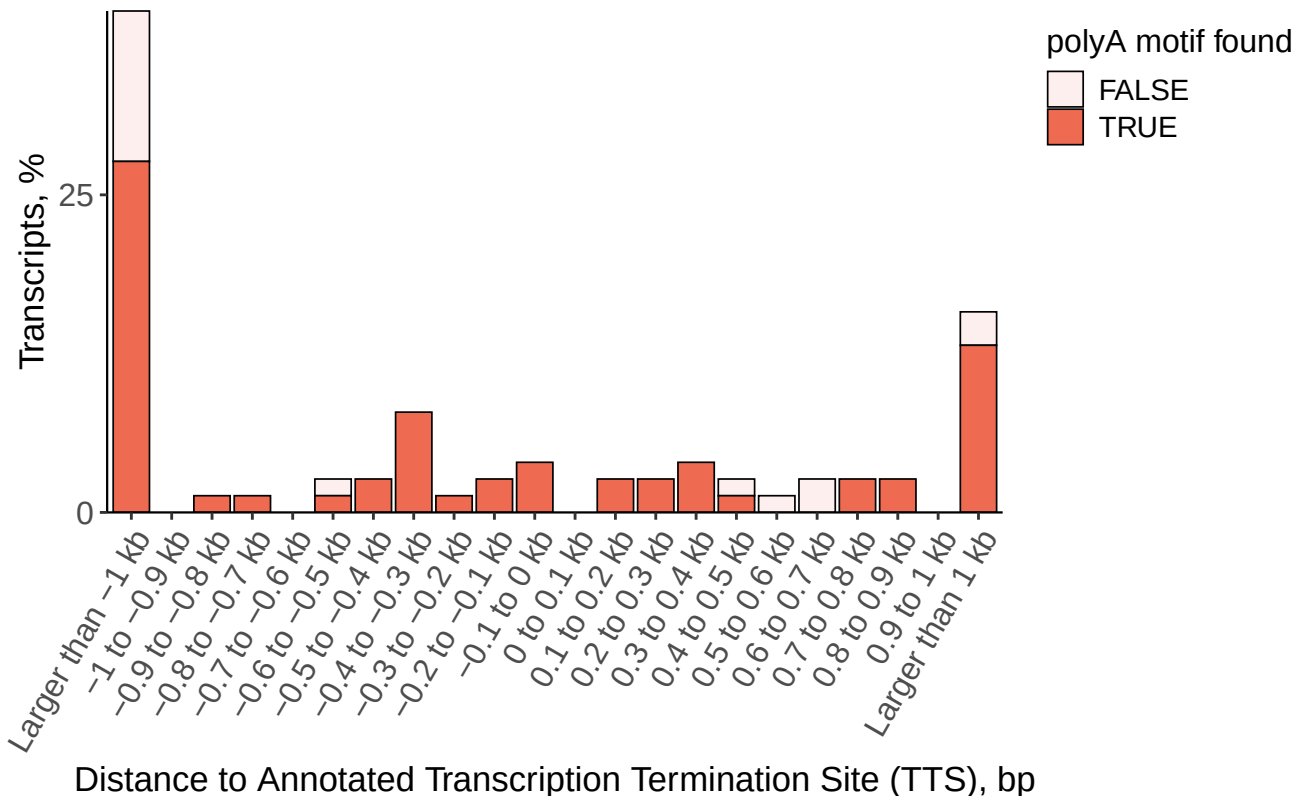
# Distance to annotated Transcription Termination Site (TTS) FSM Alternative 3'5'End

Negative values indicate upstream of annotated termination site



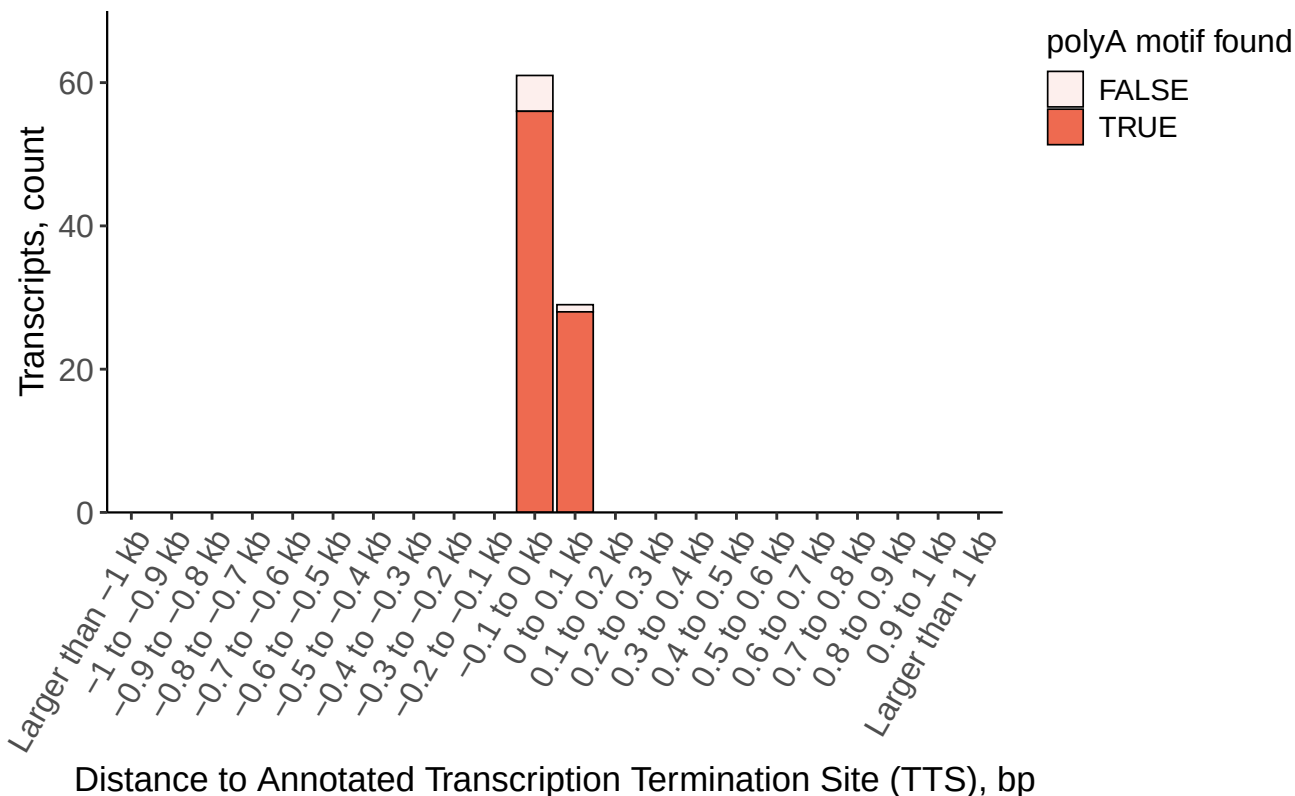
# Distance to annotated Transcription Termination Site (TTS) FSM Alternative 3'5'End

Negative values indicate upstream of annotated termination site



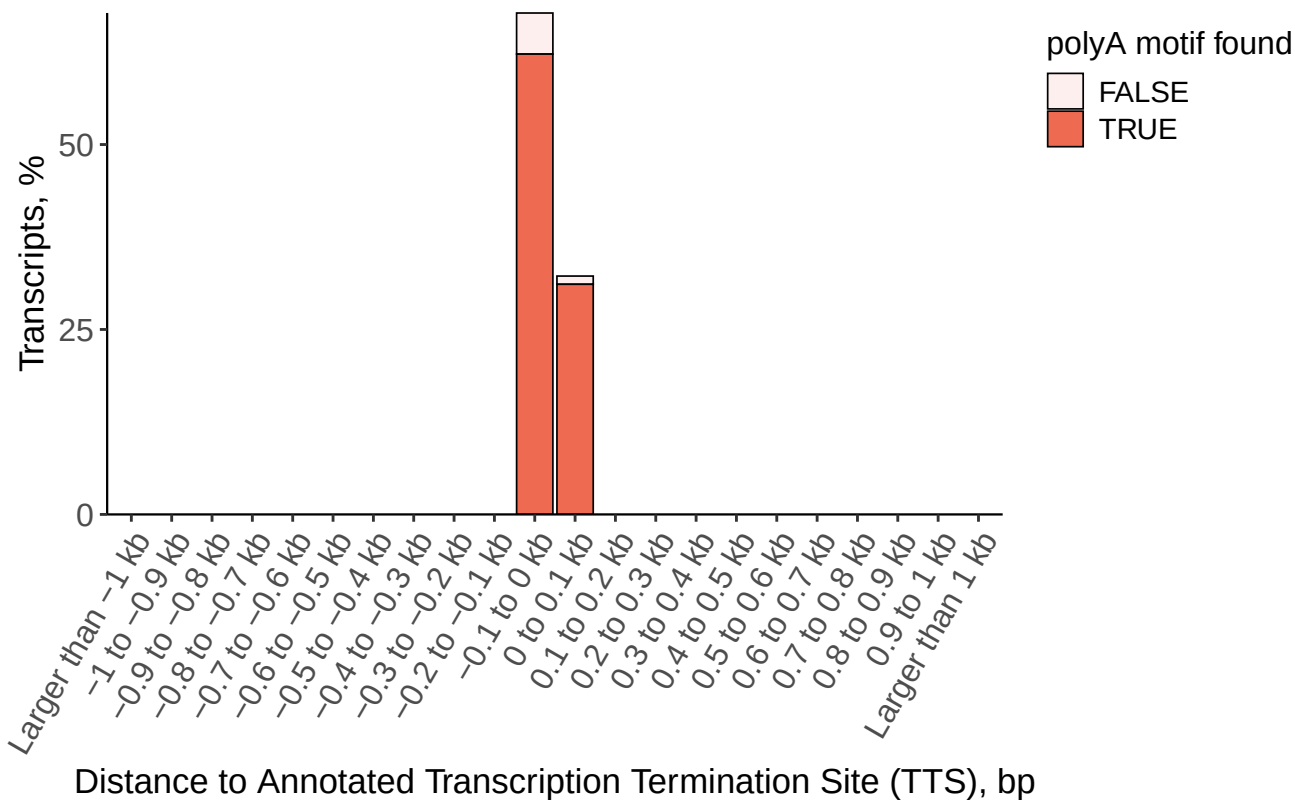
# Distance to annotated Transcription Termination Site (TTS) FSM Alternative 5'End

Negative values indicate upstream of annotated termination site



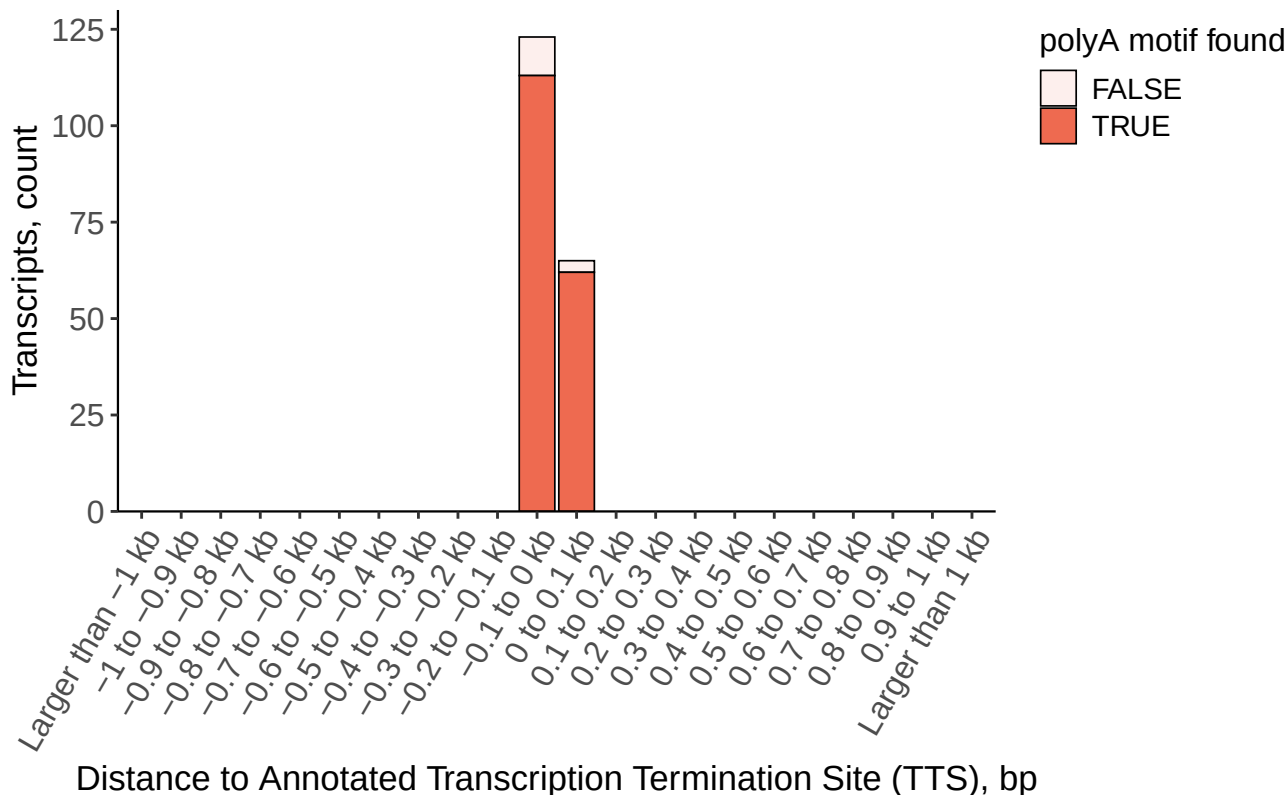
# Distance to annotated Transcription Termination Site (TTS) FSM Alternative 5'End

Negative values indicate upstream of annotated termination site



# Distance to annotated Transcription Termination Site (TTS) FSM Reference Match

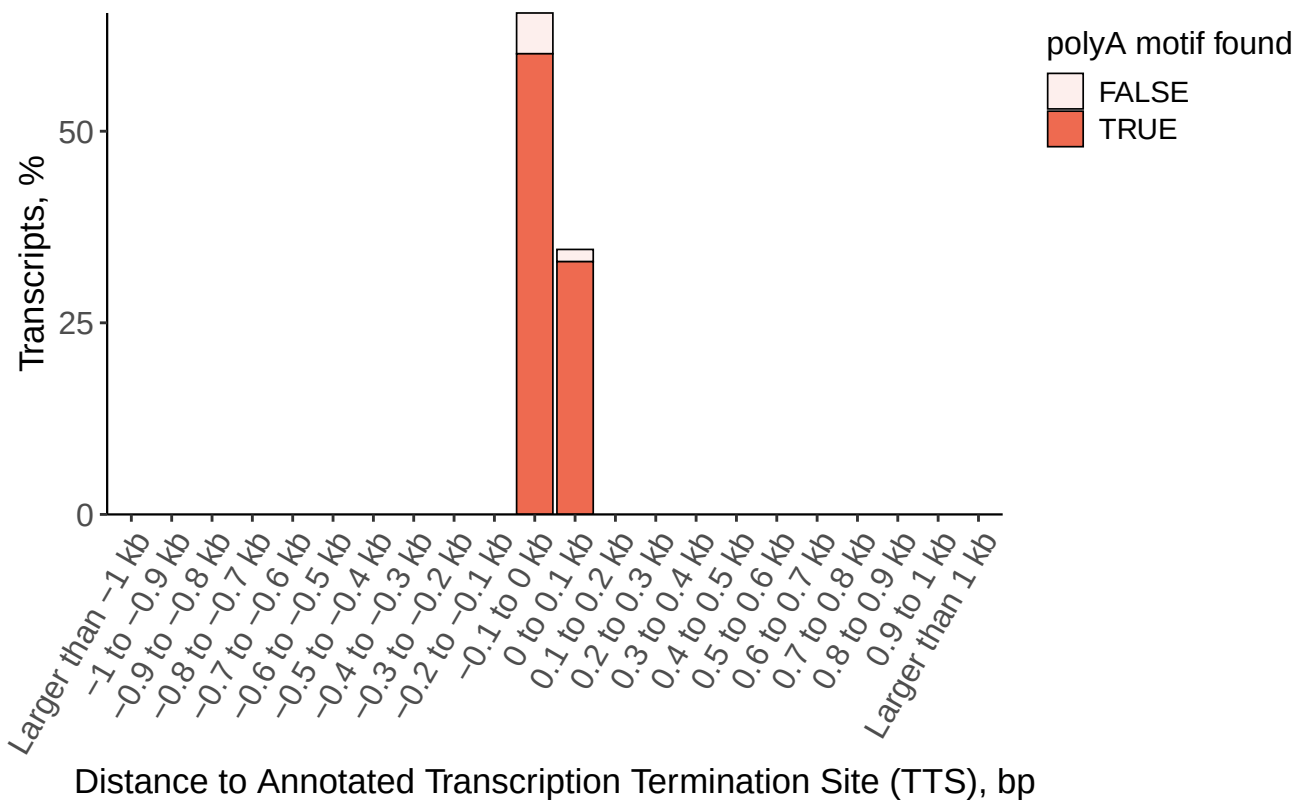
Negative values indicate upstream of annotated termination site





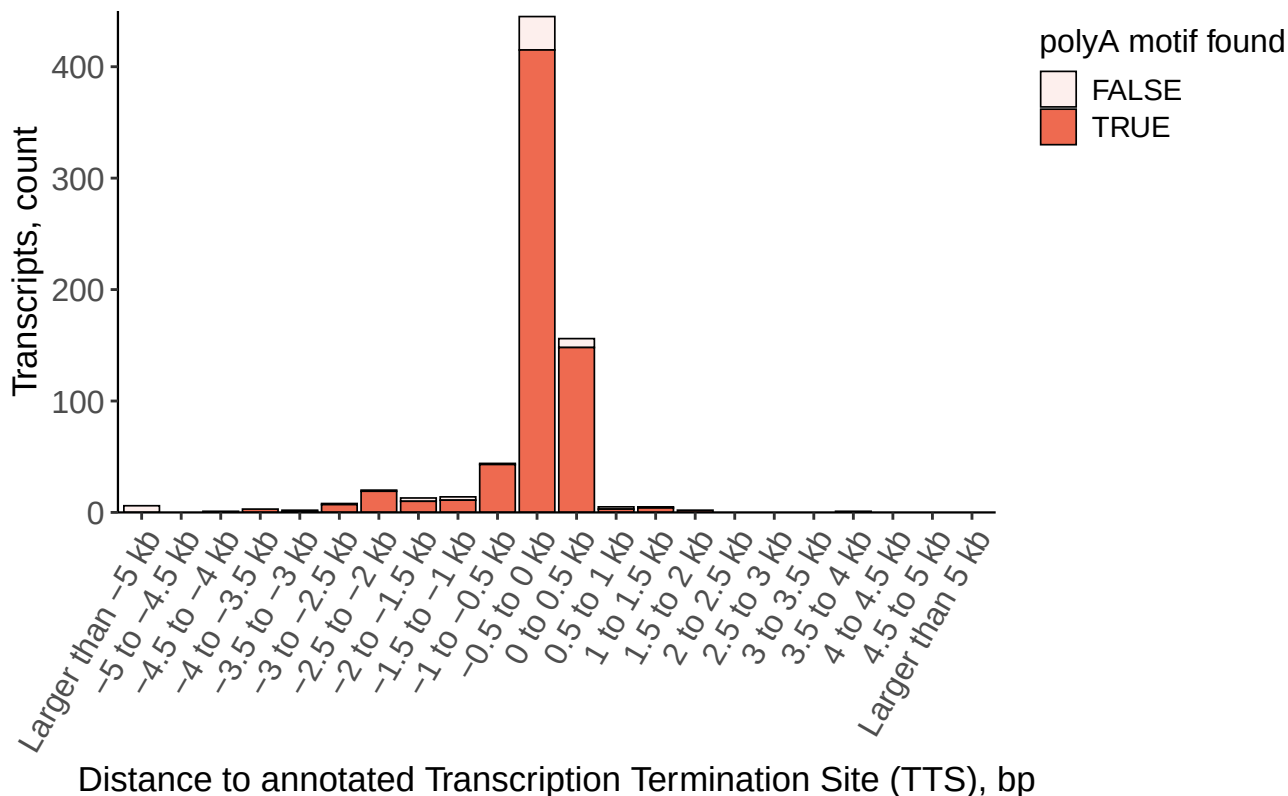
# Distance to annotated Transcription Termination Site (TTS) FSM Reference Match

Negative values indicate upstream of annotated termination site



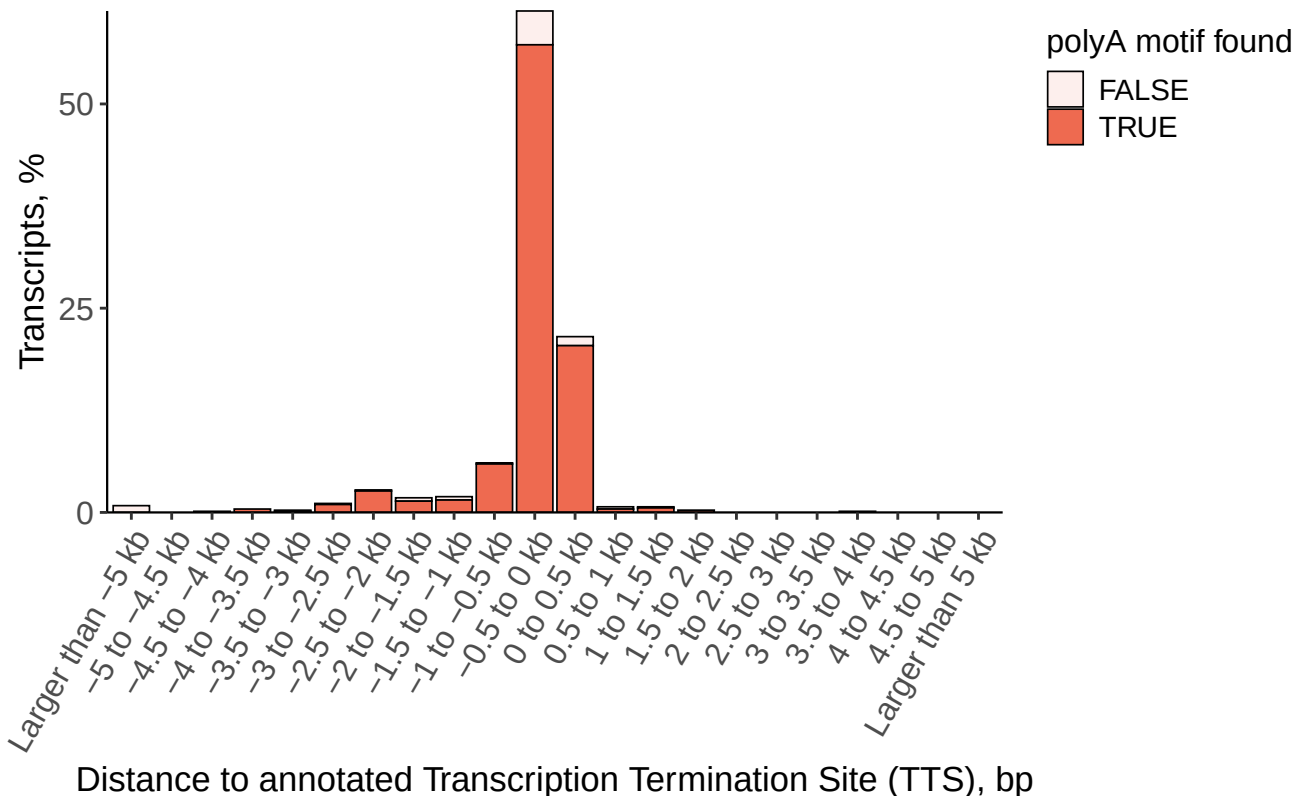
# Distance to Annotated Transcription Termination Site for ISM 3' Fragment

Negative values indicate upstream of annotated TTS



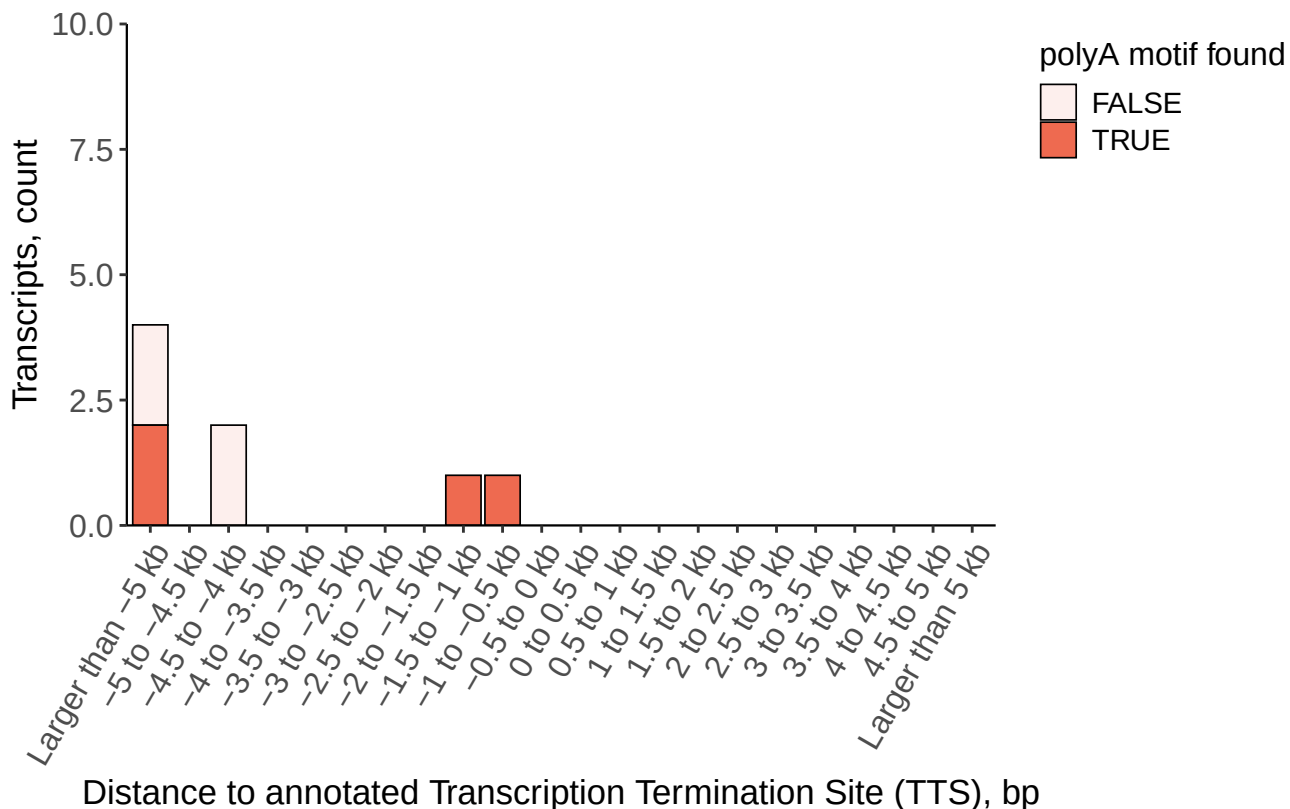
# Distance to Annotated Transcription Termination Site for ISM 3' Fragment

Negative values indicate upstream of annotated TTS



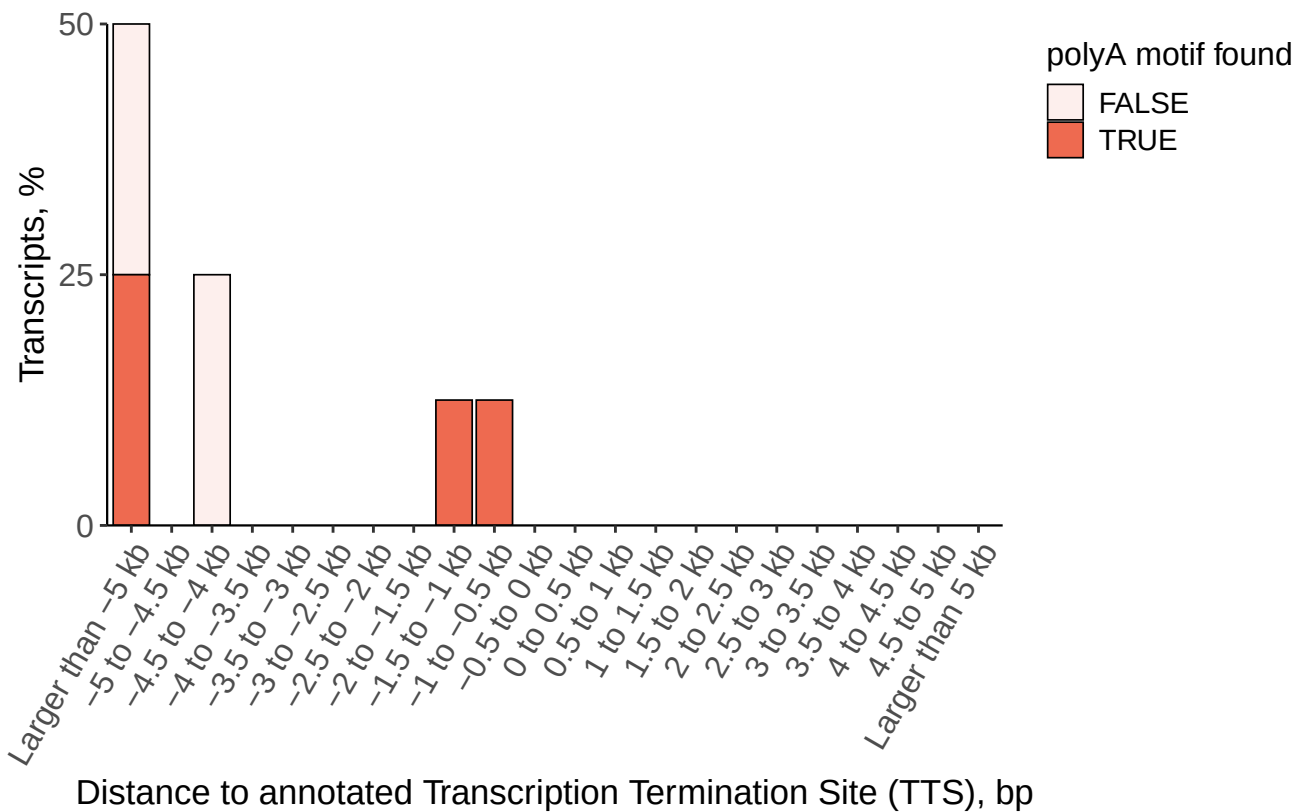
# Distance to Annotated Transcription Termination Site for ISM Internal Fragment

Negative values indicate upstream of annotated TTS



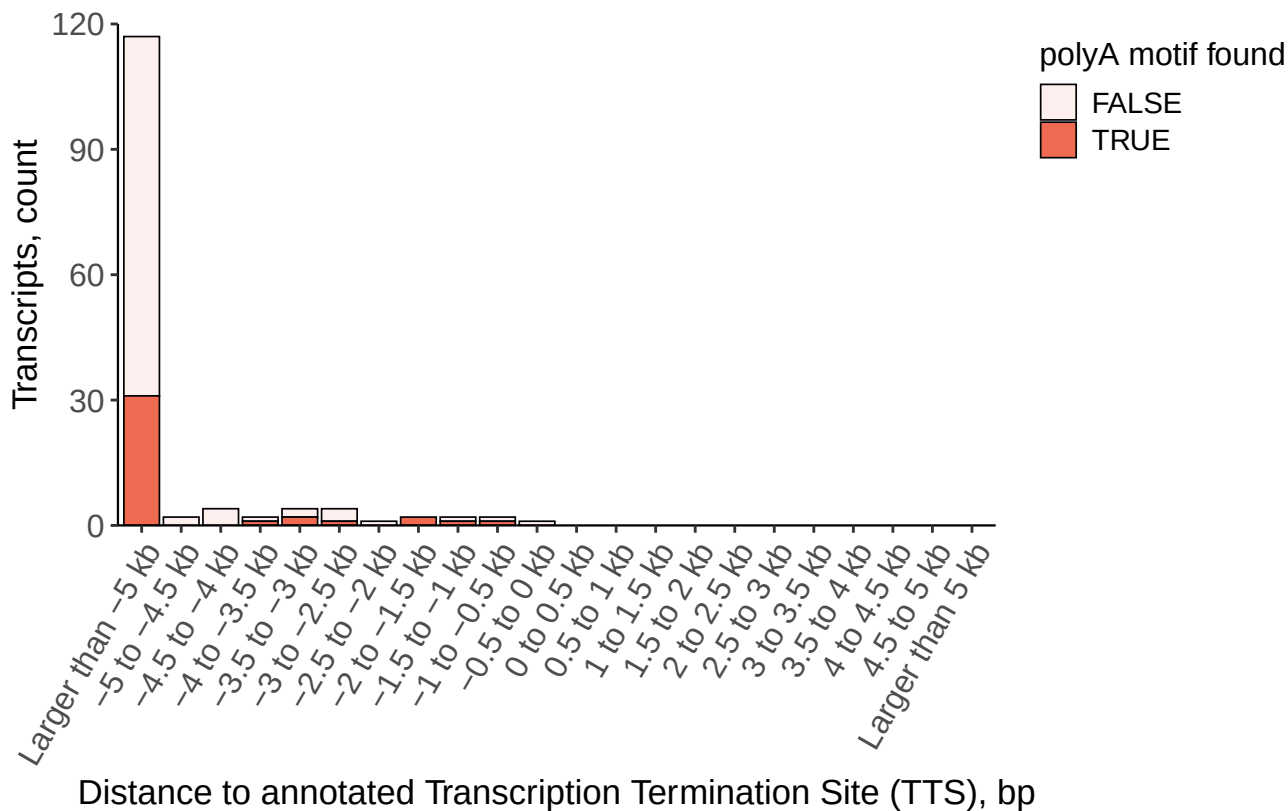
# Distance to Annotated Transcription Termination Site for ISM Internal Fragment

Negative values indicate upstream of annotated TTS



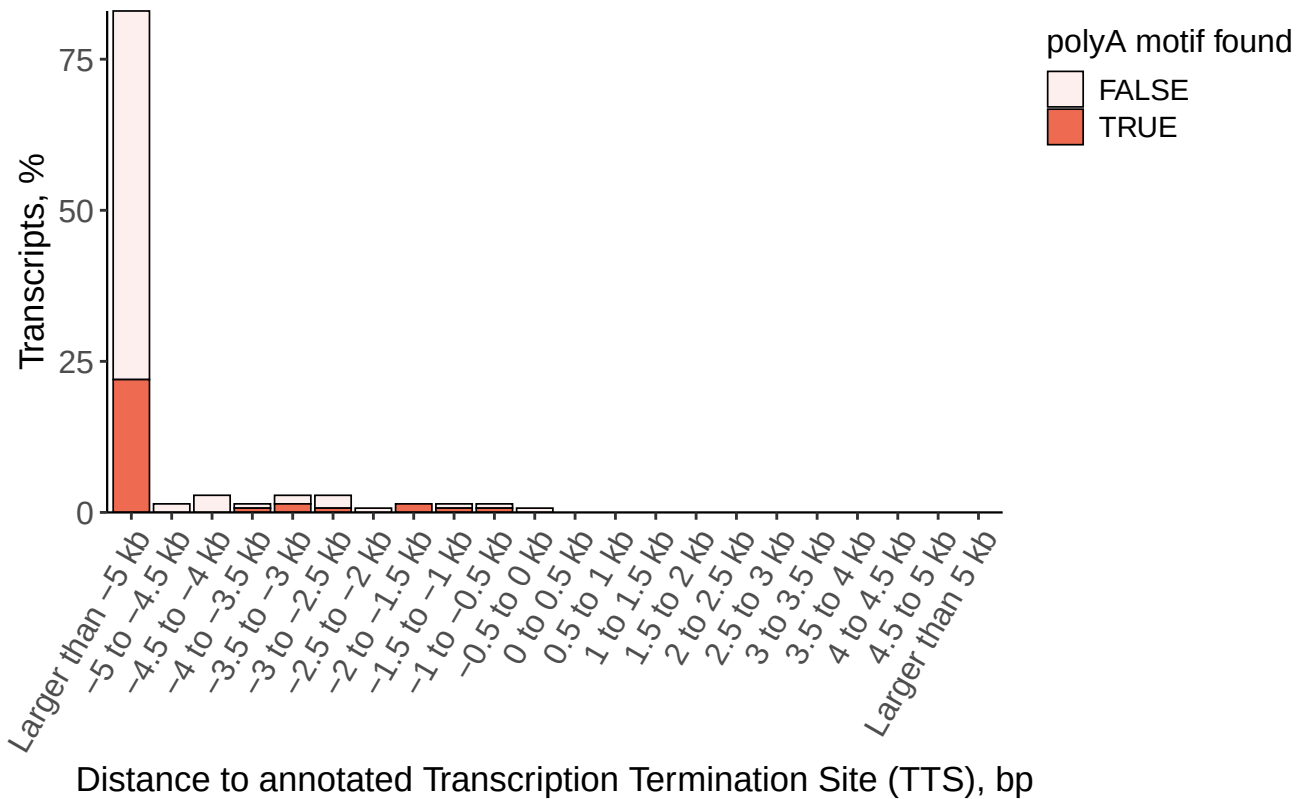
# Distance to Annotated Transcription Termination Site for ISM A5' Fragment

Negative values indicate upstream of annotated TTS



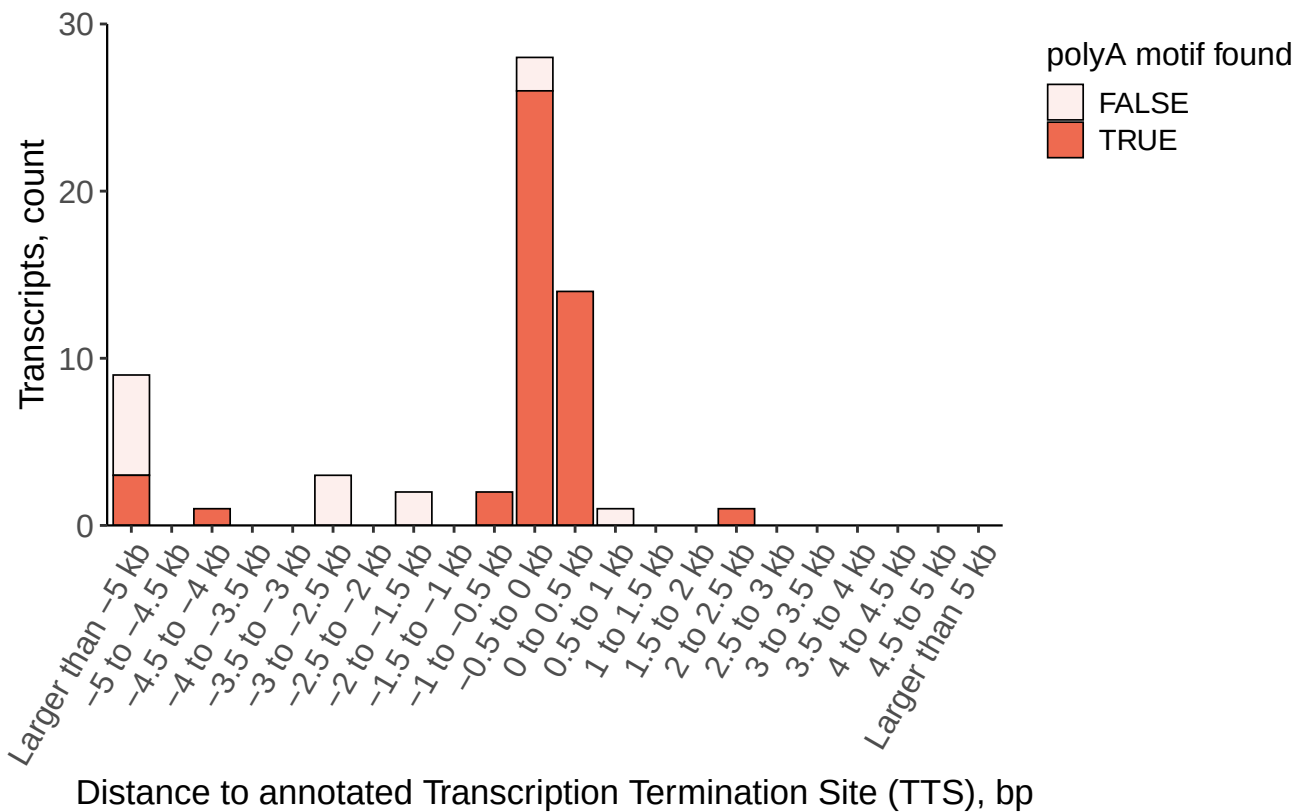
# Distance to Annotated Transcription Termination Site for ISM A5' Fragment

Negative values indicate upstream of annotated TTS



# Distance to Annotated Transcription Termination Site for ISM Intron Retention

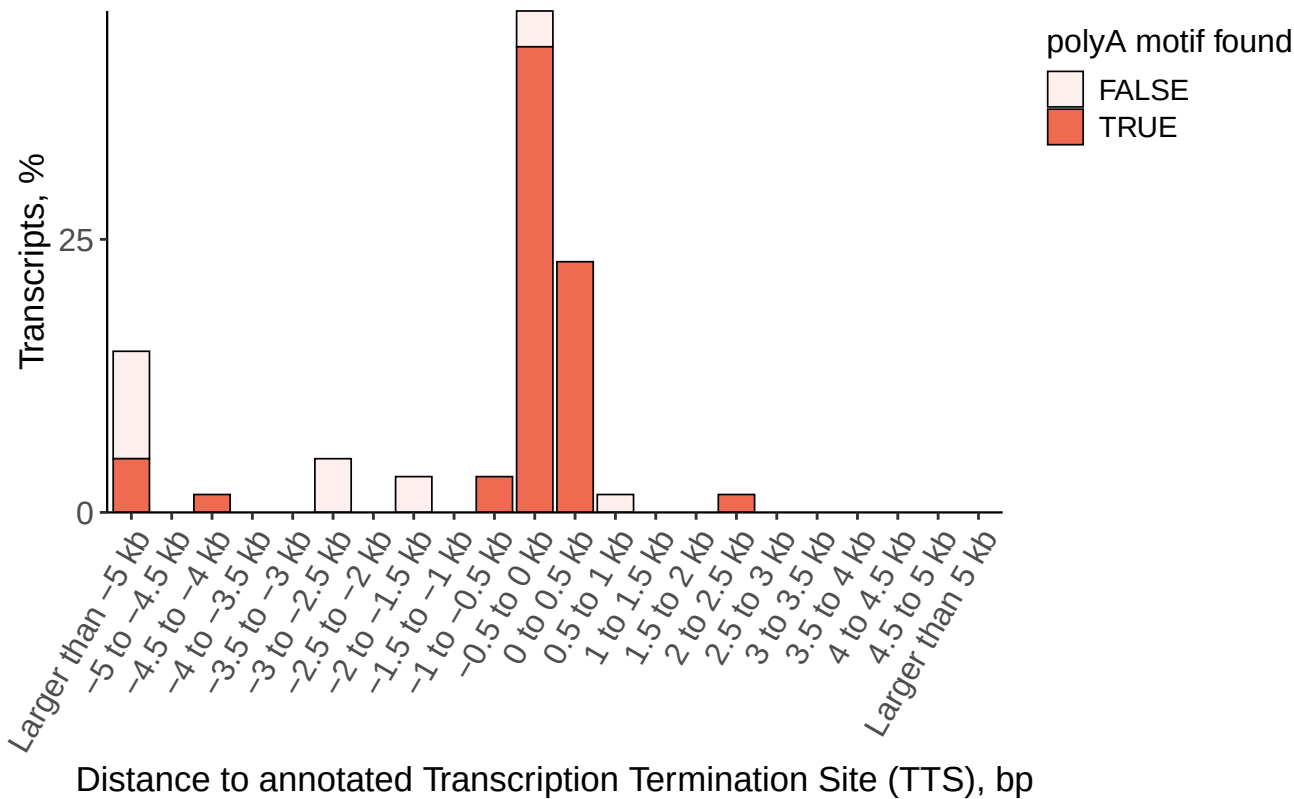
Negative values indicate upstream of annotated TTS





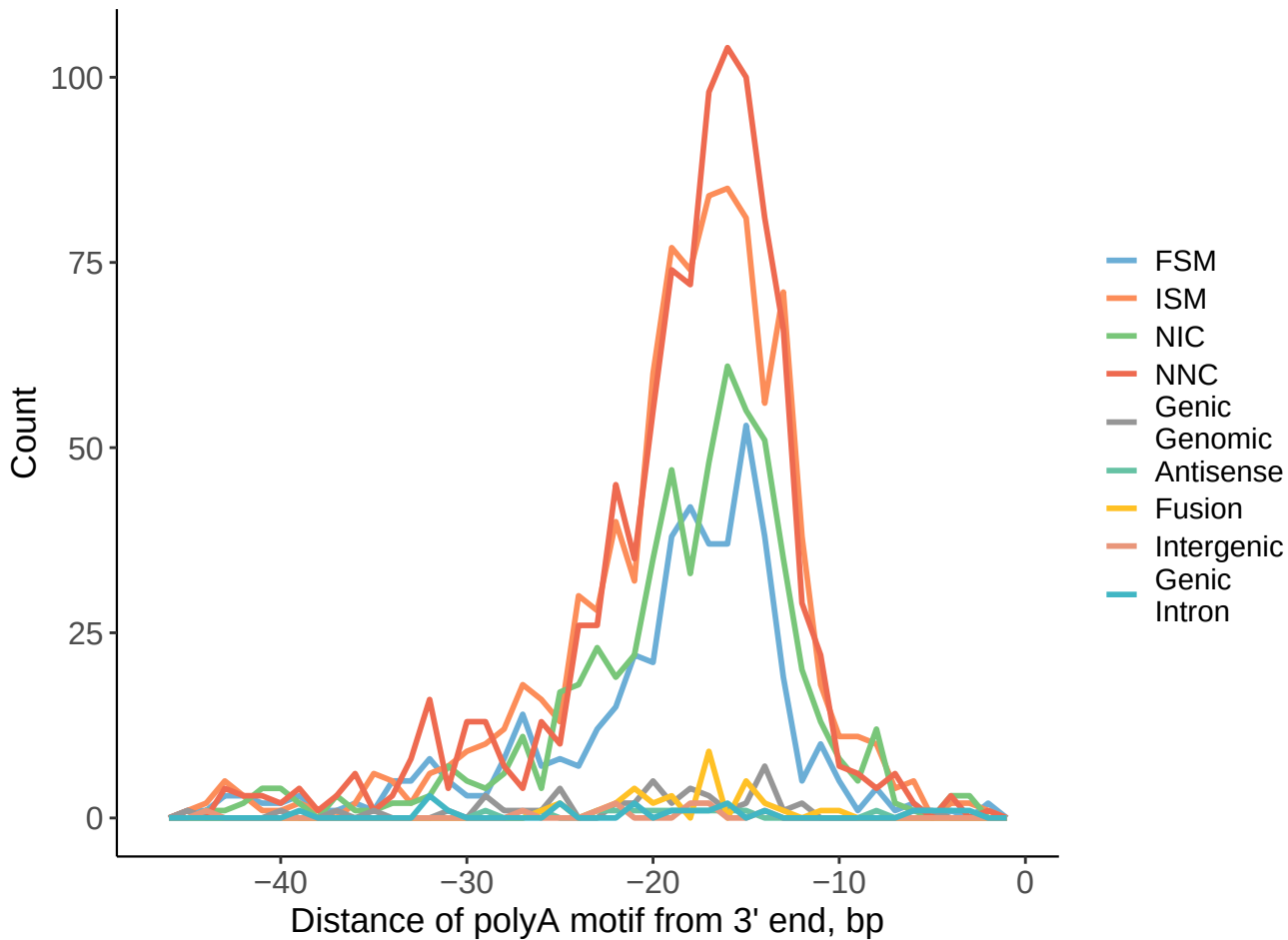
# Distance to Annotated Transcription Termination Site for ISM Intron Retention

Negative values indicate upstream of annotated TTS



## *PolyA Distance Analysis*

## Distance of Detected PolyA Motif From 3' end



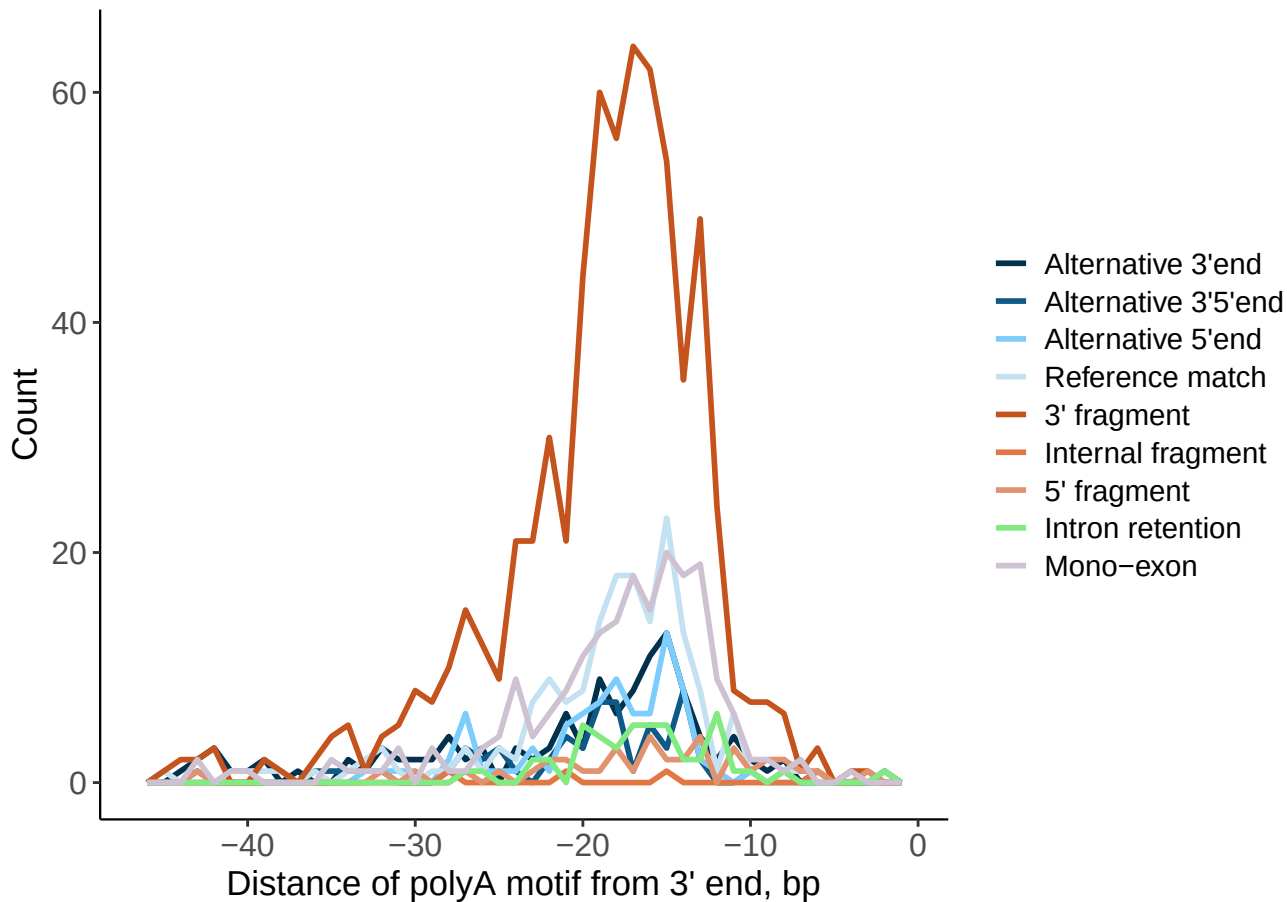
*Frequency of PolyA Motifs*

*Number of polyA Motifs Detected*

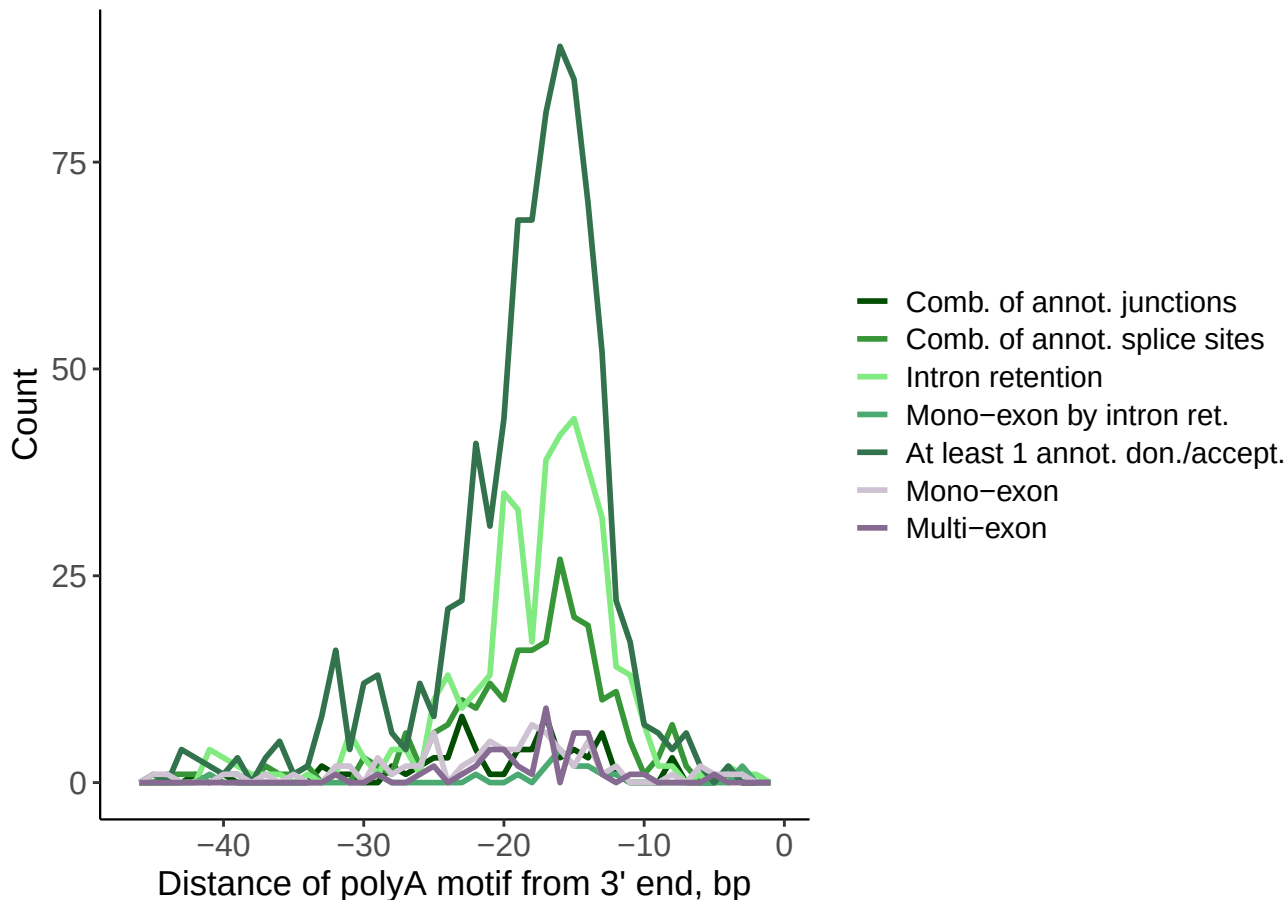
| Category         | Count | polyA<br>Detected | %  |
|------------------|-------|-------------------|----|
| FSM              | 539   | 458               | 85 |
| ISM              | 1138  | 942               | 83 |
| NIC              | 744   | 594               | 80 |
| NNC              | 1139  | 981               | 86 |
| Genic<br>Genomic | 109   | 48                | 44 |
| Antisense        | 31    | 12                | 39 |
| Fusion           | 38    | 34                | 89 |
| Intergenic       | 40    | 10                | 25 |
| Genic<br>Intron  | 147   | 19                | 13 |

| Motif  | Count | %    |
|--------|-------|------|
| AATAAA | 2014  | 65.0 |
| ATTAAA | 529   | 17.1 |
| AGTAAA | 161   | 5.2  |
| TATAAA | 90    | 2.9  |
| GATAAA | 54    | 1.7  |
| AAAACA | 41    | 1.3  |
| AATACA | 38    | 1.2  |
| AAAAAG | 32    | 1.0  |
| AATAGA | 28    | 0.9  |
| CATAAA | 26    | 0.8  |
| AAGAAA | 24    | 0.8  |
| AATATA | 20    | 0.6  |
| TTTAAA | 16    | 0.5  |
| AATGAA | 13    | 0.4  |
| ACTAAA | 10    | 0.3  |
| GGGGCT | 2     | 0.1  |

### Distance of Detected PolyA Motif From 3'End by FSM and ISM Subcategories



Distance of Detected PolyA Motif From 3'End  
by Non-FSM/ISM Subcategories



## *Number of polyA Motifs Detected*

| <b>Subcategory</b>             | <b>Count</b> | <b>polyA<br/>Detected</b> | <b>%</b> |
|--------------------------------|--------------|---------------------------|----------|
| Alternative 3'end              | 152          | 121                       | 80       |
| Alternative 3'5'end            | 76           | 60                        | 79       |
| Alternative 5'end              | 90           | 84                        | 93       |
| Reference match                | 188          | 175                       | 93       |
| 3' fragment                    | 725          | 668                       | 92       |
| Internal fragment              | 8            | 4                         | 50       |
| 5' fragment                    | 141          | 39                        | 28       |
| Comb. of annot. junctions      | 83           | 68                        | 82       |
| Comb. of annot. splice sites   | 281          | 235                       | 84       |
| Intron retention               | 554          | 459                       | 83       |
| Mono-exon by intron ret.       | 44           | 18                        | 41       |
| At least 1 annot. don./accept. | 986          | 846                       | 86       |
| Mono-exon                      | 543          | 277                       | 51       |
| Multi-exon                     | 54           | 44                        | 81       |

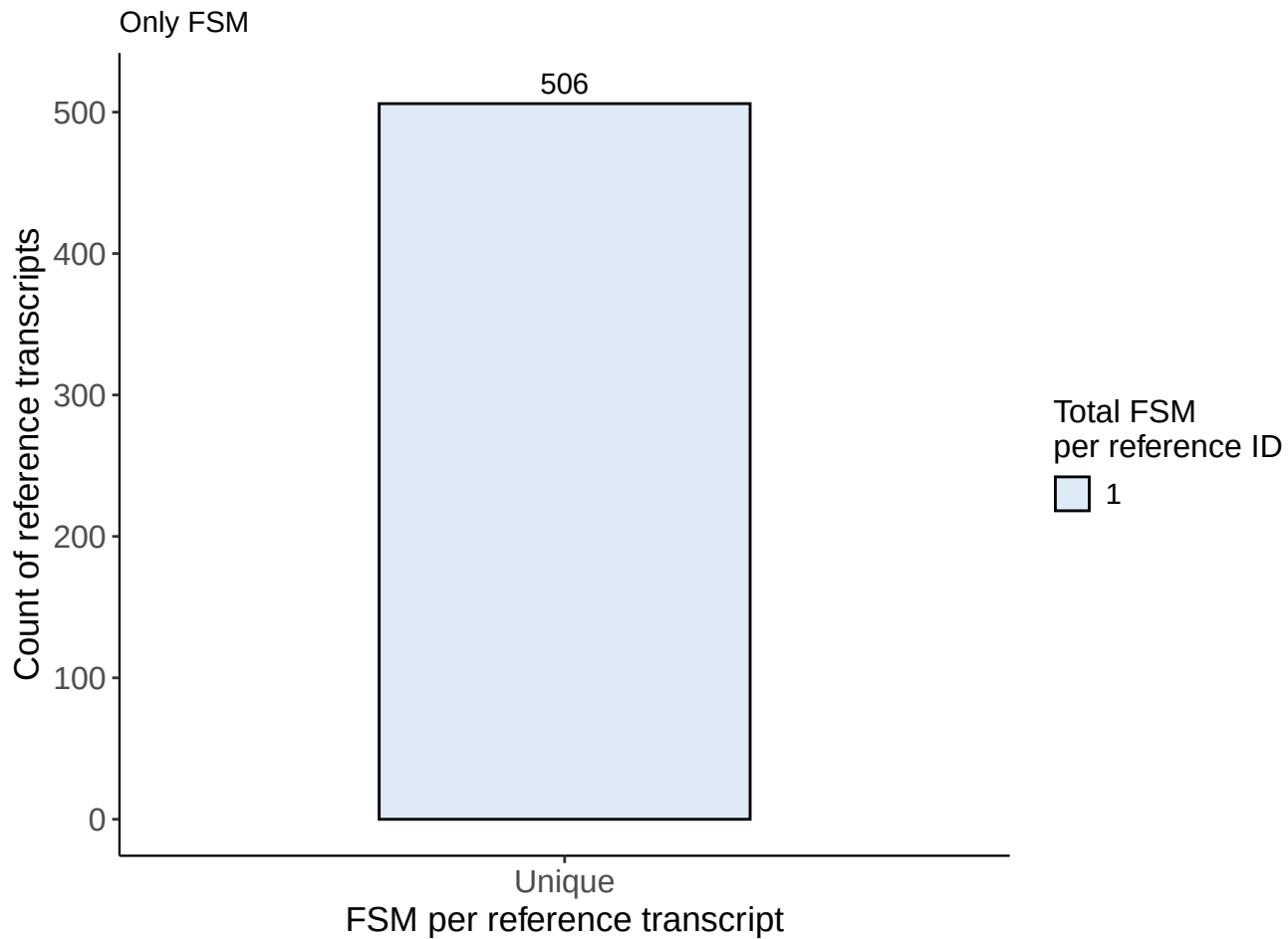
## *Frequency of PolyA Motifs*

| <b>Motif</b> | <b>Count</b> | <b>%</b> |
|--------------|--------------|----------|
| AATAAA       | 2014         | 65.0     |
| ATTAAA       | 529          | 17.1     |
| AGTAAA       | 161          | 5.2      |
| TATAAA       | 90           | 2.9      |
| GATAAA       | 54           | 1.7      |
| AAAACA       | 41           | 1.3      |
| AATACA       | 38           | 1.2      |
| AAAAAG       | 32           | 1.0      |
| AATAGA       | 28           | 0.9      |
| CATAAA       | 26           | 0.8      |
| AAGAAA       | 24           | 0.8      |
| AATATA       | 20           | 0.6      |
| TTTAAA       | 16           | 0.5      |
| AATGAA       | 13           | 0.4      |
| ACTAAA       | 10           | 0.3      |
| GGGGCT       | 2            | 0.1      |

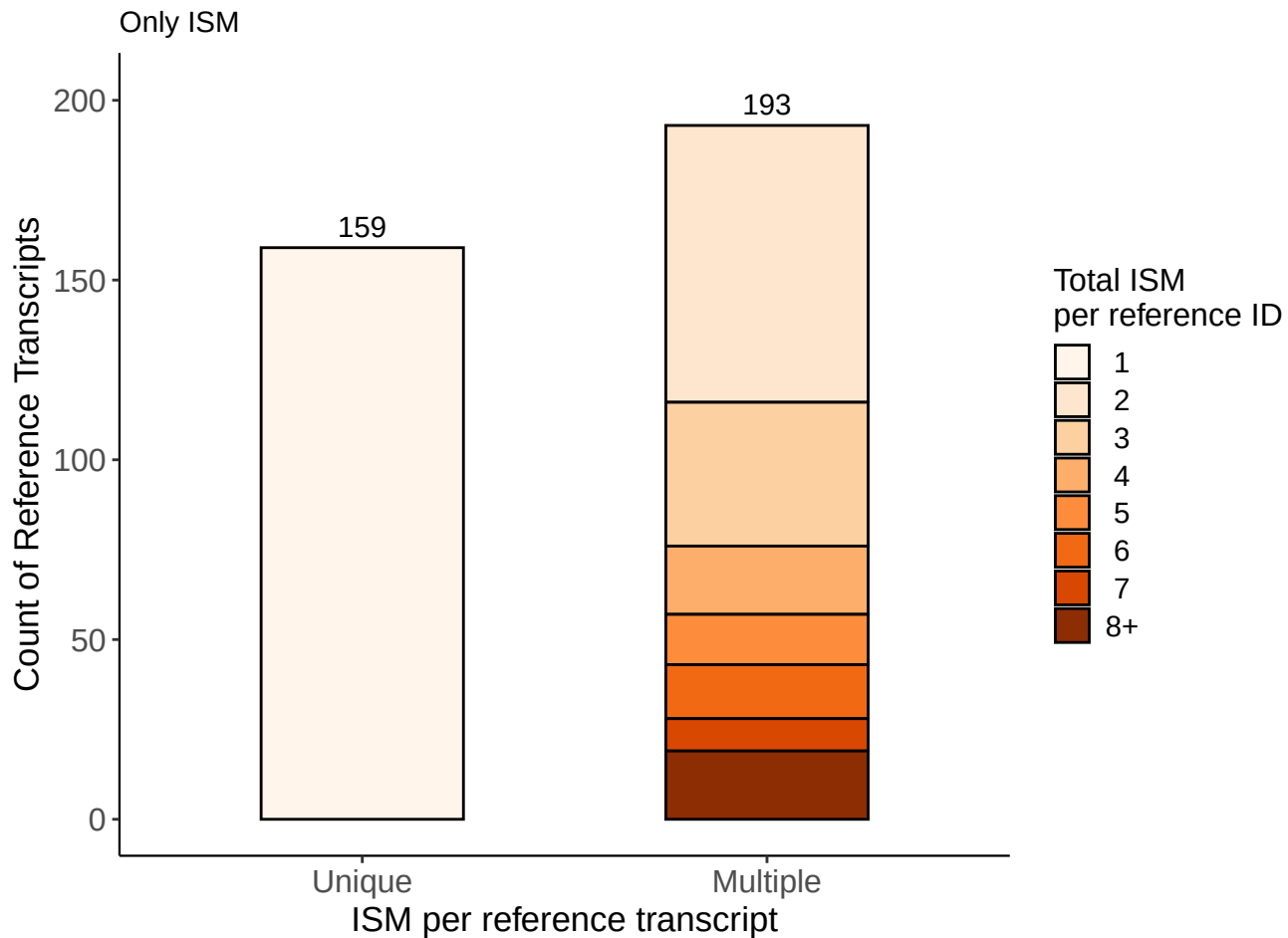


## *Redundancy Analysis*

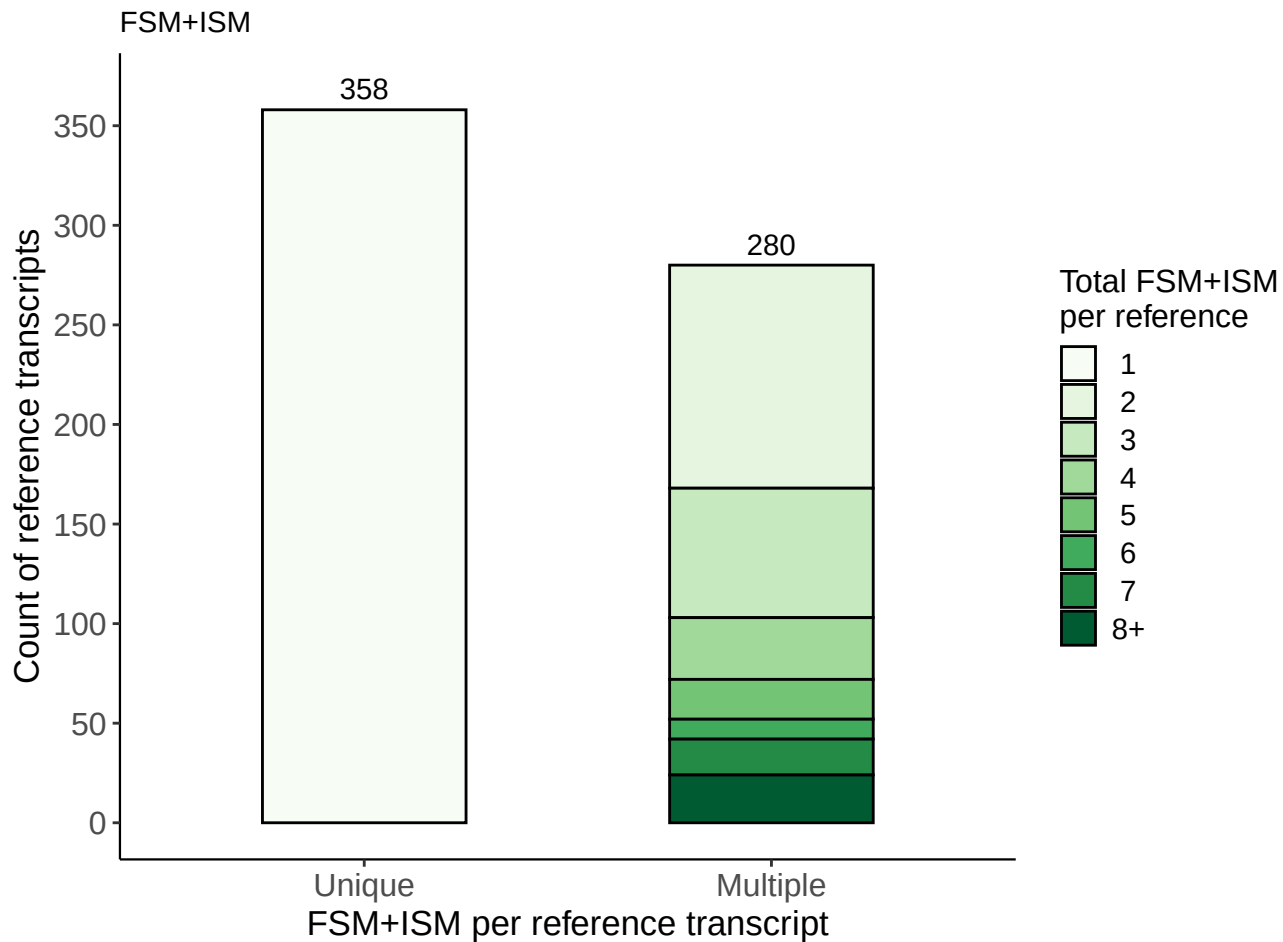
## Reference Transcript Redundancy



## Reference Transcript Redundancy

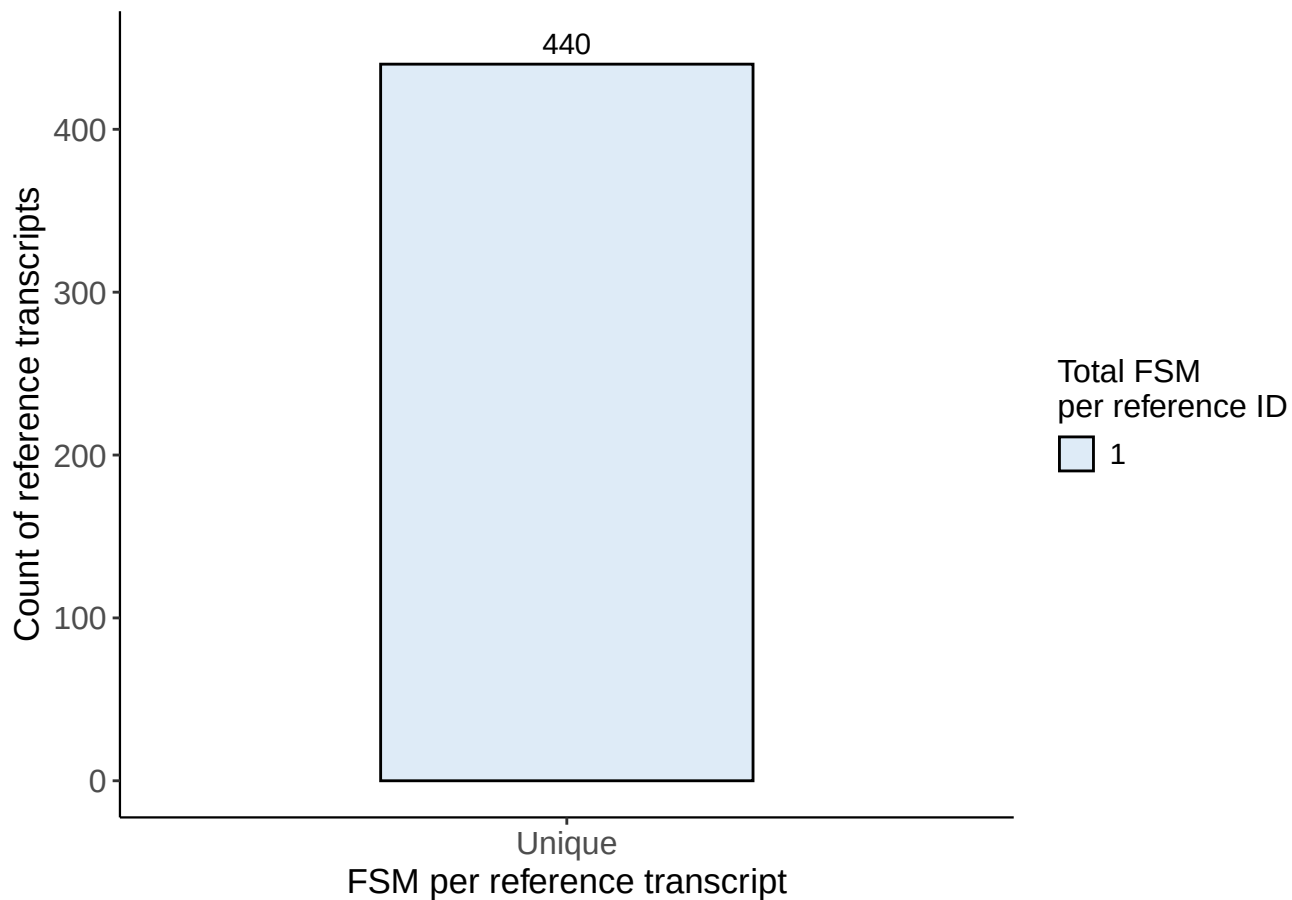


## Reference Transcript Redundancy



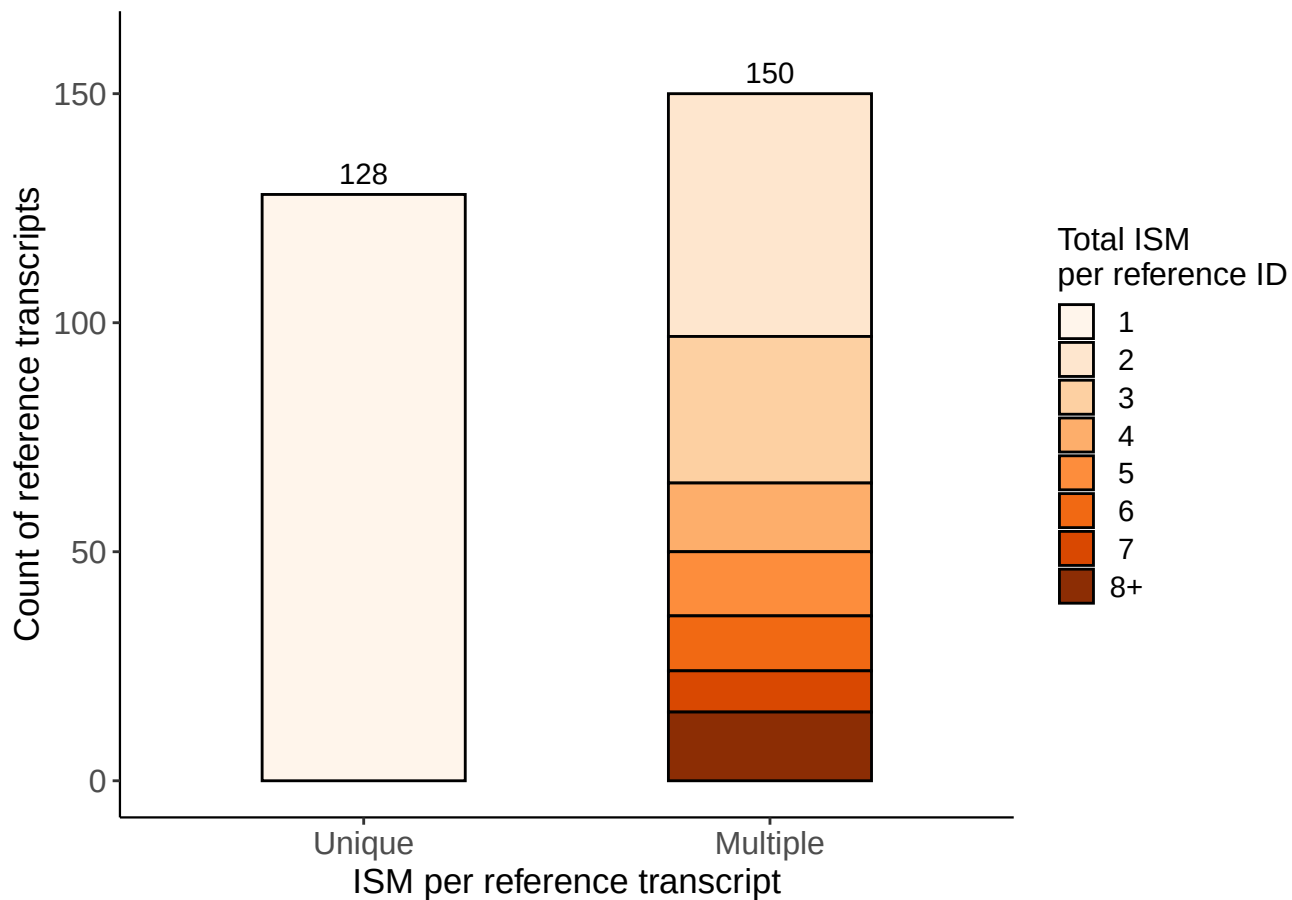
# Reference Transcript Redundancy

Only FSM with a polyA motif found



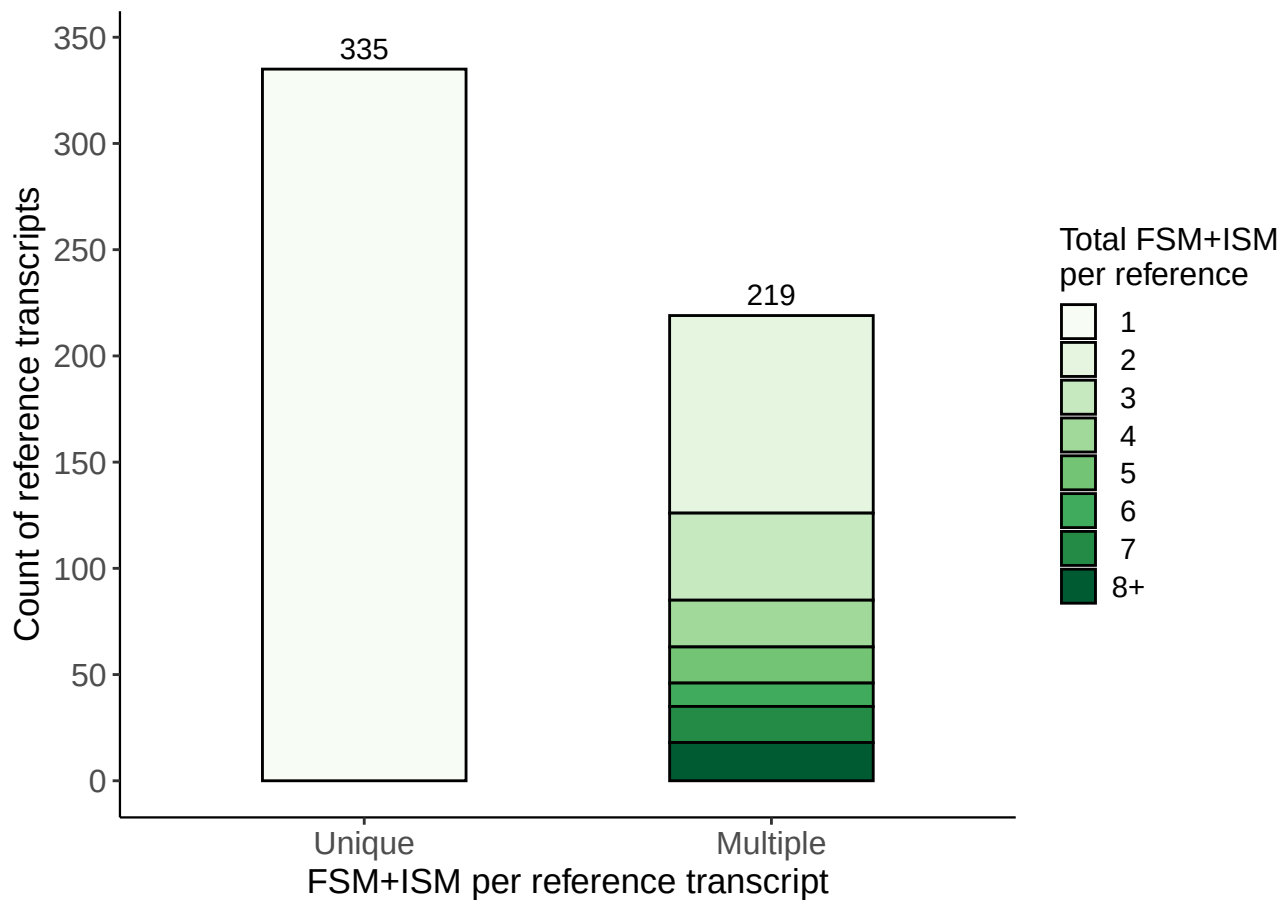
## Reference Transcript Redundancy

Only ISM with a polyA motif found



## Reference Transcript Redundancy

FSM+ISM with a polyA motif found

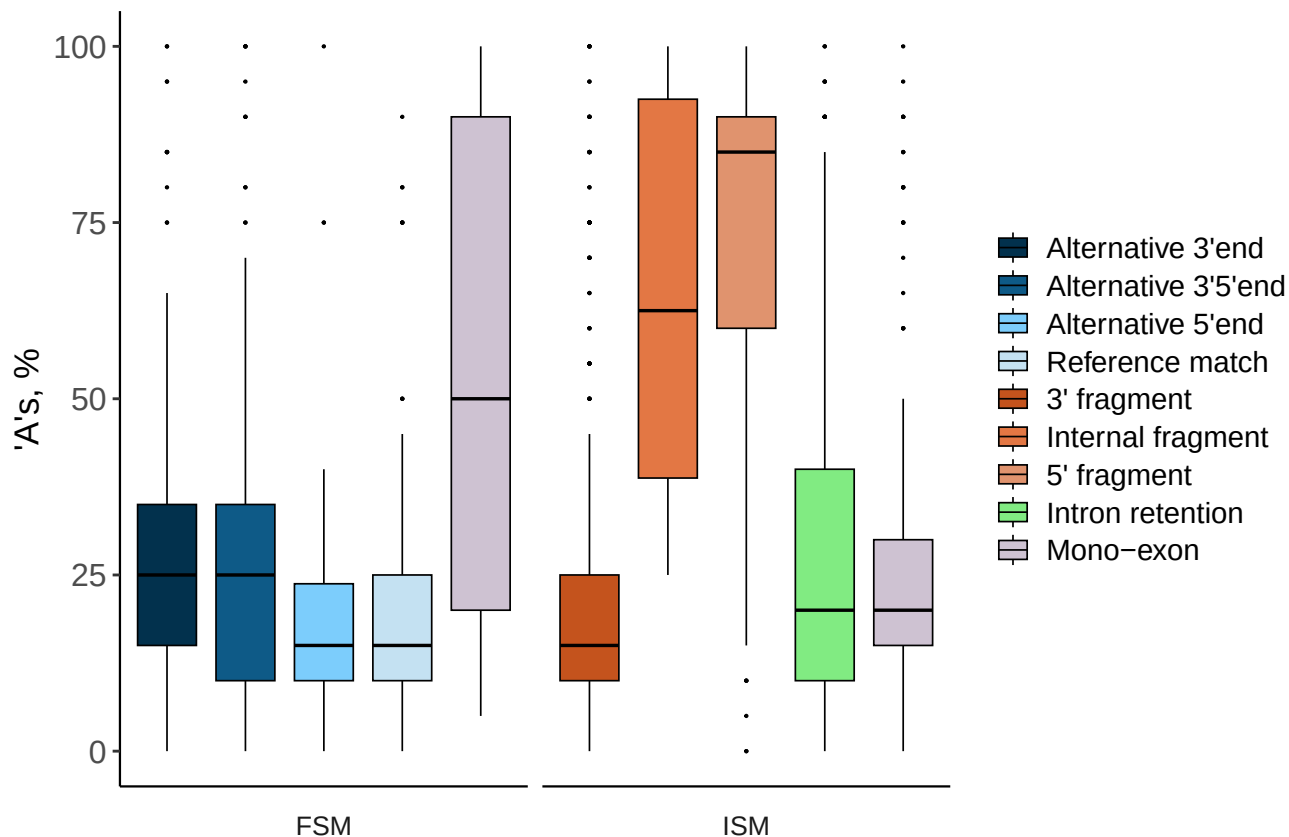


*Intra-Priming Quality Check*



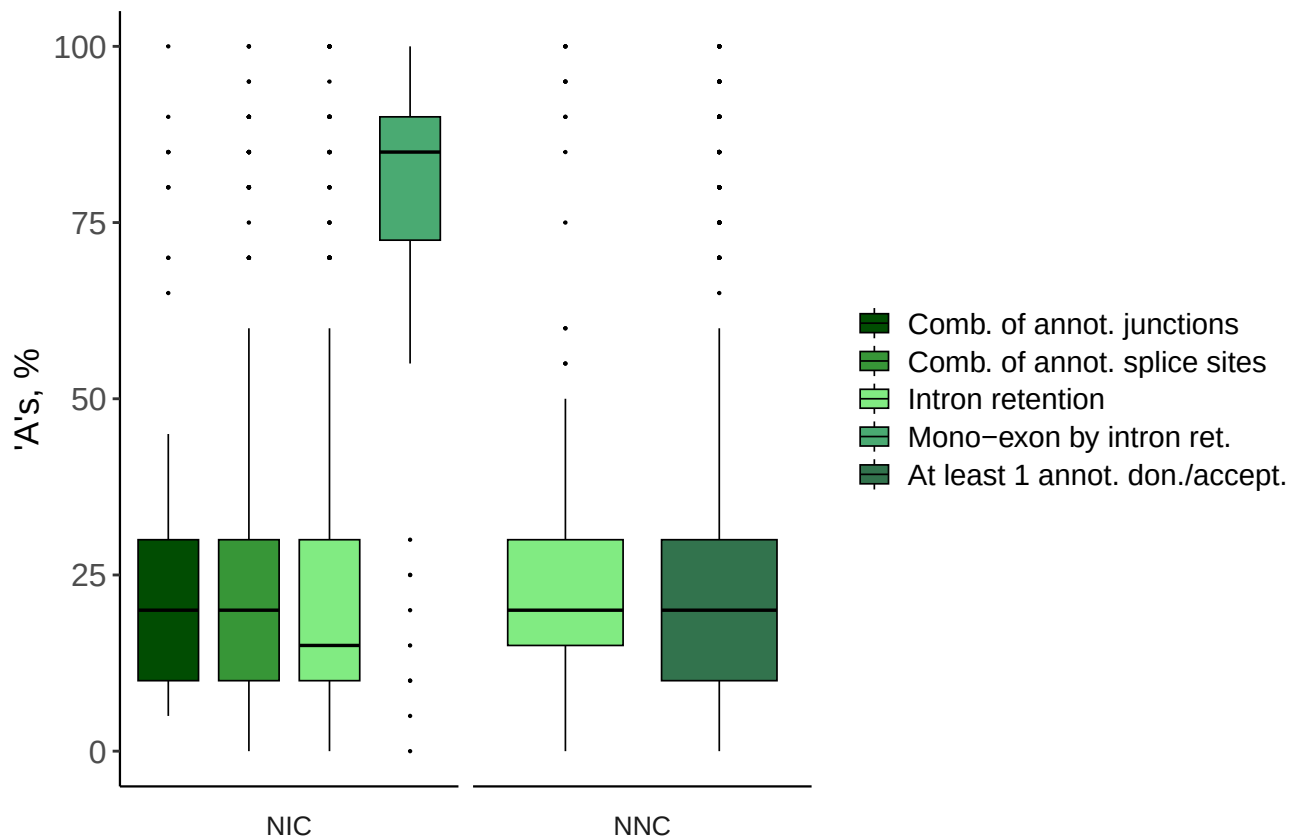
## Possible Intra-Priming by Structural Category

Percent of genomic 'A's in downstream 20 bp



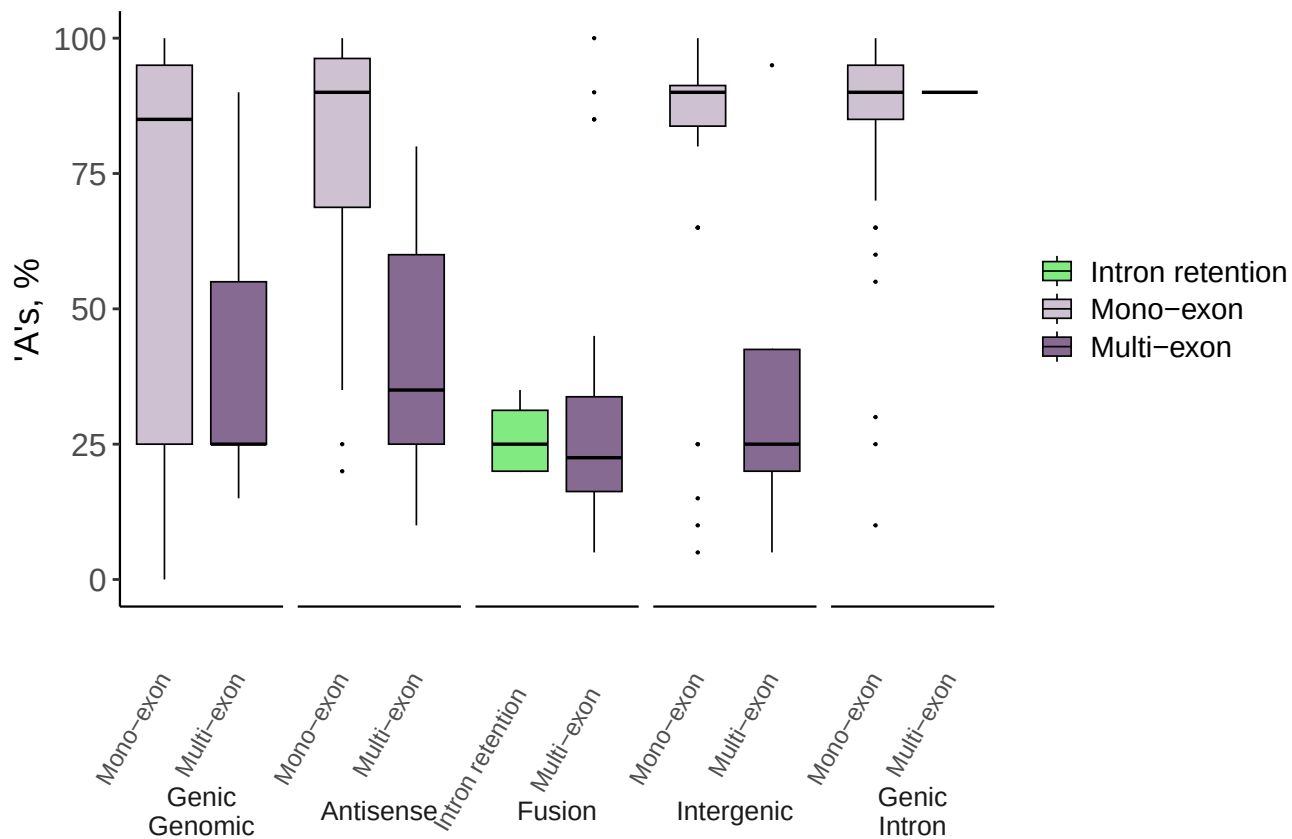
# Possible Intra-Priming by Structural Category

Percent of genomic 'A's in downstream 20 bp



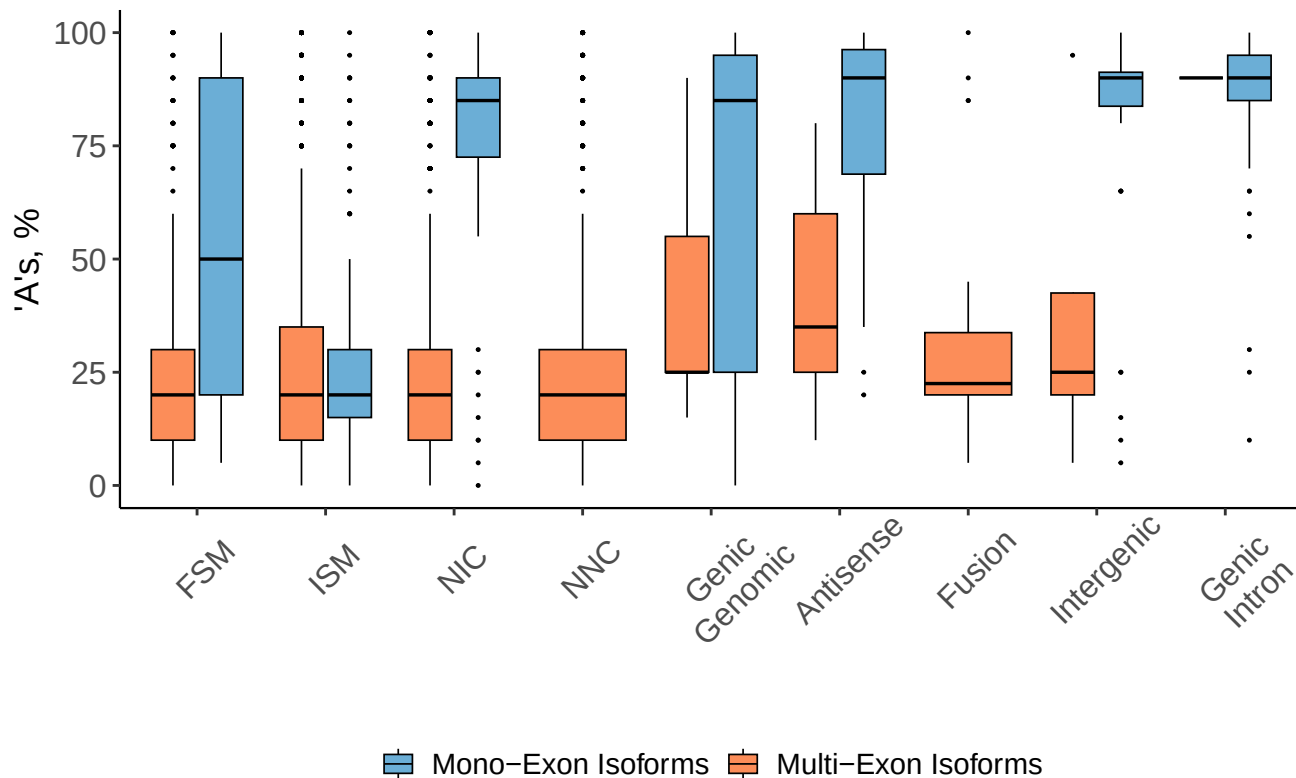
# Possible Intra-Priming by Structural Category

Percent of genomic 'A's in downstream 20 bp



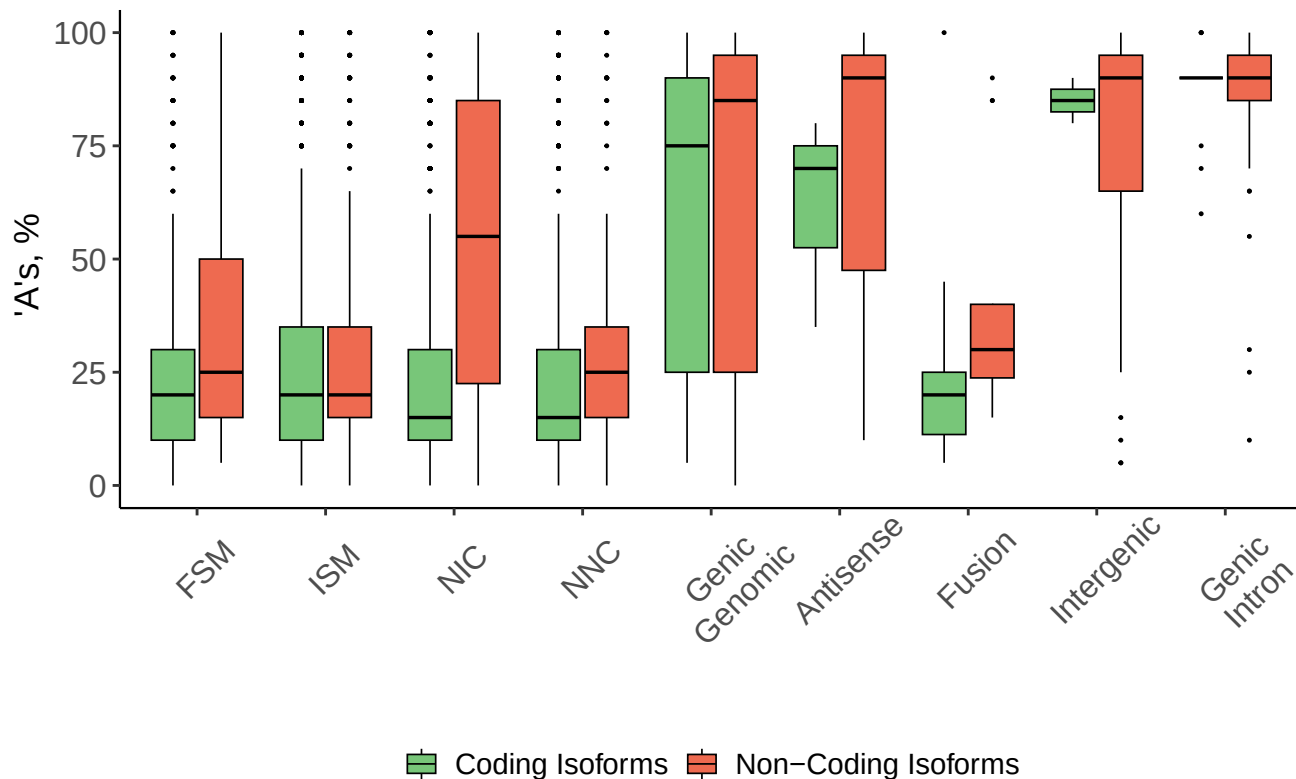
# Mono- vs Multi-Exon Possible Intra-Priming

Percent of genomic 'A's in downstream 20 bp



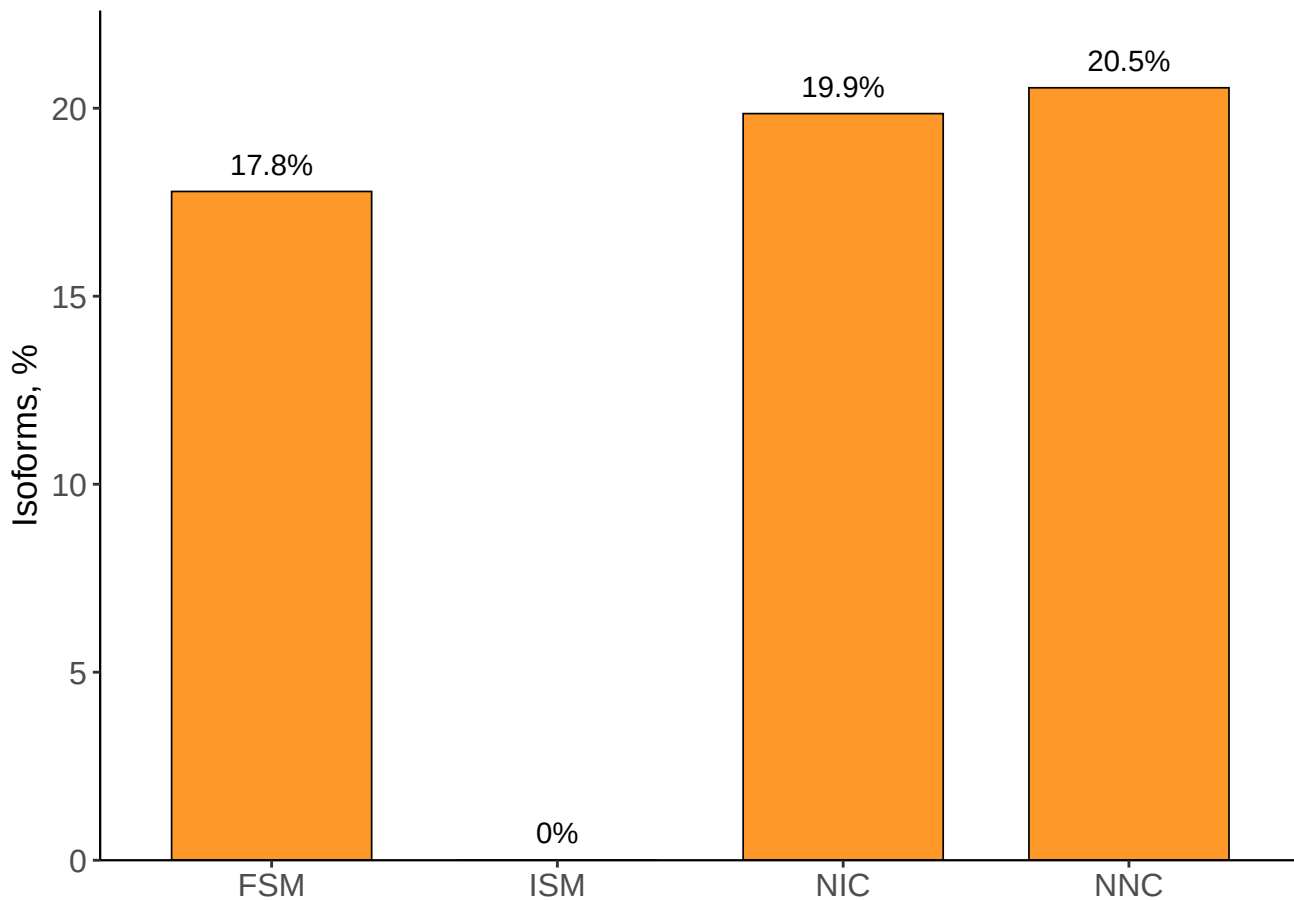
# Coding vs Non-Coding Possible Intra-Priming

Percent of genomic 'A's in downstream 20 bp

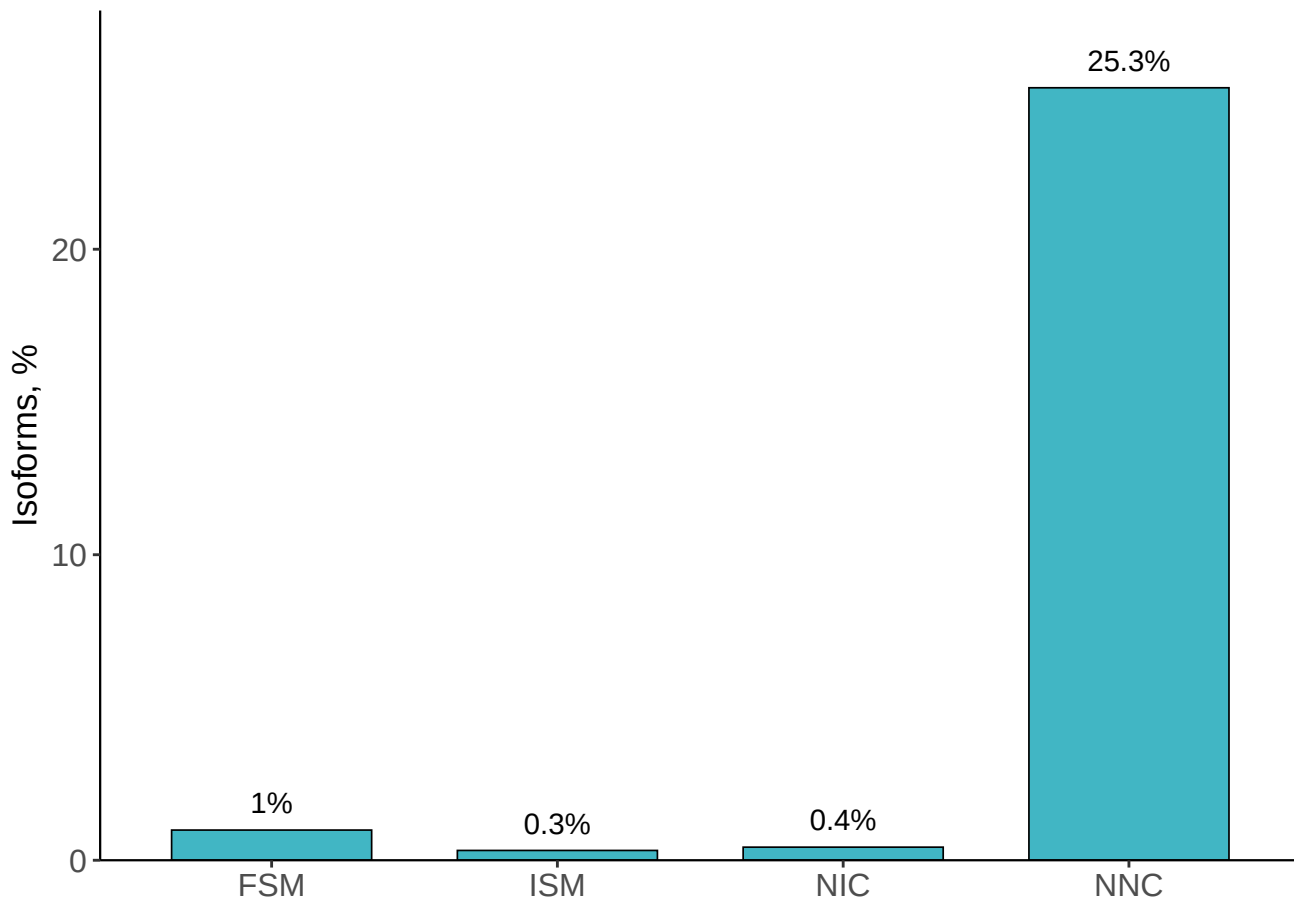


## *Features of Bad Quality*

## RT-switching

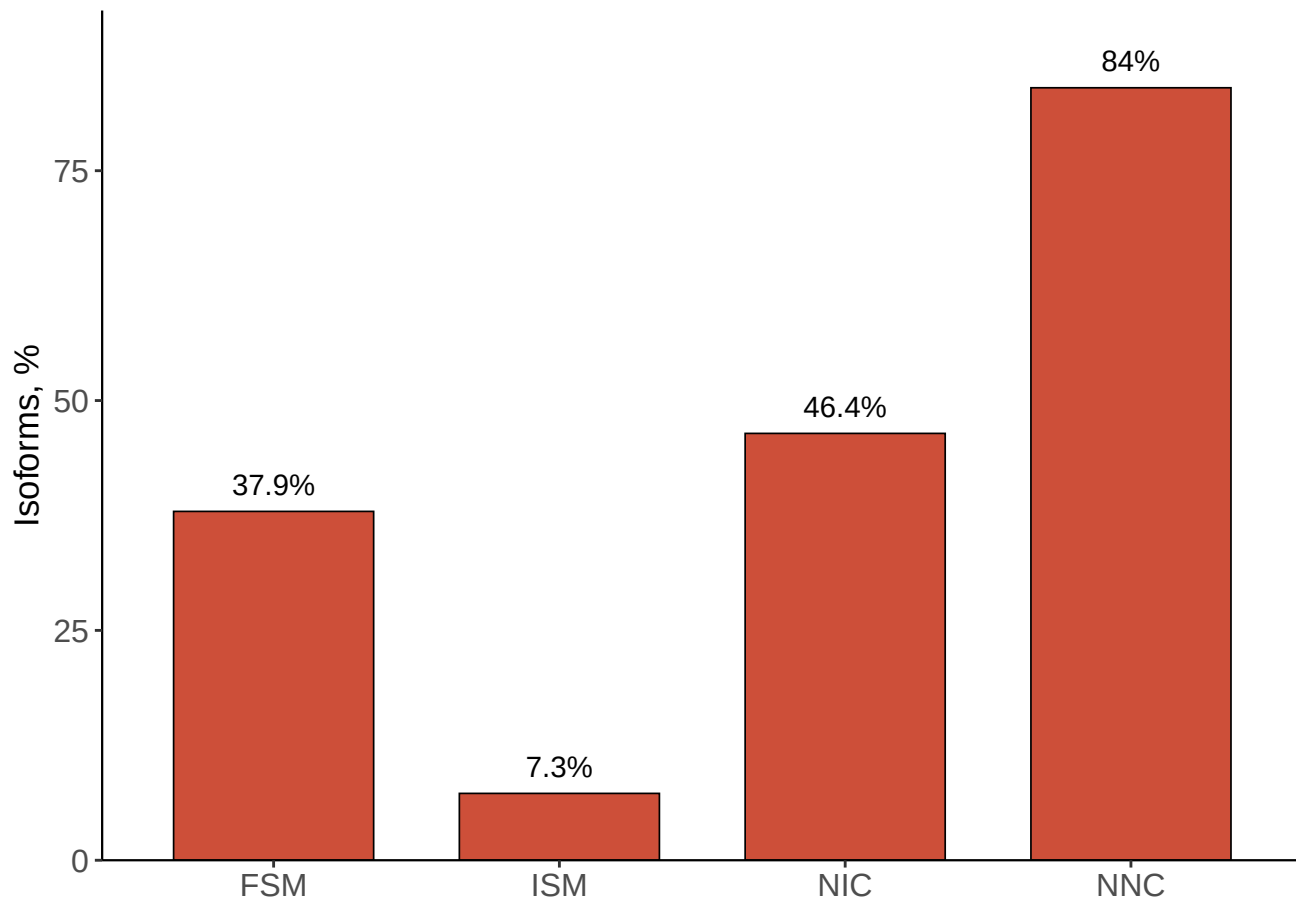


## Non-Canonical Junctions

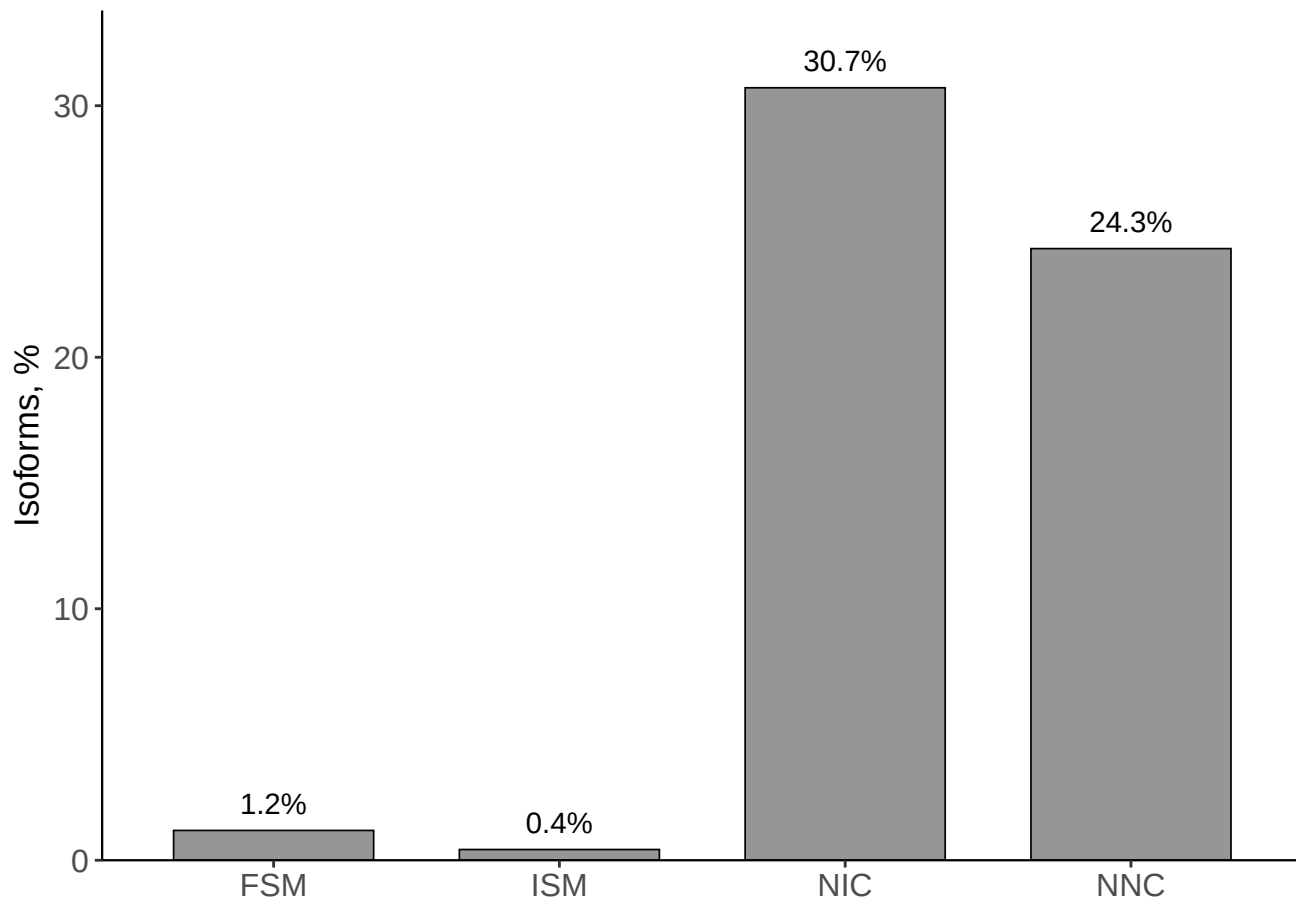




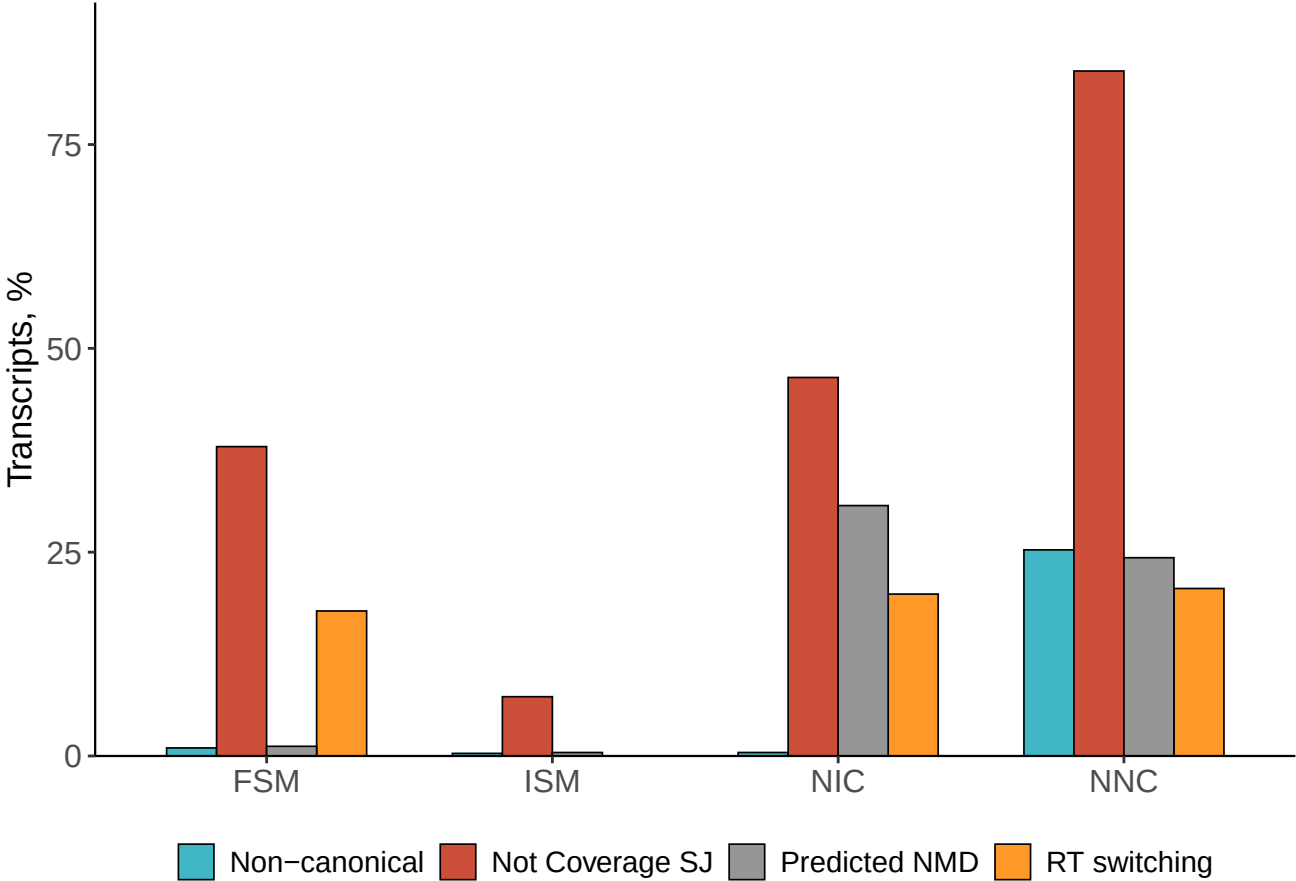
## Splice Junctions Without Short Read Coverage



## Nonsense-Mediated Decay by Structural Category

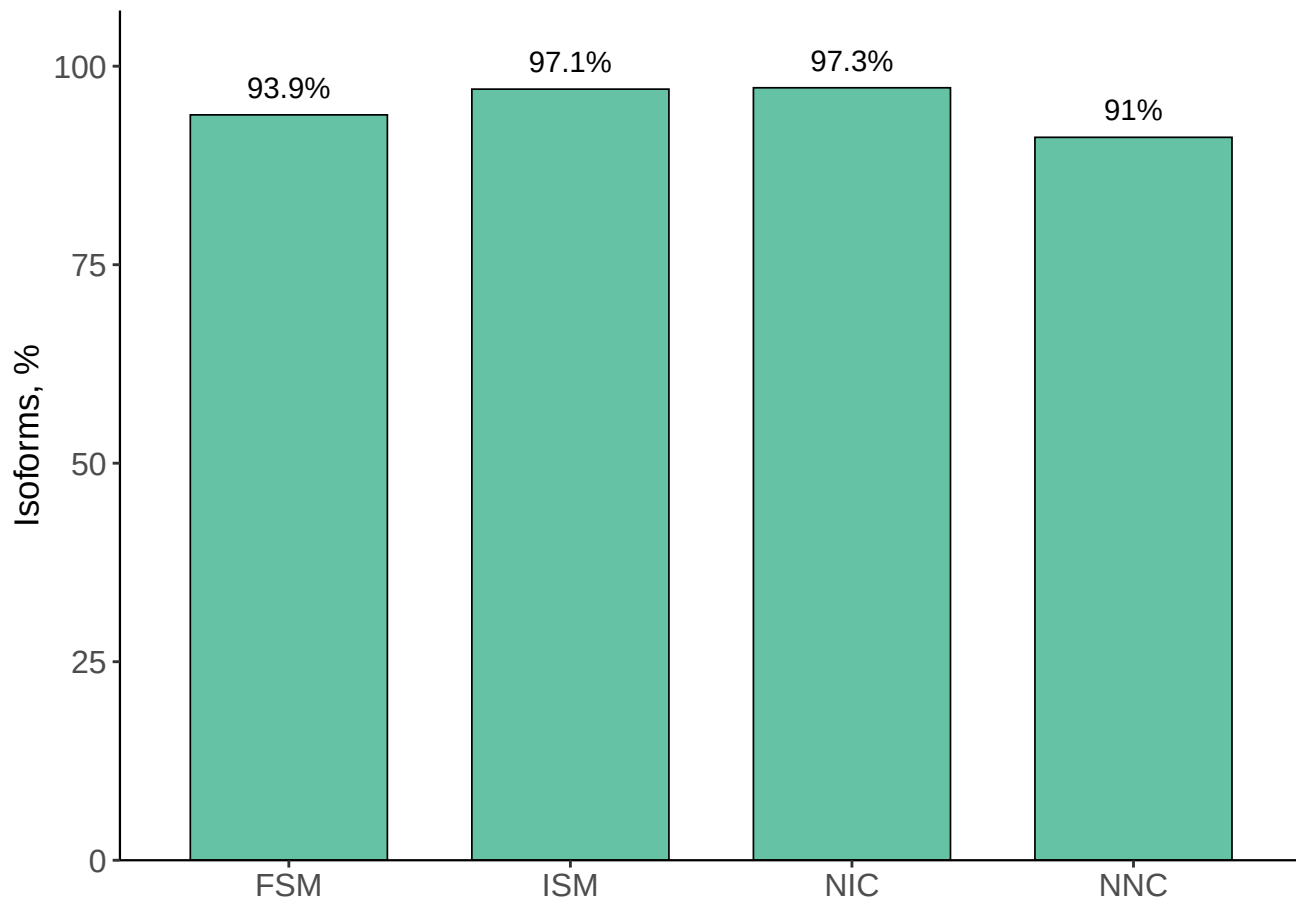


# Quality Control Attributes Across Structural Categories

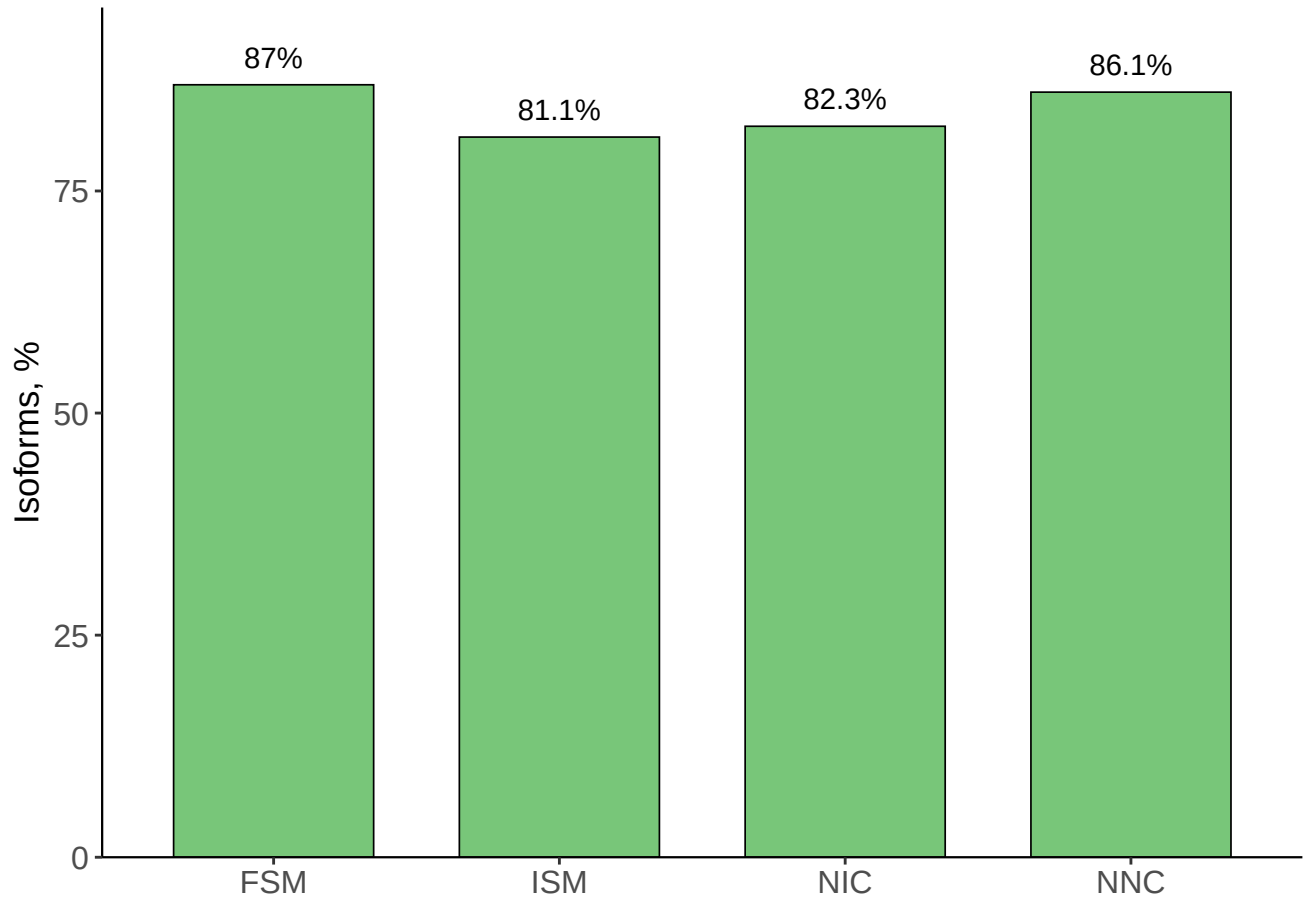


## *Features of Good Quality*

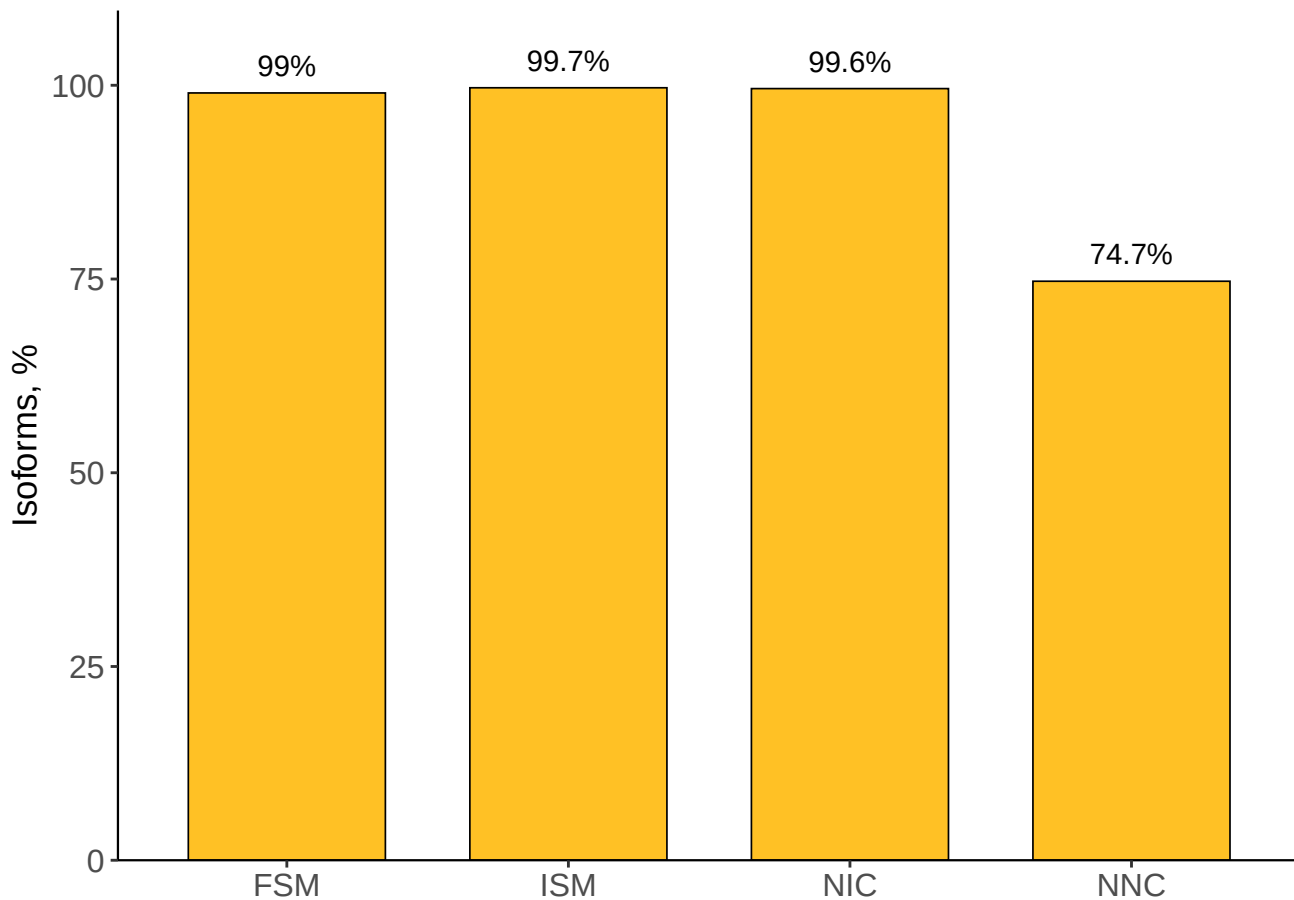
## Annotation Support



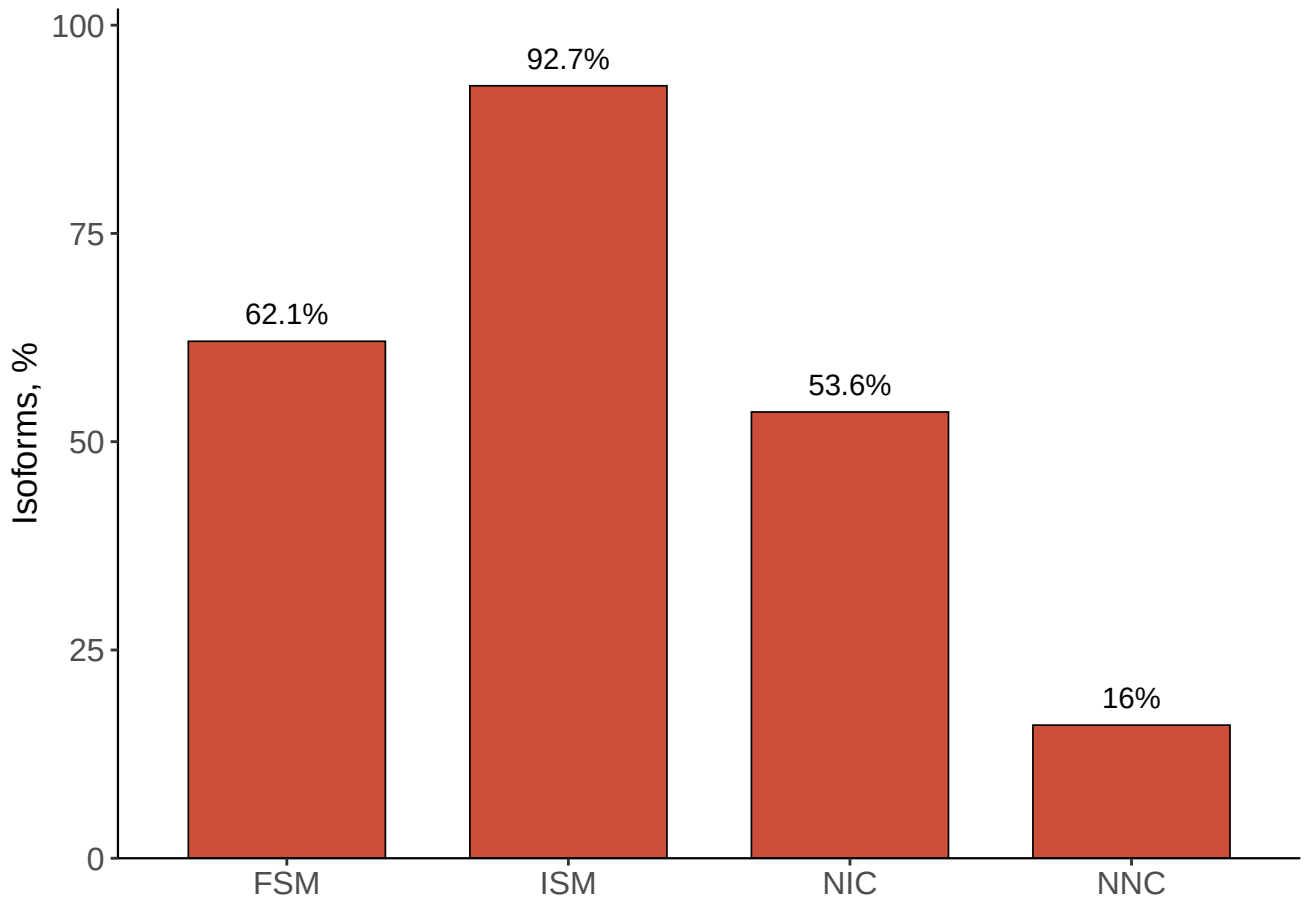
## PolyA Support



## All Canonical Junctions

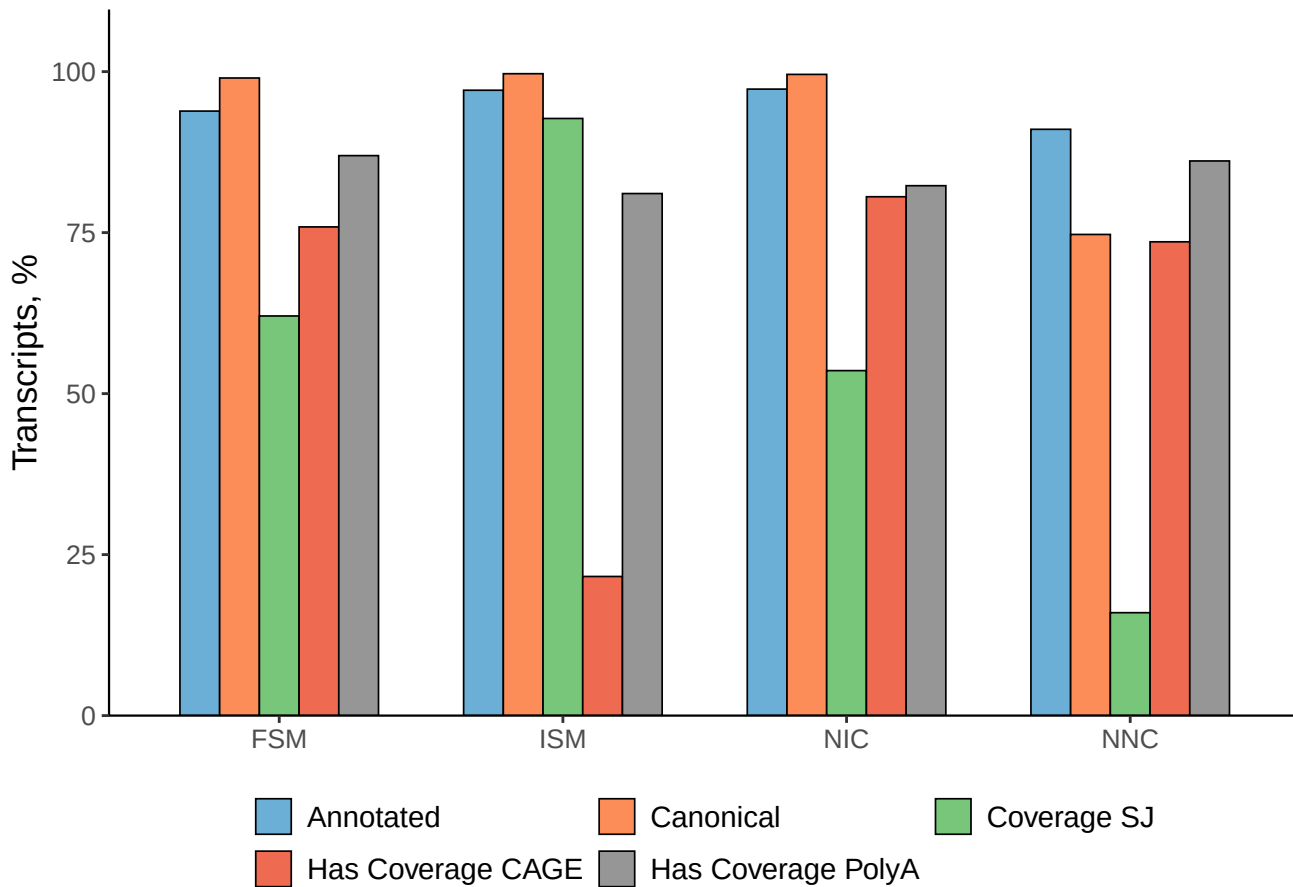


## Splice Junctions With Short Read Coverage



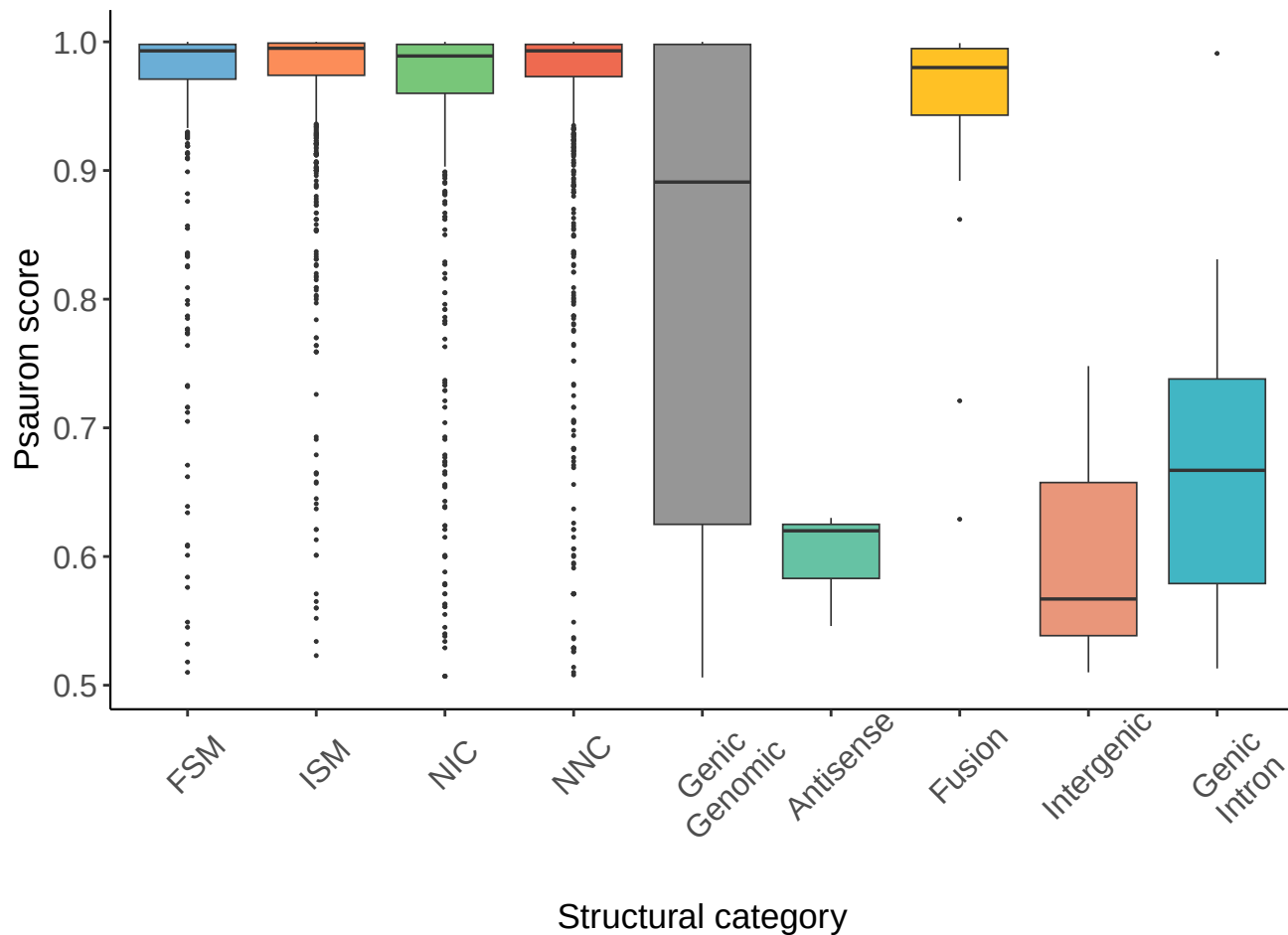


## Good Quality Control Attributes Across Structural Categories

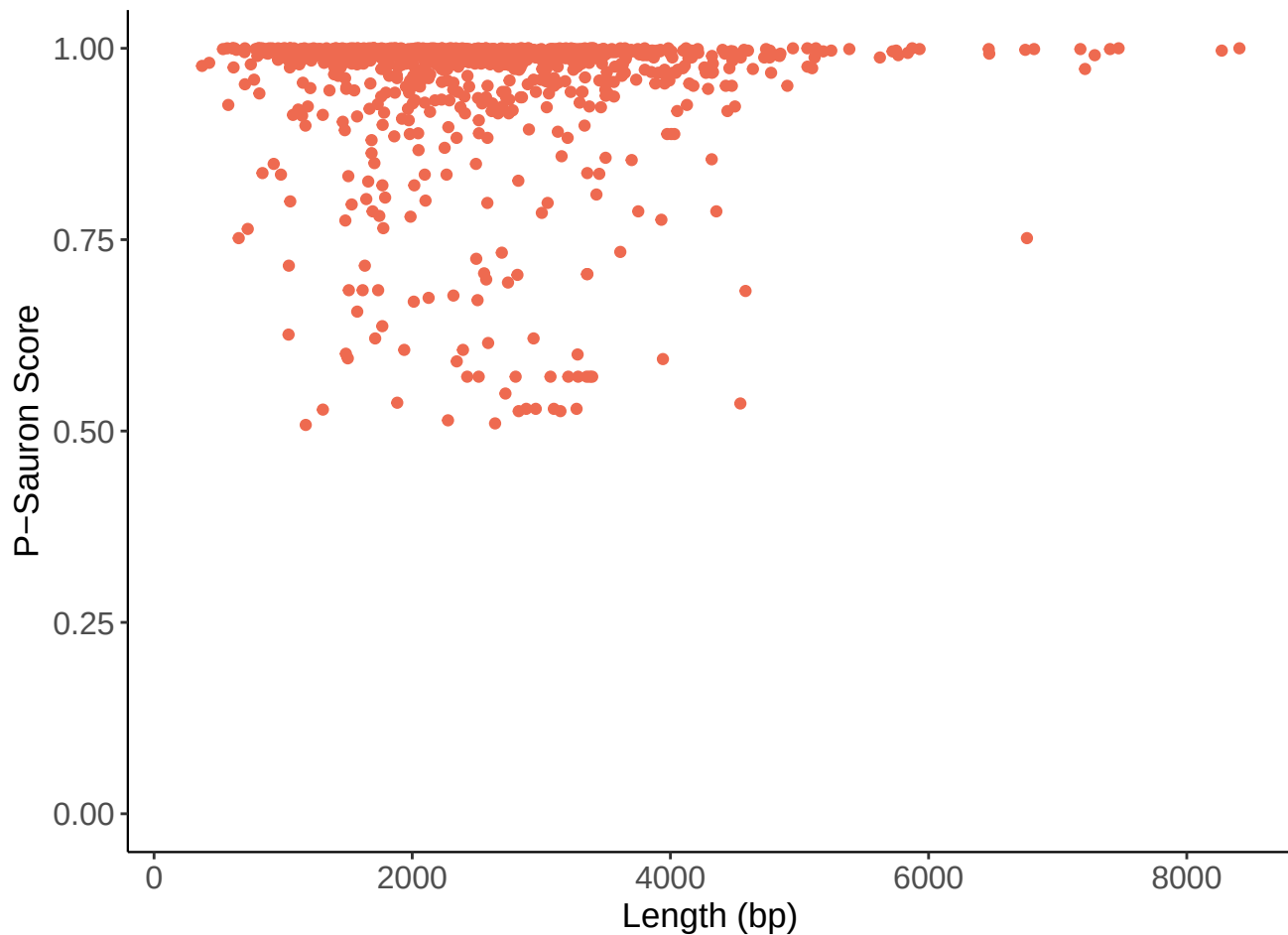


*TransDecoder2 CDS prediction statistics*

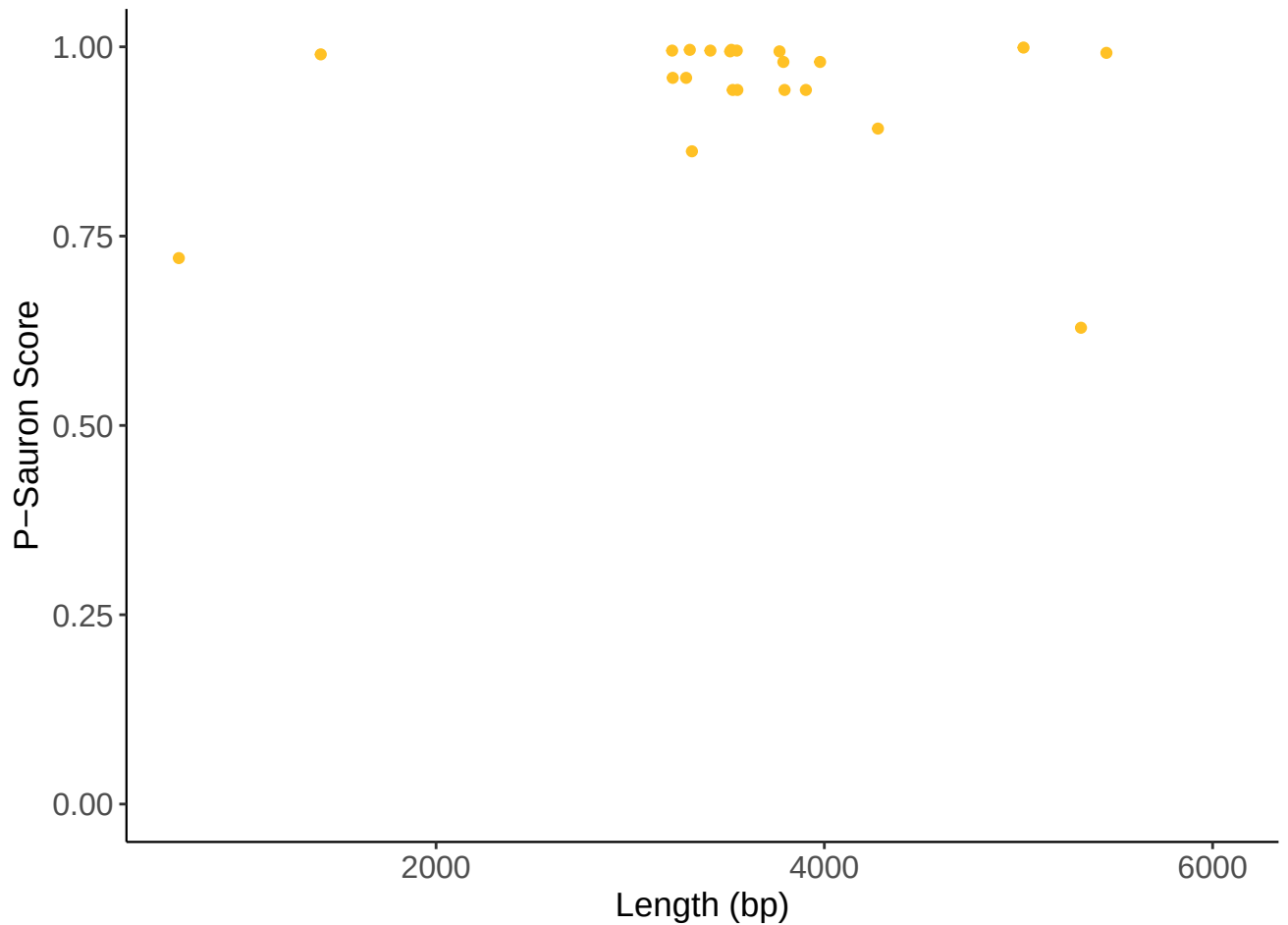
Psauron score per structural category



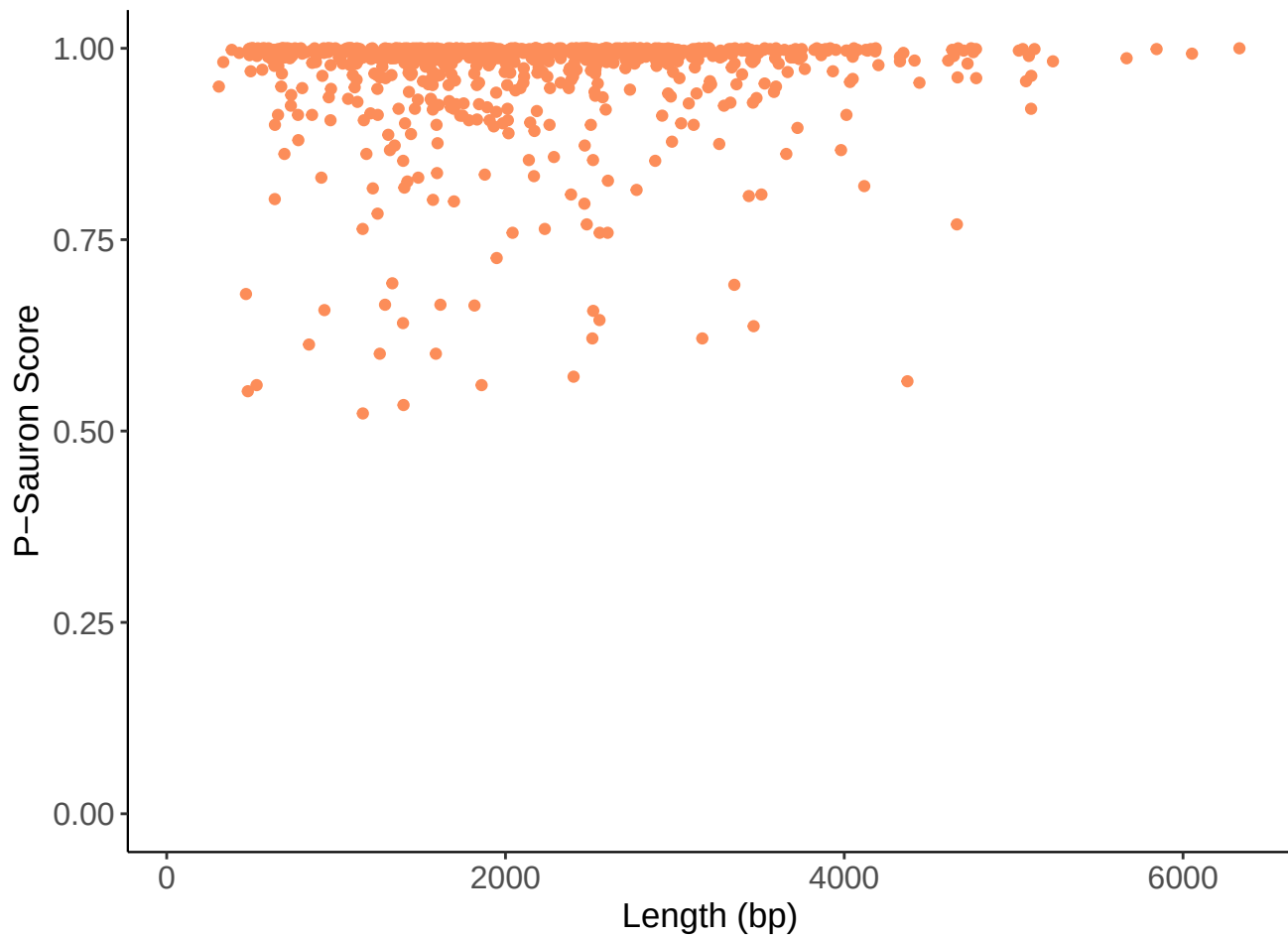
P-Sauron Score vs Length (NNC)



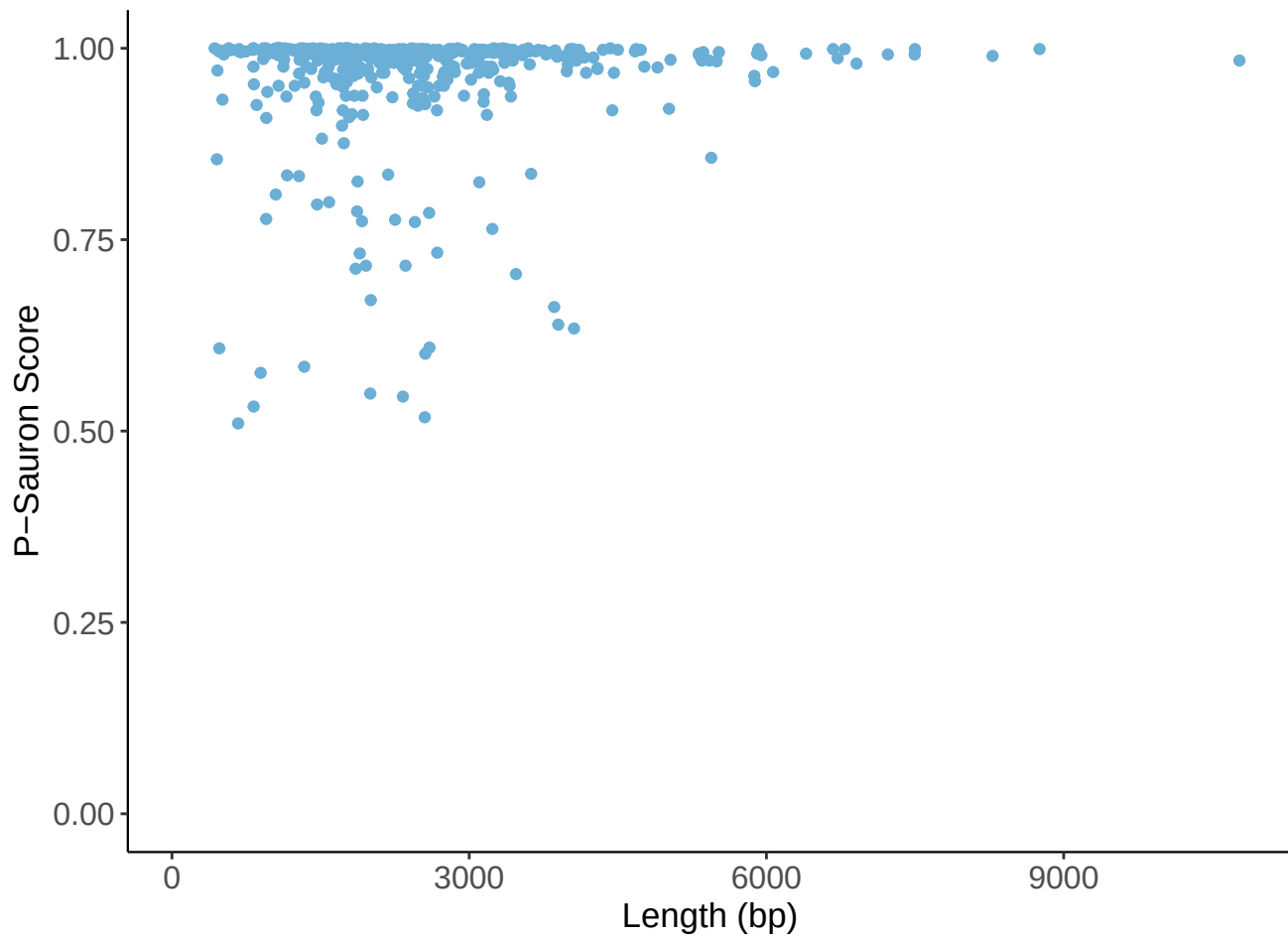
P-Sauron Score vs Length (Fusion)



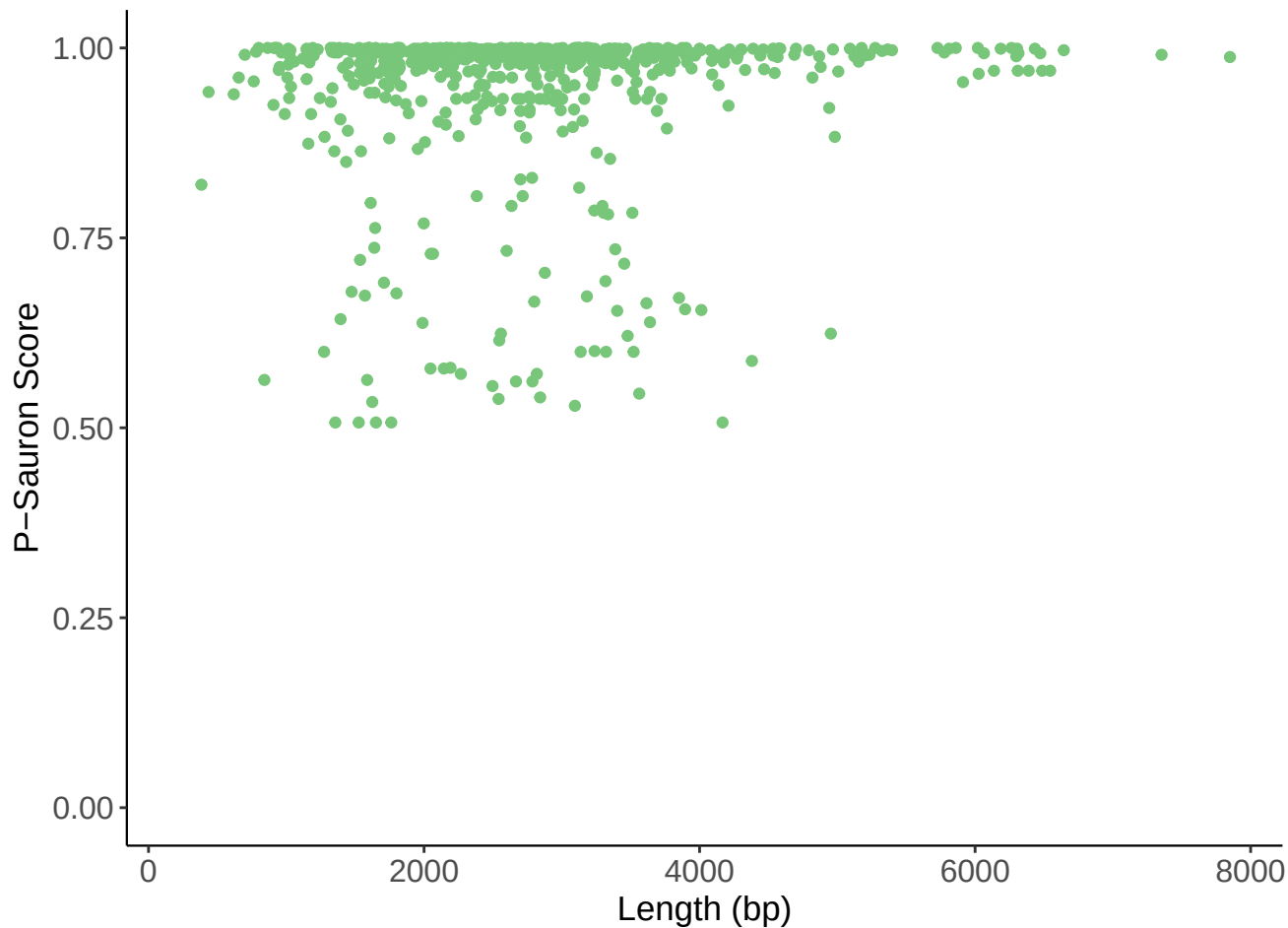
P-Sauron Score vs Length (ISM)



P-Sauron Score vs Length (FSM)

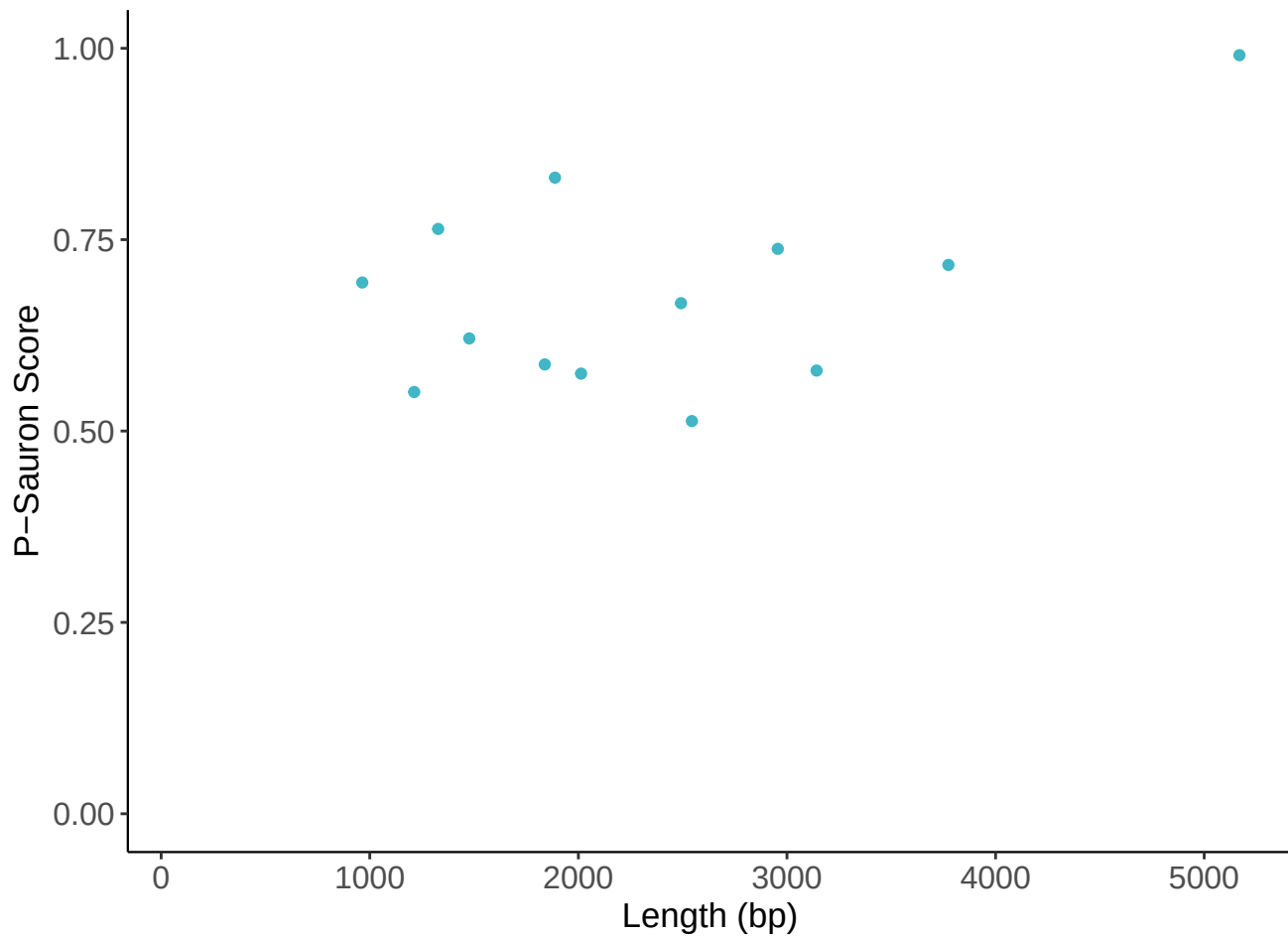


P-Sauron Score vs Length (NIC)

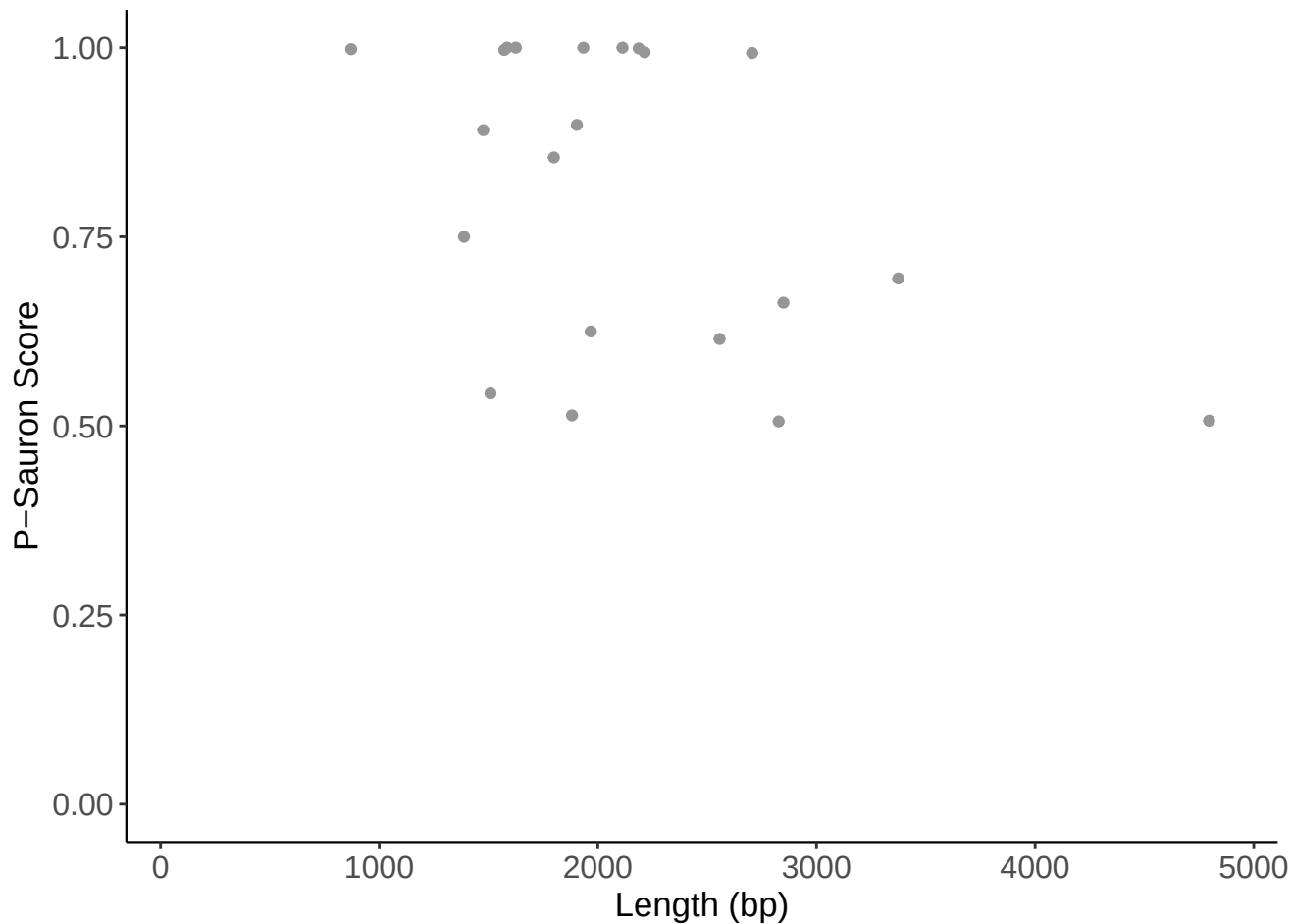




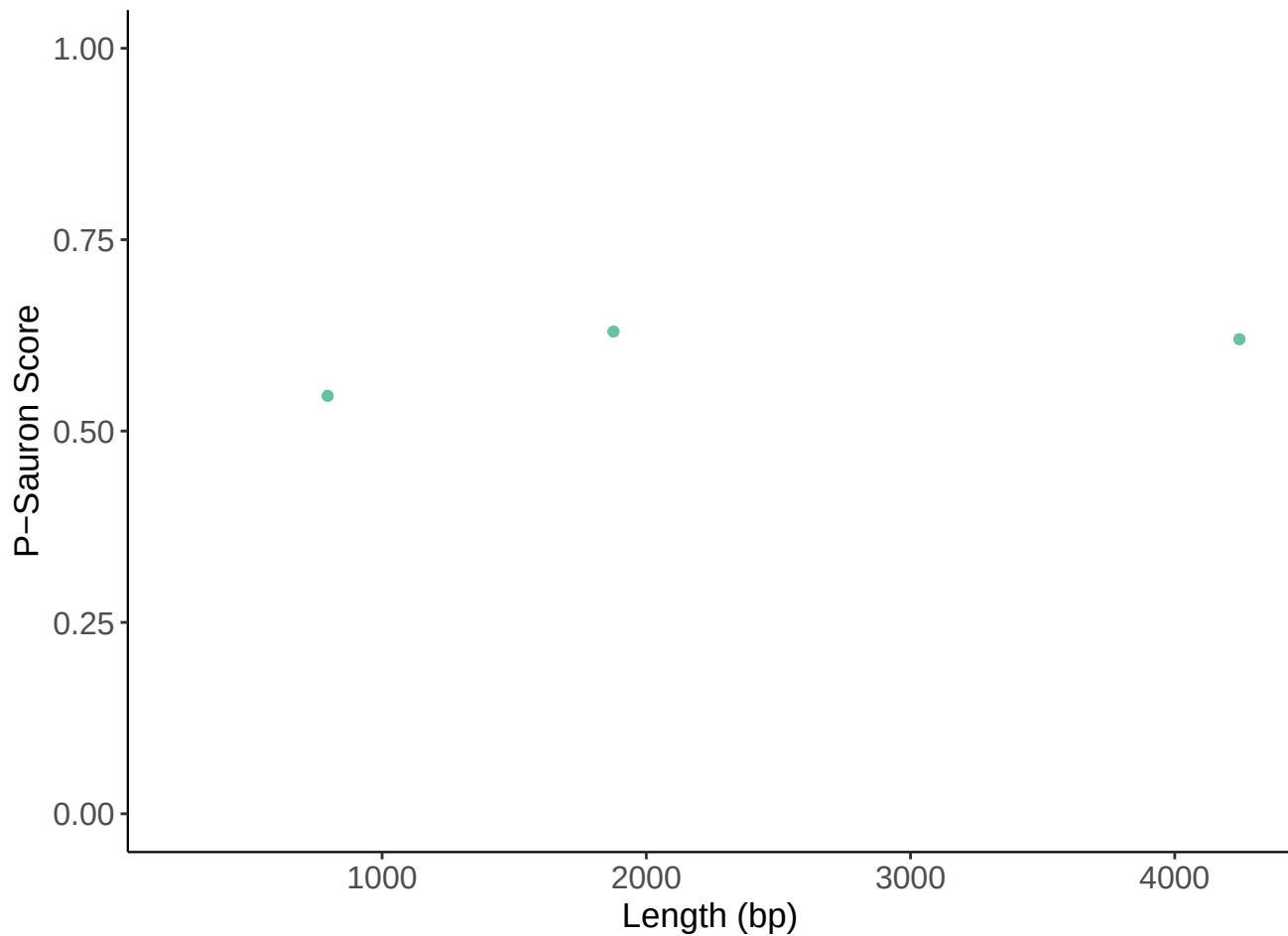
P-Sauron Score vs Length (Genic Intron)



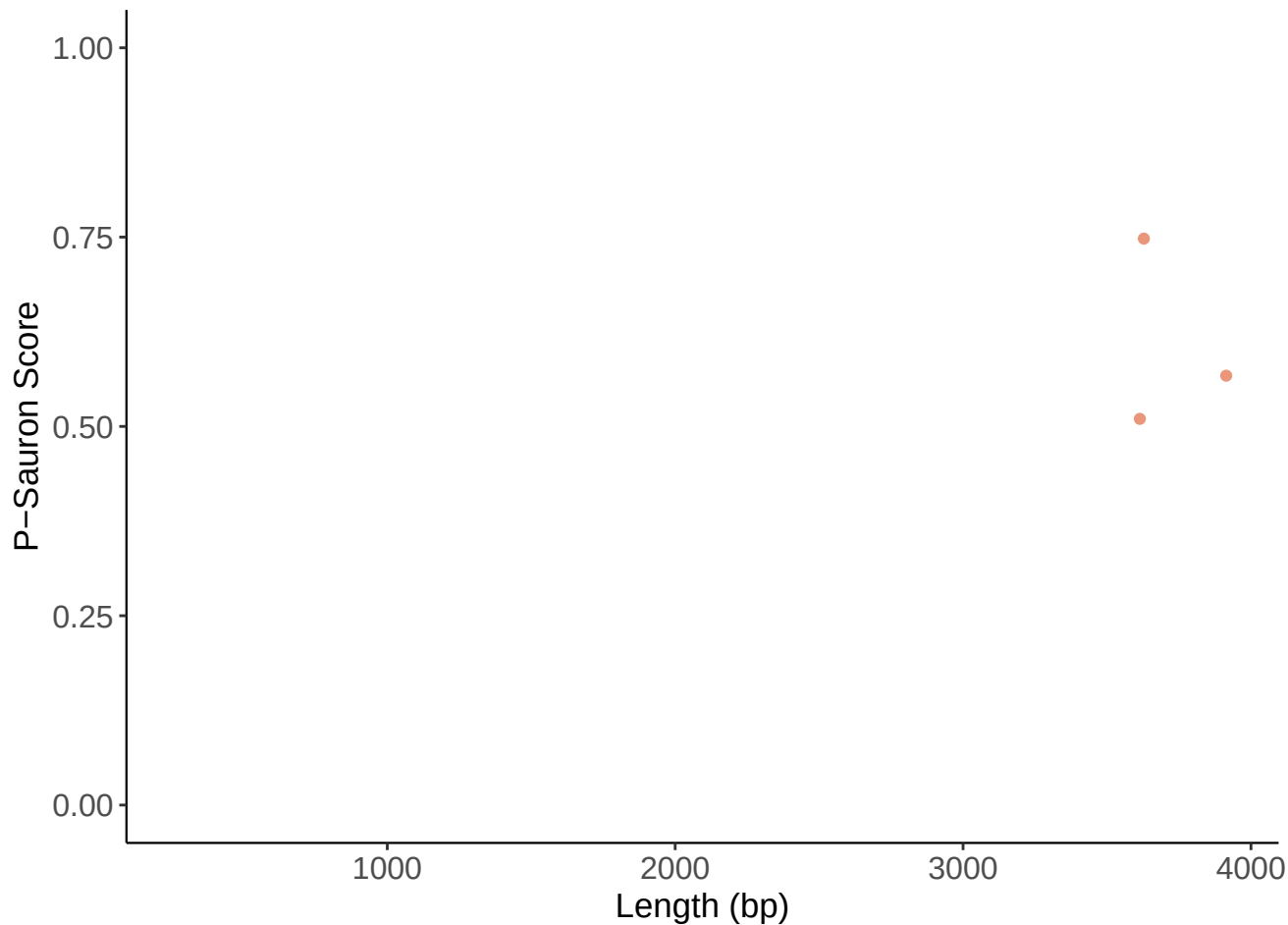
P-Sauron Score vs Length (Genic Genomic)



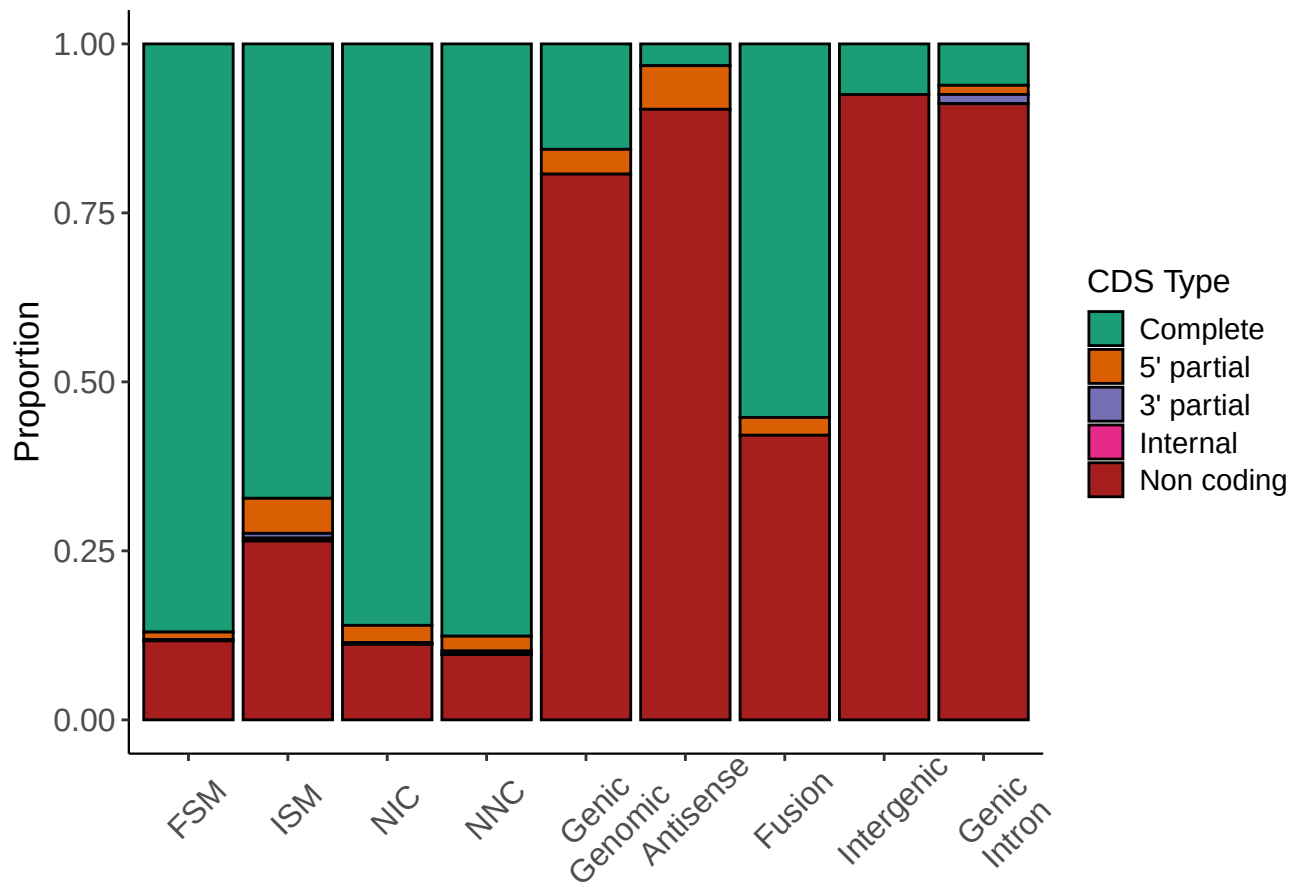
P-Sauron Score vs Length (Antisense)



P-Sauron Score vs Length (Intergenic)



# CDS type by subcategory



Psauron score per ORF type

